

Wallet Guide

The Zcash wallet layer: keys, transactions, and light-client synchronisation

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Abstract

Assuming the cryptographic primitives of the *Crypto Guide*, the best-chain model of the *Consensus Guide*, and the Orchard protocol of the *Ironwood Guide*, this volume of *The Zcash Arboretum* documents the deployed wallet layer above them: hierarchical key derivation (ZIP-32), diversified and unified addresses (ZIP-316), the conventional fee (ZIP-317), the transaction-construction pipeline of librustzcash, partially created transactions (PCZT), the transaction lifecycle from broadcast through expiry (ZIP-203), payment requests (ZIP-321), and light-client synchronisation. The synchronisation chapters cover the compact-block format (ZIP-307) and its deployed divergences, the light-client service surface as `lightwalletd`, `Zaino`, and `Zebra` implement it, the scanning pipeline of librustzcash with its verify-first reorg discipline, commitment-tree synchronisation, and what the server learns. Every derived quantity carries its exact derivation.

The compact-block wire format is deployed but underspecified; where the draft ZIP and the deployed implementations disagree, this volume documents the deployment and flags the divergence.

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1 Hierarchical key derivation (ZIP-32)

The *Ironwood Guide* constructs every Orchard key — ask , nk , r_{ivk} , the full viewing key, ivk , dk , ovk , and the diversified addresses — from a single 32-byte spending key sk , which it takes from a uniformly sampled 256-bit candidate after validity checks, deferring hierarchical derivation to ZIP-32 (§“The recipient: keys and addresses” there). This section supplies that origin. A deployed wallet does not draw an independent sk per account: it draws one *seed* and derives every account’s sk from it deterministically, so that a single backup recovers every key the wallet will ever hold. We develop ZIP-32’s Orchard instantiation: the hardened-only derivation tree, the standard account path $m_{\text{Orchard}}/32'/\text{coin_type}'/\text{account}'$, the *internal* keys that stand in for a change path level, the fingerprints that name keys and seeds, and the serialised encodings — together with where the deployed stack (orchard, zip32, zcash_keys, zcash_client_backend, zcash_client_sqlite) enforces each rule, and where its documentation and its code disagree.

1.1 Wallet seeds

The root secret of the whole wallet is a byte string S , the *seed*. ZIP-32 imposes two requirements (ZIP-32, “Wallet seeds”): the seed **MUST** be 32 to 252 bytes long, and it **MUST** be generated with at least 256 bits of entropy — a requirement that extends to any mnemonic phrase from which the seed is obtained (ZIP-315 gives further guidance). The rationale is the seed’s position in the hierarchy: every key of every account derives deterministically from S , so a lower-entropy seed is a weak link for all of them at once, and — unlike an attack on one key — a brute-force search over low-entropy seeds runs against many wallets simultaneously.

Remark 1.1 (Where the deployed stack enforces the seed bounds). Enforcement is patchier than the **MUST** suggests, and one doc comment is wrong. The derivation function itself does not check: neither `ExtendedSpendingKey::master(orchard, src/zip32.rs)` nor `HardenedOnlyKey::master(zip32 0.2.0, src/hardened_only.rs)` inspects the seed length — BLAKE2b hashes whatever it is given — although the former’s doc comment claims “Panics if the seed is shorter than 32 bytes or longer than 252 bytes”; the claimed panic does not exist. One layer up, `UnifiedSpendingKey::from_seed(zcash_keys, src/keys.rs)` panics on seeds shorter than 32 bytes and checks no upper bound. The trait documentation of `WalletWrite::create_account` and `import_account_hd(zcash_client_backend, src/data_api.rs)` declares a panic for seeds outside 32..252 bytes, and `zcash_client_sqlite` enforces both bounds — indirectly, because it computes a seed fingerprint (Definition 1.13) via `SeedFingerprint::from_seed`, which returns `None` outside 32..252 bytes (zip32 0.2.0, `src/fingerprint.rs`; `zcash_client_sqlite`, `src/lib.rs`). The ZIP requirement therefore holds at the top of the deployed stack, not in the derivation function beneath it.

1.2 Hardened-only derivation

ZIP-32 obtains Orchard keys by instantiating its generic *hardened-only key derivation* process with two domain parameters: the master-key personalisation `Orchard.MKGDomain = ``ZcashIP32Orchard``` (16 bytes) and the child-derivation lead byte `Orchard.CKDDomain = 0x81` (ZIP-32, “Orchard key derivation”). Non-hardened derivation — the BIP-32 mode in which a public key derives child public keys — is not defined for Orchard at all. That choice fixes the shape of the extended keys: an *Orchard extended spending key* is the pair (sk, c) of a normal 32-byte spending key and a 32-byte *chain code* c , and ZIP-32 defines *no* Orchard extended full viewing key, because with only hardened derivation a full viewing key confers no child-derivation capability and so has no use for a chain code (ZIP-32, “Orchard extended keys”; contrast the Sapling sections of the same ZIP, which do define extended FVKs). Deployed, `ExtendedSpendingKey` carries the pair inside a `HardenedOnlyKey` together with bookkeeping fields `depth`, `parent_fvk_tag`, and `child_index`.

Definition 1.2 (Orchard master key generation). Given a seed S of 32 to 252 bytes, compute

$$I := \text{BLAKE2b-512}(\text{'ZcashIP32Orchard'}, S), \quad I = I_L \parallel I_R$$

with I_L, I_R the first and last 32 bytes — unkeyed BLAKE2b-512 with the 16-byte personalisation above — and set $\text{sk}_m := I_L$, $c_m := I_R$. The *master node* of the Orchard tree is $m_{\text{Orchard}} := (\text{sk}_m, c_m)$ (ZIP-32, “Hardened-only master key generation” instantiated per “Orchard master key generation”). Deployed master generation additionally requires sk_m to be valid in the sense of Definition 1.9, else it fails with `Error::InvalidSpendingKey`.

Definition 1.3 (Orchard child key derivation). Write $i' := i + 2^{31}$ for the *hardened* index derived from $0 \leq i < 2^{31}$ (ZIP-32). Child derivation is defined only for hardened indices: for $i \geq 2^{31}$,

$$\begin{aligned} \text{CKDsk}((\text{sk}_{\text{par}}, c_{\text{par}}), i) : \quad I &:= \text{PRFexpand}_{c_{\text{par}}}(\text{0x81} \parallel \text{sk}_{\text{par}} \parallel \text{I2LEOSP}_{32}(i)), \\ \text{sk}_i &:= I_L, \quad c_i := I_R, \end{aligned}$$

where $\text{I2LEOSP}_{32}(i)$ is the 4-byte little-endian encoding of i and PRFexpand is the BLAKE2b-512 expansion function of the *Ironwood Guide* (personalisation `'Zcash_ExpandSeed'`); derivation for $i < 2^{31}$ fails (ZIP-32, “Hardened-only child key derivation” and “Orchard child key derivation”).

Note the division of labour in the PRF call: the parent *chain code* c_{par} is the PRF key, and the parent spending key is message data. Continuing the tree therefore requires the pair — the 32-byte sk alone is a leaf capability, which is exactly the property the account path below exploits when it discards the chain code. Deployed derivation also caps the depth: the `depth` field is a byte incremented with `checked_add`, so a child of a depth-255 key cannot exist and the attempt fails with `Error::MaxDerivationDepth`.

Remark 1.4 (The general form: triples, and sibling instantiations). ZIP-32’s hardened-only child derivation in full generality takes a triple $(i, \text{lead}, \text{tag})$ and appends $\parallel [\text{lead}] \parallel \text{tag}$ to the PRF message; each distinct triple yields an independent output. For $\text{lead} = 0$ and $\text{tag} = []$ the encoding *omits* both fields, making it byte-compatible with the earlier definition of CKDh that lacked them; Orchard child derivation always uses $\text{lead} = 0$, $\text{tag} = []$, which is why Definition 1.3 shows no such suffix (ZIP-32, “Hardened-only child key derivation” and “Orchard child key derivation”). ZIP-32 defines no path *notation* for non-zero lead or non-empty tag. The deployed encoder implements the omission literally: `PrfExpand::with` skips $[\text{lead}] \parallel \text{tag}$ when the lead is zero and the tag empty. The same hardened-only process has two sibling instantiations outside the Orchard tree: *registered* key derivation (personalisation `'ZIPRegistered_KD'`, lead byte `0xAC`, subtree rooted at *ZipNumber'* from an IKM of $[\text{len}] \parallel \text{ContextString} \parallel [\text{len}] \parallel S$) and the deprecated *ad-hoc* derivation (personalisation `'ZcashArbitraryKD'`, lead byte `0xAB`) (ZIP-32, “Specification: Registered key derivation” and “Specification: Ad-hoc key derivation (deprecated)”). They matter here only as neighbours in the domain-separation registry of Table 1.

1.3 The standard account path

Definition 1.5 (Orchard account path). The account-level key for account number $\text{account} \in \{0, \dots, 2^{31} - 1\}$ is derived along the path $m_{\text{Orchard}}/32'/\text{coin_type}'/\text{account}'$:

$$(\text{sk}_{\text{acct}}, c_{\text{acct}}) := \text{CKDsk}\left(\text{CKDsk}\left(\text{CKDsk}(\text{MKGh}^{\text{Orchard}}(S), 32'), \text{coin_type}'\right), \text{account}'\right),$$

after which the wallet keeps sk_{acct} and discards c_{acct} . The *purpose* level is $32' = 0x80000020$, registered per BIP 43; the *coin type* follows the SLIP 44 registry — 133 on Mainnet, 1 on Testnet and Regtest (all testnets share coin type 1) — and accounts are numbered sequentially from 0. Wallets MUST support this path for any account below 2^{31} (ZIP-32, “Key path levels” and “Orchard key path”).

The path terminates at the account. Addresses are not further path children: each account exposes 2^{88} diversified addresses per scope, images of the diversifier indices under FF1 (ZIP-32, “Orchard key path”; the construction is in the *Ironwood Guide*). Nor is there a change level between account and address, where BIP-44-style transparent paths put one: shielded change returning to the originating account is indistinguishable on-chain from any other output, so a change subtree would separate nothing an observer can see, and the separation the wallet itself needs — telling its own change apart from external receipts — is achieved one level lower by the *internal key derivation* of §1.5 (ZIP-32, “Key path levels”).

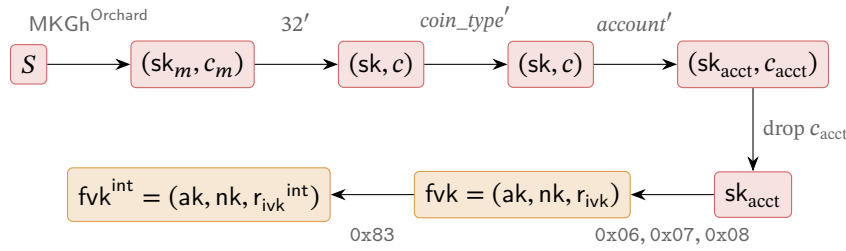


Figure 1: The deployed Orchard key path. The first horizontal arrow is master-key generation; each subsequent arrow in the top row is one CKDsk call (Definition 1.3): a PRFexpand invocation with lead byte 0x81, keyed by the parent chain code. The chain code travels only as far as the account node (sk_{acct}, c_{acct}) ; the wallet layer keeps the bare sk_{acct} , from which the external full viewing key fvk (*Ironwood Guide*, lead bytes 0x06/0x07/0x08) and the internal fvk^{int} (Definition 1.6, lead byte 0x83) both derive. Red marks spend capability, orange full-viewing capability, per the series palette.

Deployed path derivation. Function `SpendingKey::from_zip32_seed(seed, coin_type, account)` rejects $coin_type \geq 2^{31}$ with `Error::InvalidChildIndex`, runs `ExtendedSpendingKey::from_path` over the three hardened indices of Definition 1.5, and returns only the 32-byte account spending key — the chain code is discarded, so the wallet layer retains no Orchard child-derivation capability below the account level. Above it, `UnifiedSpendingKey::from_seed` stores that plain `SpendingKey`, and the unified full viewing key’s Orchard component is the `FullViewingKey` computed from it. The zip32 crate types the path levels: `AccountId` is a 31-bit integer whose `TryFrom<u32>` rejects values at or above 2^{31} and which always enters paths hardened, and `ChildIndex` is hardened-only — `from_index` returns `None` unless the hardened bit is set, and `hardened(v)` asserts $v < 2^{31}$ and stores $v + 2^{31}$. A distinguished index `ChildIndex::PRIVATE_USE = 0x7fffffff'` is reserved for private-use subtrees; `zcashd`’s legacy Sapling account uses it, and we find no Orchard use (ZIP-32, “Sapling key path”; zip32 0.2.0, `src/lib.rs`).

1.4 From the account key to addresses: the layer already built

Everything below sk_{acct} is the hierarchy the *Ironwood Guide* constructs in “The recipient: keys and addresses”, and we do not re-derive it: ask, nk, r_{ivk} from PRFexpand lead bytes 0x06, 0x07, 0x08 with the `ToScalar/ToBase` reductions and their bias analysis; the sign-canonicalised $ak = [ask]^{SpendAuth}$; $fvk = (ak, nk, r_{ivk})$; $ivk = \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}(ak), nk)$; the (dk, ovk) split of a single PRFexpand output with lead byte 0x82; and the FF1 diversified addresses with default index 0 (protocol specification §4.2.3, “Orchard Key Components”).

What ZIP-32 adds at the wallet level is the capability boundary of the full viewing key. The holder of fvk obtains, for *both* the external and the internal scope of §1.5: the incoming viewing keys, the diversifier keys and outgoing viewing keys, and hence every diversified address of the account; the *index* behind any of its addresses, since the diversifier map is a keyed permutation and $FF1\text{-}AES_{256}^{-1}_{dk}(d)$ — $FF1$ decryption under the empty tweak — recovers the index; and the scope of any of its addresses, by trying external then internal. Two contrasts with Sapling sharpen the picture (ZIP-32, “Orchard diversifier and OVK derivation”): the diversifier key derives from the *full viewing key* rather than the extended spending key, so viewing capability suffices to locate an address’s position in the sequence; and all 2^{88} diversifiers are valid, so no rejection loop exists and the default diversifier is d_0 . What the full viewing key can *not* do: derive ask, or derive any child account — the former by DL hardness for $ak = [ask]G^{\text{SpendAuth}}$, the latter because only hardened derivation exists and no extended FVK is defined (§1.2). Finally, no mechanism derives an outgoing viewing key per *diversified address*: payments are sent from accounts, with external or internal scope, not from individual addresses (ZIP-32, “Orchard diversifier and OVK derivation”, citing the ZIP-316 rationale “Usage of Outgoing Viewing Keys”).

1.5 Internal keys: change and auto-shielding

The path of Definition 1.5 reserved no change level, so the separation between externally visible receiving and wallet-internal operations — change outputs, and the auto-shielding of transparent funds — happens below the account instead: every account key yields a second, *internal* full viewing key. The mechanism was implemented in `zcashd` as part of ZIP-316 support and applies to keys that are children of the account level, i.e. account keys (ZIP-32, “Orchard internal key derivation”).

Definition 1.6 (Orchard internal key derivation). Given an external full viewing key $fvk = (ak, nk, r_{ivk})$, let $K := LE_{256}(r_{ivk})$ and

$$\begin{aligned} r_{ivk}^{\text{int}} &:= \text{ToScalar}(\text{PRFexpand}_K(0x83 \parallel I2LEOSP_{256}(ak) \parallel I2LEOSP_{256}(nk))) \in \mathbb{F}_{p_{\text{Vesta}}}, \\ fvk^{\text{int}} &:= (ak, nk, r_{ivk}^{\text{int}}), \end{aligned}$$

i.e. $\text{DeriveInternalFVK}^{\text{Orchard}}$ modifies *only* r_{ivk} (ZIP-32, “Orchard internal key derivation”; protocol specification §4.2.3). The downstream keys then follow the standard FVK-level rules of the *Ironwood Guide*, applied to fvk^{int} : $ivk^{\text{int}} := \text{Commit}_{r_{ivk}^{\text{int}}}^{\text{ivk}}(\text{Extract}(ak), nk)$ and $(dk^{\text{int}}, ovk^{\text{int}})$ from the $0x82$ split keyed by $LE_{256}(r_{ivk}^{\text{int}})$. The *expanded internal spending key* is $(ask, nk, r_{ivk}^{\text{int}})$ — the same ask — and no 32-byte internal spending key exists (ZIP-32, “Orchard internal key derivation”).

The shape is deliberately that of the (dk, ovk) derivation: the same key K , the same message body $I2LEOSP_{256}(ak) \parallel I2LEOSP_{256}(nk)$, a different lead byte. Its consequences: ak and nk are unchanged (the protocol specification notes $ak^{\text{int}} = ak$ and $nk^{\text{int}} = nk$, §4.2.3), so spend authorisation and nullifier derivation are common to both scopes, while ivk , dk , and ovk are re-derived from r_{ivk}^{int} . The construction does not explicitly enforce that the two scopes’ derived keys or addresses differ; equality would be a negligible cryptographic collision, not an impossible case. Except with that negligible probability, a note addressed under the internal scope does not trial-decrypt under the external ivk (*Ironwood Guide*, “Delivery: the note reaches its owner”), which is precisely the separation a wallet wants when it hands its external incoming viewing key to a payment processor. Note that the internal FVK is *computable from* the external FVK — the derivation consumes only (ak, nk, r_{ivk}) — so disclosing the full viewing key disclosed both scopes all along; the boundary runs at the incoming-viewing tier, not the full-viewing tier. Unlike its Sapling counterpart, the construction does not rest on non-hardened derivation, which is undefined for Orchard (ZIP-32, “Orchard internal key derivation”).

Deployed, the `zip32` crate’s `Scope` enum `{External, Internal}` names the two scopes and or-

chard re-exports it; `FullViewingKey::rivk(Scope)`, `to_ivk`, `to_ovk`, and `derive_internal` implement Definition 1.6. The change path is concrete: `zcash_client_backend` sends Orchard change to `ufvk.orchard().address_at(0u32, Scope::Internal)` — the internal-scope address at diversifier index 0.

The transparent pool needs the same external/internal ovk split — its auto-shielding transactions carry shielded outputs whose outgoing ciphertexts want a viewing key — but P2PKH keys (pay-to-public-key-hash: the output script pays to the hash of a single public key) have no protocol-level ovk to scope, so ZIP-316 synthesises the pair.

Definition 1.7 (Transparent internal and external OVK). For a transparent P2PKH account with BIP-32 account-level extended public key (c, pk) — chain code c , keying PRFexpand as a 256-bit key, and $\text{ser}(pk)$ the 33-byte compressed public key —

$$I_{\text{ovk}} := \text{PRFexpand}_c(0\text{xd}0 \parallel \text{ser}(pk)), \quad \text{ovk}_{\text{external}} := I_{\text{ovk}}[0..32], \quad \text{ovk}_{\text{internal}} := I_{\text{ovk}}[32..64]$$

(ZIP-316, *Deriving Internal Keys*; lead byte in Table 1).

Remark 1.8 (Byte-level normalisation between ZIP and code). ZIP-32 writes the PRF key as the bit sequence $\text{I2LEBSP}_{256}(r_{\text{ivk}})$ where PRFexpand consumes bytes; the code passes `rivk.to_repr()`, the 32-byte little-endian encoding. The two are identical under the LE_{256} convention this series uses, and we write LE_{256} throughout, as the *Ironwood Guide* already does. Likewise the ak bytes fed to the 0x82 and 0x83 messages are `SpendValidatingKey::to_bytes()`, the 32-byte RedPallas point encoding — which, because key generation forces the sign bit to zero (the canonicalisation of the *Ironwood Guide*), coincides with $\text{I2LEOSP}_{256}(\text{Extract}(ak))$, matching the specification’s $\text{I2LEOSP}_{256}(ak)$ for $ak \in \{0, \dots, p_{\text{Pallas}} - 1\}$ (protocol specification §4.2.3 and the `AuthSignPublic` type). The nk bytes are the base-field representation.

1.6 Key validity and derivation failure

Definition 1.9 (Valid Orchard spending key). A 32-byte string sk is a *valid* spending key iff all three hold: (i) $\text{ask} \neq 0$; (ii) the external $\text{ivk} = \text{Commit}_{r_{\text{ivk}}}^{\text{ivk}}(\text{Extract}(ak), nk) \notin \{0, \perp\}$; (iii) the internal $\text{ivk}^{\text{int}} = \text{Commit}_{r_{\text{ivk}}^{\text{int}}}^{\text{ivk}}(\text{Extract}(ak), nk) \notin \{0, \perp\}$. Function `SpendingKey::from_bytes` returns `None` exactly when one of the three fails, and the specification’s key-generation algorithm mandates exactly these three discard conditions — “discard this key and repeat with a new sk ” (protocol specification §4.2.3, “Orchard Key Components”). `FullViewingKey::from_bytes` re-checks both scopes when parsing an FVK from bytes.

Remark 1.10 (Derivation failure has no specified recovery). ZIP-32 notes that a child spending key may yield an invalid external or internal full viewing key “with small probability” (ZIP-32, “Orchard child key derivation”) but prescribes no wallet behaviour when it does; the specification’s “repeat with a new sk ” addresses a freshly random key, not a fixed HD path, where there is nothing to redraw. The deployed stack hard-errors: `master` and `derive_child` return `Err(Error::InvalidSpendingKey)`, `from_path` propagates it, and `zcash_keys` surfaces it as `DerivationError::Orchard(orchard, src/zip32.rs; zcash_keys, src/keys.rs)`; no layer implements a skip-to-next-index convention. This behaviour is normative in practice: an account index at which derivation fails simply has no key, and no deployed convention reassigns it. The event is negligibly rare — neither ZIP-32 nor the specification quantifies “small probability”, so Proposition 1.11 derives a bound — and the gap is theoretical.

Proposition 1.11 (Probability of an invalid key). *Model the PRFexpand outputs and the GroupHash-derived Sinsemilla generators as uniform and independent (the random-oracle model in which the Ironwood Guide analyses both primitives). A uniformly random 32-byte sk then fails the conditions of Definition 1.9 with probabilities*

$$\Pr[\text{ask} = 0] < 2^{-254}, \quad \Pr[\text{ivk} \in \{0, \perp\}] < 307 \cdot 2^{-254} < 2^{-245} \quad (\text{per scope}),$$

and is invalid with probability below 2^{-244} .

Proof. The scalar ask is the ToScalar image of the 0x06 expansion (the conditional negation that canonicalises ak preserves zeroness), so ask = 0 iff a uniform 512-bit integer reduces to 0 modulo p_{Vesta} . Exactly $\lfloor 2^{512}/p_{\text{Vesta}} \rfloor$ of the 2^{512} inputs do, and $p_{\text{Vesta}} > 2^{254}$ gives $\lfloor 2^{512}/p_{\text{Vesta}} \rfloor < 2^{258}$, hence $\Pr[\text{ask} = 0] < 2^{258} \cdot 2^{-512} = 2^{-254}$.

For ivk, the two memberships separately. *Zero:* conditioned on the Sinsemilla hash not returning $\perp$, the commitment point is the hash point plus $[r_{\text{ivk}}]H_D$ — the blinding term of the *Ironwood Guide*’s “Sinsemilla: hash, commitment, and short forms” — uniform on the p_{Vesta} -element Pallas group when the blinding scalar is uniform and $H_D \neq \mathcal{O}$. Here r_{ivk} is the ToScalar image of a uniform 512-bit string, so the probability of any particular scalar is at most $\lfloor 2^{512}/p_{\text{Vesta}} \rfloor / 2^{512} < 2^{-254}$. The extractor sends only the identity to 0, because $y^2 = 0^3 + 5$ has no solution in $\mathbb{F}_{p_{\text{Pallas}}}$ — 5 is a non-square there (protocol specification §5.4.9.7, note) — so $\Pr[\text{ivk} = 0] < 2^{-254}$. *Bottom:* the $\perp$ output arises only inside the Sinsemilla hash, since the blinding addition is complete (§5.4.8.4). On the 510-bit Commit^{ivk} message — two 255-bit field encodings, $\lceil 510/10 \rceil = 51$ ten-bit chunks — the hash performs $2 \cdot 51 = 102$ incomplete additions (§5.4.1.9). Each fails only when one operand is the identity or shares the other’s x -coordinate. To avoid assuming uniformity after conditioning on prior success, consider each round’s unconditioned complete-law expression $(A + P) + A$. Conditional on the message and table points, its accumulator is $A = [2^i]Q + C_i$, uniform as the independent generator Q varies. For $P \neq \mathcal{O}$, exceptional incomplete additions require $A \in \{\mathcal{O}, P, -P, -P/2\}$; the separate event $P = \mathcal{O}$ costs at most $1/p_{\text{Vesta}}$. Thus the 51 rounds cost at most $255/p_{\text{Vesta}}$ by a union bound, without independence between rounds. Including the negligible event that the blinding base is the identity still fits the conservative bound $\Pr[\text{ivk} \in \{0, \perp\}] < 307 \cdot 2^{-254} < 2^{-245}$ per scope. The internal scope repeats the argument with the independent $r_{\text{ivk}}^{\text{int}}$ (the 0x83 expansion). In total, $\Pr[\text{sk invalid}] < (1 + 2 \cdot 307) \cdot 2^{-254} < 2^{10} \cdot 2^{-254} = 2^{-244}$. $\square$

The bound makes ZIP-32’s “small probability” concrete: below 2^{-244} per derived key, and below 2^{-242} for the union over the four nodes of the standard account path (master included, Definition 1.2). This is a random-oracle-model bound, including randomness over the generators, not a measured frequency for the one deployed set of fixed generators. Moreover, a concrete input exhibiting $\perp$ would constitute a nontrivial discrete-log relation among the Sinsemilla generators (*Ironwood Guide*, the Sinsemilla collision-resistance proof), so none is expected ever to be observed.

1.7 Fingerprints and tags

Two identifiers name objects of this section without revealing them; both are BLAKE2b-256 digests with dedicated personalisations.

Definition 1.12 (Orchard FVK fingerprint and tag). For a full viewing key $\text{fvk} = (\text{ak}, \text{nk}, r_{\text{ivk}})$, the *fingerprint* is

$$\begin{aligned} \text{FVFP}(\text{fvk}) &:= \text{BLAKE2b-256}(\text{``ZcashOrchardFVFP''}, \text{fvk}_{\text{raw}}), \\ \text{fvk}_{\text{raw}} &:= \text{I2LEOSP}_{256}(\text{ak}) \parallel \text{I2LEOSP}_{256}(\text{nk}) \parallel \text{I2LEOSP}_{256}(r_{\text{ivk}}), \end{aligned}$$

the digest of the 96-byte raw FVK encoding fvk_{raw} (protocol specification §5.6.4.4). The *tag* is its first 4 bytes and MUST NOT be assumed unique; the master key’s parent tag is the 4 zero bytes (ZIP-32, “Orchard Full Viewing Key Fingerprints and Tags”). Deployed as `FvkFingerprint` and `FvkTag`; child derivation stamps each child with `parent_fvk_tag` = the tag of the parent’s FVK.

Definition 1.13 (Seed fingerprint). For a seed S of 32 to 252 bytes, with $[\text{length}(S)]$ its single length byte,

$$\text{SeedFP}(S) := \text{BLAKE2b-256}(\text{'`Zcash_HD_Seed_FP' '}, [\text{length}(S)] \| S);$$

the string form is Bech32m (the checksummed base-32 string encoding of BIP-350: a human-readable part, a separator 1, and base-32 data characters ending in a six-character checksum) with human-readable part `zip32seedfp`, and no short tag is defined; the length prefix was chosen to match `zcashd`’s implementation (ZIP-32, “Seed Fingerprints”). Deployed, `SeedFingerprint::from_seed` returns `None` outside the 32..252-byte bounds, and `zcash_client_sqlite` uses the fingerprint to bind each stored account to the seed that produced it.

1.8 Extended-key encoding and test vectors

Definition 1.14 (Orchard extended spending key encoding). An extended spending key at depth depth , child index i (hardened offset included), with parent tag tag_{par} , serialises as the 73-byte string

$$\text{xsk} := \text{I2LEOSP}_8(\text{depth}) \| \text{tag}_{\text{par}} \| \text{I2LEOSP}_{32}(i) \| c \| \text{sk} \quad (1 + 4 + 4 + 32 + 32 \text{ bytes}),$$

with the master key at depth 0, tag `0x00000000`, index 0. The Bech32 human-readable parts are `secret-orchard-extsk-main` (Mainnet) and `secret-orchard-extsk-test` (Testnet) (ZIP-32, “Orchard extended spending keys”).

ZIP-32 defines this encoding “for completeness” and expects most wallets to expose only the plain `sk` encoding of the protocol specification (§5.6.4.5) — consistent with the deployed stack, which discards the chain code at the account level (§1.3). Indeed we find *no* deployed `librustzcash` code path that emits or parses the Bech32 form: the 73-byte layout is exercised only by the orchard crate’s embedded test-vector checks, and `zcash_keys` stores the raw spending key inside the unified spending key encoding (`orchard`, `src/zip32.rs`, `src/test_vectors/zip32.rs`; `zcash_keys`, `src/keys.rs`). We classify the encoding as specified but unused in the scanned stack.

Remark 1.15 (Test vectors). The ZIP-32 Orchard vectors live in `zcash-test-vectors` under `orchard_zip32`; the orchard crate embeds four of them — the nodes m , $m/1'$, $m/1'/2'$, $m/1'/2'/3'$ from the 32-byte seed `0x00, ..., 0x1f` — each giving `sk`, `c`, the 73-byte `xsk`, and the FVK fingerprint (ZIP-32, “Test Vectors”; `orchard`, `src/test_vectors/zip32.rs`, `src/zip32.rs`). They do *not* cover the standard path of Definition 1.5. Separate key-component vectors ($\text{sk} \rightarrow \text{ask}, \text{ak}, \text{nk}, r_{\text{ivk}}, \text{ivk}$ and the internal-scope components) sit in `src/test_vectors/keys.rs`, exercised by the `keys.rs` tests.

Example 1.16 (The standard path, worked). No published vector covers the path of Definition 1.5, so we compute one, from the embedded vectors’ seed $S = 0x00, \dots, 0x1f$, for Mainnet (`coin_type = 133`) and account 0:

m	sk	7eee3c1017870990a3dd6891b82f80be8976c1e7dc20d60817a5e88e8b2cd4b8
	c	ab8b7a00509ef20e469b5292b61d474b7cfffcb1657924cda720250ae40526677
$m/32'$	sk	93af0a87c2e4704d8825e145557c6a5d4dcbc9016170e276a2080a2dc9eb0f87
	c	6e1e808d410d6d1f84e62e67c8b60e80e5ee4de5021d575c0433a625d609c45c
$m/32'/133'$	sk	149fcee594a2a03de277ab919f22611d0ef6aa8e8e9a0bef32f01a0bcb0e8c4c
	c	07d13fdeccd21da16b467f8e785edc32f74ceeb541bede0394647f10fdb7114f
$m/32'/133'/0'$	sk	b67d8d87cab9189500afb45dbca9f92c924c1de9d9ee1451be4a78313cb223b4
	c	7ff0a6392e144d4c8f45bca21fb7827b508446ac069890aefbebbe388b7a84c0

The bottom sk is exactly what `SpendingKey::from_zip32_seed` returns for $(S, 133, 0)$; the bottom c is what it discards. The account key’s FVK fingerprint (Definition 1.12) is

da8dab741dc4f6ff8964fbbaa8840270caf47d286e9a0722e6c89e7f2c38189d5

(tag da8dab74), its parent’s fingerprint begins 7010c19b, and the account’s 73-byte encoding of Definition 1.14 is therefore

$$xsk = 03 \parallel 7010c19b \parallel 00000080 \parallel c \parallel sk, \quad 00000080 = I2LEOSP_{32}(0'),$$

with c and sk the bottom table row. The numbers are script-computed per the series conventions: a standalone reimplementation of `MKGhOrchard` and `CKDsk` first reproduces all four embedded vectors byte for byte (sk, c, xsk); the orchard crate, driven as a library, reproduces their FVK fingerprints, returns the same account sk from `from_zip32_seed`, and validates every node above per Definition 1.9.

1.9 The PRFexpand lead-byte registry

One PRF underlies every derivation above and in the *Ironwood Guide*, but domain separation is per PRF *key*, not protocol-wide: under a fixed key, distinct lead bytes provide domain-separated pseudorandom outputs, while separate consumers may reuse a byte under different keys. Sapling derives rcm and esk from lead bytes 0x04 and 0x05 with no ρ suffix — the assignment swapped relative to Orchard’s, which the specification attributes to an oversight (protocol specification §4.7.3, note) — and the Sapling instantiation of ZIP-32 intentionally reuses 0x00–0x02 from the Sapling key components under a different key. Table 1 collects the accounting for the Orchard, Orchard-ZIP-32, and ZIP-316 bytes this series constructs; the Sapling-only bytes (0x00–0x03, 0x10–0x18, of which §2.2 cites 0x10 and 0x16) are accounted in the specification’s list (protocol specification, the `PRFexpand` accounting under “Pseudo Random Functions” and §4.2.3; ZIP-32). The Sprout instantiation of ZIP-32 reserves the lead byte 0x80, the personalisation `“ZcashIP32_Sprout”`, and the HRP `zxsprout/zxtestsprout` — values the ZIP says SHOULD NOT be used in future Zcash-related specifications (ZIP-32, “Values reserved due to previous specification for Sprout”). All Orchard and ZIP-32 constants of those derivations match between the specification text and `zcash_spec` 0.2.1 — the version locked by both the orchard and `librustzcash` workspaces (`Cargo.lock` in each; likewise `zip32` 0.2.0). The additional Ironwood 0x0B derivation is implemented directly in orchard 0.15.0, `src/note.rs`, not in that older `zcash_spec` registry.

2 Diversified and unified addresses (ZIP-316)

An address layer must reconcile two demands that pull apart. Each payee should receive their own address, mutually unlinkable with every other — yet all of them spendable by one key and recoverable from one seed; each shielded protocol solves this with *diversified addresses*. And Zcash is several protocols at once: transparent, Sapling, and Orchard receivers coexist, each with its own format, so a recipient wants to hand a sender *one* string from which the sender’s wallet picks the best receiver both sides support; ZIP-316 solves this above the protocols with the *unified* container encodings. The *Ironwood Guide* (“The recipient: keys and addresses”) already constructs the Orchard-side machinery

Lead byte	PRF key	Message after the lead byte	Output
0x04	rseed	ρ	esk
0x05	rseed	ρ	legacy rcm
0x06	sk	—	ask
0x07	sk	—	nk
0x08	sk	—	r_{ivk}
0x09	rseed	ρ	ψ
0x0B	rseed	pre_rcm[1..137)	Ironwood rcm
0x80	Sprout ZIP-32 child (reserved; SHOULD NOT reuse)		
0x81	c_{par}	$sk_{\text{par}} \parallel I2LEOSP_{32}(i)$	$sk_i \parallel c_i$
0x82	$LE_{256}(r_{ivk})$ (per scope)	$ak \parallel nk$	$dk \parallel ovk$
0x83	$LE_{256}(r_{ivk})$	$ak \parallel nk$	r_{ivk}^{int}
0xAB	ad-hoc ZIP-32 child (deprecated)		
0xAC	registered ZIP-32 child		
0xD0	c (chain code)	$ser(pk)$	$ovk_{\text{ext}} \parallel ovk_{\text{int}}$

Table 1: Lead-byte registry of PRFexpand messages. Field arguments use $I2LEOSP_{256}$; points use their compressed encodings (the canonical ak encoding coincides with its x -coordinate encoding). Here ρ is the 32-byte nullifier of the Action’s spent note (*Ironwood Guide*, “Delivery: the note reaches its owner”). Rows 0x04–0x09 are constructed in the *Ironwood Guide* and carry the Orchard meanings of 0x04/0x05. The 0x0B row uses the 137-byte pre_rcm defined in that volume’s “The note seed: Ironwood derivations (lead byte 0x03)”, with its first byte removed because the column already lists it (ZIP-2005). Sapling reuses both bytes, without the ρ suffix and with the outputs swapped (0x04 $\rightarrow$ rcm, 0x05 $\rightarrow$ esk), under its own per-note rseed keys (protocol specification §4.7.3, note). Rows 0x81 and 0x83 are constructed in Definitions 1.3 and 1.6; 0xD0 in Definition 1.7; 0x80, 0xAB, 0xAC belong to other subsystems (ZIP-32) and appear for the completeness of the accounting.

— the key tiers, the FF1-AES256 diversifier map $d := \text{FF1-AES256}_{\text{dk}}(d_{\text{plain}})$, the diversified base $g_d = \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-gd}, d)$, and the rationale for a keyed permutation over the index space — and we cite it throughout. This section adds what the wallet layer builds on top: the index discipline shared across Sapling, Orchard, and transparent derivation, the raw and unified encodings, viewing-key containers, and the deployed address-generation code.

At zips ad30e59, ZIP-316 is organised into revisions: Revision 0 is Active and is what the deployed stack implements; Revision 1 was withdrawn; Revision 2 is a Draft (ZIP-316, *Revisions*). This volume documents Revision 0 and states Revision 2 in §2.8, classified *specified, not deployed*; its reference implementation is librustzcash PR #2199, absent from both the baseline and the 2026-08-30 91f448b5 snapshot. Deployed-behaviour claims below are verified against librustzcash commit e30517e4 — the volume’s baseline (§7.7) — and the crates-io releases zcash_spec 0.2.1, zip32 0.2.0, sapling-crypto 0.7.0, and orchard 0.15.0 (exact versions pinned by the librustzcash workspace Cargo.lock); ZIP text is quoted from the zips repository at commit c6b7035.

2.1 Accounts: ZIP-32 key paths and wallet seeds

Every address below hangs off an *account*, a node of the ZIP-32 hierarchical-deterministic key tree. Sapling and Orchard share the path shape

$$m / \text{purpose}' / \text{coin_type}' / \text{account}',$$

with $\text{purpose} = 32$ (hardened index 0x80000020), the SLIP-44 coin type (133 on mainnet, 1 on all testnets), the account index counting from 0, and every level hardened (ZIP-32, *Key path levels*; *Sapling key path*; *Orchard key path*; deployed constants in zcash_protocol, constants/). Two absences are deliberate. There is no change level: shielded change pays the wallet’s own address and is indistinguishable from any other output, so the transparent external/internal split buys nothing (ZIP-32, *Key*

path levels). And there is no non-hardened tail: Orchard ZIP-32 defines hardened-only derivation, instantiated with $\text{MKGDomain} = \text{"ZcashIP32Orchard"}$ and $\text{CKDDomain} = 0x81$ over the extended spending key (sk, c) ; a child may, with small probability, yield a spending key whose external or internal full viewing key is invalid, in which case derivation at that index simply fails — ZIP-32 prescribes no recovery, and the deployed stack surfaces an error (§1.6; ZIP-32, *Specification: Orchard key derivation*). Wallets MUST support this path for accounts 0 to $2^{31} - 1$ and MUST support generating each account's default payment address; a stream of diversified addresses MAY be supported (ZIP-32, *Key path levels; Sapling key path; Orchard key path*). The seed feeding the tree MUST be 32–252 bytes with at least 256 bits of entropy: an attack on a lower-entropy mnemonic can be run against many wallets at once, by matching derived addresses and viewing keys against observed ones (ZIP-32, *Specification: Wallet seeds*).

2.2 Diversified addresses

A diversified address multiplies addresses without multiplying keys: one incoming viewing key ivk , one diversifier key dk , and one 88-bit *diversifier* d per address. The *Ironwood Guide* (“The recipient: keys and addresses”) constructs the map and its design rationale — a keyed permutation of the index space, rather than a hash or a random draw; the wallet layer fixes the index discipline that ZIP-32 shares between Sapling and Orchard.

Definition 2.1 (Diversifier derivation, Sapling and Orchard). For an account's diversifier key dk , the diversifier at *diversifier index* j is

$$d_j := \text{FF1-AES256}.\text{Encrypt}(dk, "", \text{LE}_{88}(j)), \quad j \in \{0, \dots, 2^{88} - 1\},$$

where FF1-AES256 is NIST SP 800-38G FF1 with a 256-bit AES key, radix 2, $\text{minlen} = \text{maxlen} = 88$, the tweak always the empty string, and $\text{LE}_{88}(j)$ (the ZIP's I2LEBSP_{88}) the index's 88-bit little-endian bit encoding; input and output are 88-bit sequences. The protocol specification abbreviates the keyed permutation as $\text{PRP}_{dk}^d(d) := \text{FF1-AES256}_K("", d)$ (ZIP-32, *Sapling diversifier derivation; Orchard diversifier and OVK derivation; protocol specification §5.4.4, Pseudo Random Permutations*).

The Orchard diversified payment address at index j is then

$$d := \text{PRP}_{dk}^d(\text{LE}_{88}(j)), \quad g_d := \text{DiversifyHash}^{\text{Orchard}}(d), \quad \text{pk}_d := [ivk] g_d,$$

the pair (d, pk_d) , with the address at index 0 the *default* diversified payment address; the last step is the key-agreement $\text{KA}^{\text{Orchard}}.\text{DerivePublic}$ of the protocol specification, and the *Ironwood Guide* constructs all three maps. Diversifier indices SHOULD NOT be chosen at random, and an address generator MAY encode recoverable information in the index — the dk holder recovers it by FF1 decryption (protocol specification §4.2.3, *Orchard Key Components*).

The deployed cipher is the fpe crate's $\text{FF1}\langle\text{Aes256}\rangle$ at radix 2; the orchard crate pins fpe 0.6 in its `Cargo.toml`. Both protocols encrypt the index bytes $j.\text{as_bytes}()$ wrapped as a `BinaryNumeralString` via `from_bytes_le` — the 11-byte little-endian index with bits least-significant-first within each byte, which is exactly $\text{I2LEBSP}_{88}(j)$ — under the empty tweak, and convert the 88-bit output back with `to_bytes_le` into the 11-byte diversifier, exactly LEBS2OSP_{88} .

The permutation earns its keep quantitatively. An 88-bit diversifier is too short to be cryptographically unguessable at the 128-bit security level, and diversifiers chosen at random begin to collide by the birthday bound as the number of draws approaches 2^{44} ; a keyed permutation reaches the full 2^{88} range with no repetition, provided the input index never repeats for a given node (protocol specification §5.4.1.6, notes). The remaining rationale — per-account sequences pseudorandom and mutually uncorrelated, and the issue-count leak a bare counter would cause — is the *Ironwood Guide*'s diversifier construction, cited above.

Remark 2.2 (FF1 parameter security). The property the design requires is that FF1-AES256, with the tweak fixed to the empty string, is a secure pseudo-random permutation. As parameterized here — radix 2, an 88-bit domain, 10 rounds per NIST SP 800-38G — it is unaffected by the DKLS2020 attacks on small-domain FF1: at their parameters $r = 5$ (half the number of rounds) and $n = 44$ (half the domain size in bits), the distinguishing attack has data complexity $2^{2n((r-1)-1/2)-n} = 2^{264}$ and time complexity $2^{2n((r-1)-1/2)} = 2^{308}$, no better than brute-forcing the 256-bit key (protocol specification §5.4.4).

Definition 2.3 (Diversifier validity and default diversifiers). For Sapling,

$$\text{DiversifyHash}^{\text{Sapling}}(d) := \text{GroupJHash}^{\text{URS}}(\text{"Zcash_gd"}, \text{LEBS2OSP}_{88}(d)),$$

valued in the prime-order subgroup of Jubjub — Sapling’s elliptic curve, carrying its keys and commitments as Pallas carries Orchard’s — excluding zero, or $\perp$; the diversifier d_j is *valid* iff the result is not $\perp$, which for a given dk holds for approximately half of all indices j . The Sapling *default diversifier* is d_j for the least nonnegative j yielding a valid diversifier, and a Sapling account has approximately 2^{87} payment addresses (ZIP-32, *Sapling diversifier derivation*; *Sapling key path*). For Orchard, with $P := \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-gd}, \text{LEBS2OSP}_{88}(d))$,

$$\text{DiversifyHash}^{\text{Orchard}}(d) := \begin{cases} \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-gd}, \text{"}) & \text{if } P = \mathcal{O}, \\ P & \text{otherwise,} \end{cases}$$

which never returns $\perp$: every one of the 2^{88} Orchard diversifiers is valid, the default diversifier is d_0 , and an account has exactly 2^{88} payment addresses (protocol specification §5.4.1.6; ZIP-32, *Orchard diversifier and OVK derivation*; *Orchard key path*).

The totality of the Orchard hash — and why it removes Sapling’s try-the-next-index rejection loop — is discussed in the *Ironwood Guide* (“The recipient: keys and addresses”). The residual Sapling failure rate of $\frac{1}{2}$ per index is not a curiosity: it resurfaces when a single index must serve several protocols at once (§2.7).

The two protocols also place dk at different depths of the key tree. In Sapling the diversifier key is a component of the *extended spending key*, derived down the ZIP-32 tree beside the spending authority:

$$\text{dk}_m := \text{truncate}_{32}(\text{PRFexpand}_{\text{sk}_m}([0x10])), \quad \text{dk}_i := \text{truncate}_{32}(\text{PRFexpand}_{I_L}([0x16] \parallel \text{dk}_{\text{par}})),$$

where I_L is the ZIP-32 child-derivation entropy (ZIP-32, *Sapling master key generation*; *Sapling child key derivation*). Orchard instead derives dk directly from the full viewing key ($\text{ak}, \text{nk}, \text{r}_{\text{ivk}}$), as one half of a single PRFexpand output whose other half is ovk:

$$R := \text{PRFexpand}_{\text{LE256}(\text{r}_{\text{ivk}})}(0x82 \parallel \text{I2LEOSP}_{256}(\text{ak}) \parallel \text{I2LEOSP}_{256}(\text{nk})), \\ \text{dk} := R[0..32), \quad \text{ovk} := R[32..64),$$

the derivation the *Ironwood Guide* types in full ($\text{PRFexpand}_K(t)$ is BLAKE2b-512 with personalisation $\text{Zcash_ExpandSeed over } K \parallel t$; ZIP-32, *Conventions*; protocol specification §4.2.3). The consequence is capability parity with a simpler key structure: any holder of the Orchard full viewing key can determine the position of any diversifier in the account’s sequence, as in Sapling only a holder of the *extended* full viewing key can (ZIP-32, *Orchard diversifier and OVK derivation*).

Remark 2.4 (Reuse of r_{ivk}). The randomizer r_{ivk} serves twice: as the $\text{Commit}^{\text{ivk}}$ trapdoor and as the PRFexpand key deriving dk and ovk. If dk or ovk is known to an adversary, this reuse prevents proving *perfect* hiding for $\text{Commit}^{\text{ivk}}$ in this context, and modelling PRFexpand merely as a PRF is insufficient; the specification instead makes a joint assumption on Pallas scalar multiplication and PRFexpand (protocol specification §4.2.3, security note).

Inverting the permutation recovers the index:

$$j = \text{FF1-AES256.Decrypt}(\text{dk}, "", d).$$

Decryption alone proves nothing about ownership: every valid diversifier decrypts to *some* index under *every* diversifier key (sapling-crypto 0.7.0, `src/zip32.rs`, doc comment: “all valid diversifiers can be produced by all diversifier keys”), so the ownership test re-derives the address at the decrypted index under the candidate ivk and compares it with the given address. Deployed: the method `DiversifierKey::diversifier_index` performs the raw FF1 decryption in both protocols, and the checked variants are the Orchard `IncomingViewingKey::diversifier_index` and the Sapling `IncomingViewingKey::decrypt_diversifier`.

Remark 2.5 (Unlinkability, and an oracle to avoid). The security requirement: given two randomly selected shielded payment addresses of *different* spend authorities and a third address derivable from either, all three using different diversifiers, it is not possible to tell from which of the two authorities the third address derives. For Sapling this holds under DDH on the Jubjub prime-order subgroup when GroupJHash is modelled as a random oracle, via IK-CPA key privacy of ElGamal; the specification applies the same property and notes to Orchard (protocol specification §5.4.1.6). One pitfall is normative: an implementation SHOULD avoid providing a *chosen-diversifier* oracle — deriving a new address for a given ivk at an adversary-supplied diversifier — since such an oracle trivially breaks unlinkability for the other diversified addresses of that ivk (same section, final note).

2.3 Raw encodings

An Orchard shielded payment address encodes raw in 43 bytes,

$$\text{LEBS2OSP}_{88}(d) \parallel \text{pk}_d^*,$$

the 11-byte diversifier followed by the 32-byte star-encoding of pk_d ; a decoder rejects the encoding if the point fails to parse or is the zero point (protocol specification §5.6.4.2, *Orchard Raw Payment Addresses*). A Sapling address is likewise $11 + 32 = 43$ bytes under Jubjub ctEdwards compression, rejected if parsing fails, the encoding is non-canonical, or pk_d lies outside the prime-order subgroup minus zero — the exclusion of zero dates from specification version 2025.6.0 (protocol specification §5.6.3.1, *Sapling Payment Addresses*).

The Orchard raw *full viewing key* is 96 bytes,

$$\text{I2LEOSP}_{256}(\text{ak}) \parallel \text{I2LEOSP}_{256}(\text{nk}) \parallel \text{I2LEOSP}_{256}(\text{r}_{\text{ivk}}),$$

invalid if any component is a non-canonical field element, if ak is not a valid affine x -coordinate of a Pallas point, or if either the external or the internal ivk derived from it is 0 or $\perp$; the deployed `FullViewingKey::from_bytes` enforces exactly these checks, including both scopes’ ivk validity (protocol specification §5.6.4.4, *Orchard Raw Full Viewing Keys*). The Orchard raw *incoming viewing key* is 64 bytes, $\text{dk} \parallel \text{I2LEOSP}_{256}(\text{ivk})$, where ivk MUST be a nonzero canonical Pallas base field element; deployed as `IncomingViewingKey::to_bytes`, `from_bytes` over a `NonZeroPallasBase` (protocol specification §5.6.4.3, *Orchard Raw Incoming Viewing Keys*).

None of these raw Orchard encodings has a string form of its own: no Bech32 or Bech32m encoding is defined for an individual Orchard address, incoming viewing key, or full viewing key — the unified encodings below are their only string carrier (protocol specification §5.6.4.2). Sapling, by contrast, retains standalone address strings (mainnet HRP `zs`, testnet `ztestsapling`; protocol specification §5.6.3.1), and the deployed renderer honours the asymmetry: the method `Receiver::to_zcash_address` emits a Sapling receiver as a bare `zs` address and a transparent receiver as a `t`-address, but an Orchard receiver only as a single-receiver unified address.

2.4 Unified addresses

A unified address (UA) bundles at most one *receiver* per type into one string, so the recipient distributes a single address and the sender’s wallet chooses the pool. The choice is normative, by the *priority list* of Table 2: the Sender of a payment to a UA MUST use the receiver of the most preferred type it supports — Orchard over Sapling over P2SH (pay-to-script-hash: the output pays the hash of a script, revealed and satisfied when spending) over P2PKH. A wallet MAY let its user configure which receiver types it can send to, but MUST NOT allow the priority order itself to be changed, except via experiments (ZIP-316, *Encoding of Unified Addresses*; mirrored in protocol specification §5.6.4.1, *Unified Payment Addresses and Viewing Keys*).

Typecode	Item type	Receiver	UFVK item	UIVK item
0x00	transparent P2PKH	20	65	65
0x01	transparent P2SH	20	—	—
0x02	Sapling	43	128	64
0x03	Orchard	43	96	64

Table 2: Revision 0 item typecodes with byte lengths of the receiver, UFKV-item, and UIVK-item encodings. *Priority* is descending typecode among these four — 0x03 is most preferred — while encoding order is *ascending* typecode, deliberately not priority order. P2SH viewing-key items do not exist in Revision 0 (ZIP-316, *Encoding of Unified Addresses*; *Encoding of Unified Full/Incoming Viewing Keys*).

Definition 2.6 (Unified container, raw encoding). The raw encoding of a UA, UFKV, or UIVK is the concatenation, over its items in *ascending* typecode order, of

$$\text{compactSize}(\text{typecode}) \parallel \text{compactSize}(\ell_{\text{item}}) \parallel \text{item},$$

where `compactSize` is Bitcoin’s variable-length integer encoding — one byte for values below 253, otherwise a marker byte followed by the value in 2, 4, or 8 little-endian bytes. The deployed decoder additionally rejects typecode or length values above 0x02000000 (`MAX_COMPACT_SIZE` in `zcash_encoding, src/lib.rs`); ZIP-316 does not state that implementation limit as a normative container rule. A transparent receiver is the bare 20-byte hash — the P2PKH validating-key hash or the P2SH script hash — excluding the two version bytes of the base58check raw encoding (ZIP-316, *Encoding of Unified Addresses*).

Encoding in a fixed, non-priority order makes the byte encoding canonical; the deployed container preserves the parse order in `items_as_parsed` for round-trip serialization and separately offers `items()` sorted by preference (ZIP-316, *Rationale for Item ordering*).

Revision 0 imposes containment rules. A UA or UVK MUST contain at least one non-metadata item; a Revision 0 UA MUST contain at least one *shielded* item (typecode 0x02 or 0x03), and a Revision 0 UVK likewise (Revision 2 drops the requirement for UVKs). A consumer MUST reject: duplicate typecodes; both P2SH and P2PKH present; only metadata items; items out of ascending typecode order; trailing bytes; or any constituent item failing its own validation. A consumer MUST ignore items with unrecognized typecodes (except the MUST-understand metadata of §2.8), and every wallet must still parse a container holding unrecognized items (ZIP-316, *Requirements for both Unified Addresses and Unified Viewing Keys; Receivers*). There is intentionally no typecode for Sprout addresses or viewing keys: ZIP-211 disabled sending funds into the Sprout pool at Canopy (ZIP-316; protocol specification §5.6.4.1).

The deployed parser implements these rules once, identically for all three container types (`Address`, `Ufvk`, `Uivk`): the function `try_from_items_internal` rejects out-of-order typecodes

(InvalidTypecodeOrder), duplicates (DuplicateTypecode), the P2PKH/P2SH combination (BothP2phkAndP2sh), and only-transparent containers (OnlyTransparent); typecodes 0x04 through 0x02000000 parse as Typecode::Unknown, and larger values fail as InvalidTypecodeValue.

Remark 2.7 (Unknown items versus the shielded requirement). The deployed code renders “at least one shielded item” as “not only transparent items”, and deliberately counts unknown typecodes as *not* transparent (comment in `zcash_address, kind/unified.rs`): a UA containing only a P2PKH receiver plus one unknown-typecode item parses successfully, although it contains no item of typecode 0x02 or 0x03. The ZIP’s parenthetical “currently Typecodes 0x02 and 0x03” suggests this forward-compatible reading is intended, but the ZIP nowhere states that an unknown item may satisfy the shielded requirement. We flag the divergence and do not resolve it.

Within a single container, all items obtained by hierarchical deterministic derivation SHOULD represent the same account — the ZIP-32 and BIP-44 account level (ZIP-316, *Requirements for both Unified Addresses and Unified Viewing Keys*). Above the parser, the shielded requirement is enforced at construction as well: `UnifiedAddress::from_receivers` returns `None` without a Sapling or Orchard receiver, and address generation fails with `ShieldedReceiverRequired`.

The deployed sendable-address surface is the enum `zcash_keys::address::Address`: a bare Sapling PaymentAddress, a TransparentAddress (P2PKH or P2SH), a Unified address, or a Tex address — the ZIP-320 transparent-source-only form wrapping a 20-byte P2PKH hash. A unified address can receive a memo iff it has a Sapling or Orchard receiver.

2.5 String encoding: F4Jumble and Bech32m

The raw container still needs a string form, and the obvious choice — Bech32m directly over the raw bytes — has a human-factors flaw at UA lengths. The Bech32m checksum does not *cascade*: a user who visually verifies only the beginning and end of a long string can miss an adversarial substitution of an internal receiver. ZIP-316 therefore passes the whole payload through *F4Jumble*, an unkeyed four-round Feistel construction approximating a random permutation of the entire byte string, before the checksummed encoding: any local change diffuses over the whole string, giving near-second-preimage resistance and nonmalleability — an adversary cannot cheaply alter an internal receiver while preserving a sufficiently long independently checked prefix. This is not authentication: anyone can jumble and checksum a wholly substituted address; a user must still obtain or verify it through a trusted channel. Three rounds would not suffice: a controlled xor-difference attack fixes the input to H_0 . The HRP, zero-padded to 16 bytes, is appended *inside* the jumbled payload, extending the same anti-malleation protection to the HRP itself and defending against cross-network confusion and address-versus-viewing-key confusion (ZIP-316, *Jumbling*).

Definition 2.8 (Unified string encoding). Let *raw* be the encoding of Definition 2.6 and *padding* the container’s HRP in US-ASCII, padded to exactly 16 bytes with zero bytes; the total length must satisfy $\ell(\text{raw}) \leq \ell_M^{\text{MAX}} - 16$. The string encoding is

$$\text{Bech32m}(\text{HRP}, \text{F4Jumble}(\text{raw} \parallel \text{padding})),$$

ignoring the BIP-350 length restrictions (Bech32m was chosen over Bech32 for better checksum behaviour at variable lengths). Decoding **MUST**: Bech32m-decode, rejecting bad checksums; reject out-of-range lengths; apply F4Jumble^{-1} ; verify and strip the trailing 16-byte padding; then parse the items, rejecting the whole encoding on any failure (ZIP-316, *Encoding of Unified Addresses*).

Definition 2.9 (F4Jumble). Let M be a message of ℓ_M bytes with $38 \leq \ell_M \leq \ell_M^{\text{MAX}} := (2^{16} + 1) \cdot 64 = 4194368$, and let $\ell_H := 64$. Set $\ell_L := \min(\ell_H, \lfloor \ell_M/2 \rfloor)$ and $\ell_R := \ell_M - \ell_L$, and split

Container	Mainnet	Testnet	Regtest
Unified address	u	utest	uregtest
Unified FVK	uview	uviewtest	uviewregtest
Unified IVK	uivk	uivktest	uivkregtest

Table 3: Revision 0 human-readable parts, as deployed. ZIP-316 defines the mainnet and testnet columns (ZIP-316, *Revisions*); the regtest column is a librustzcash extension, present only in `zcash_protocol`, `constants/regtest.rs`.

$M = a \parallel b$ with $\ell(a) = \ell_L$. Then, with all letters local to this definition,

$$x := b \oplus G_0(a), \quad y := a \oplus H_0(x), \quad d := x \oplus G_1(y), \quad c := y \oplus H_1(d),$$

and $\text{F4Jumble}(M) := c \parallel d$. The round functions are

$$H_i(u) := \text{BLAKE2b}(8\ell_L)(\text{"UA_F4Jumble_H"} \parallel [i, 0, 0], u),$$

$$G_i(u) := \text{first } \ell_R \text{ bytes of } \left\|_{j=0}^{\lceil \ell_R/\ell_H \rceil - 1} \text{BLAKE2b-512}(\text{"UA_F4Jumble_G"} \parallel [i] \parallel \text{I2LEOSP}_{16}(j), u),$$

each personalisation filling the 16-byte BLAKE2b personal field. The inverse applies the four XOR rounds in reverse order (ZIP-316, *Jumbling*).

The deployed implementation keeps the state as *left* $\parallel$ *right* with *left* the first $\min(64, \lfloor \ell_M/2 \rfloor)$ bytes, and runs $\text{g_round}(0)$ (*right* $\oplus= G_0(\text{left})$), emulating the XOF (extendable-output function; *Crypto Guide*, § “Hash functions and the random oracle model”) chunkwise in 64-byte blocks), $\text{h_round}(0)$, $\text{g_round}(1)$, $\text{h_round}(1)$, with the inverse running the same rounds in reverse; this matches the formulas above under the identification of (a, b, x, y, c, d) with the successive half-states. One bound differs: the crate accepts message lengths $48 \leq \ell_M \leq 4194368$ (`VALID_LENGTH`) — the Revision 0 minimum of 48 rather than Revision 2’s 38. ZIP-316 states the difference is unobservable for Revision 0-valid item sets, since no such set parses to 22–31 content bytes, and Revision 2 consumers MAY apply either minimum to Revision 0 containers (ZIP-316, *Rationale for length restrictions*; `f4jumble`, `src/lib.rs`). On the checksum side, the deployed encoder uses a custom checksum type `Bech32mZip316`, identical to `Bech32m` except that `CODE_LENGTH = 4194368 = \ell^{\text{MAX}}` lifts the BIP-350 length cap; decoding rejects any string whose HRP does not match a known network (main, test, regtest) for the container type.

The jumble is cheap at address lengths: 4 BLAKE2b compressions for $\ell_M \leq 128$ bytes — which covers both common UA shapes, since a Sapling-plus-Orchard UA has $\ell_M = 45 + 45 + 16 = 106$ bytes and a transparent-plus-Sapling-plus-Orchard UA $\ell_M = 22 + 45 + 45 + 16 = 128$ — rising to 6 for $\ell_M \leq 192$, 10 for $\ell_M \leq 256$, and 196608 at $\ell_M = \ell_M^{\text{MAX}}$. A consumer unable to handle maximum-length inputs MUST still support $\ell_M \geq 256$ bytes: two conformance levels (ZIP-316, *Efficiency*).

Display is normative too: if a UA or UVK is abridged, at least the first 20 characters MUST be shown. The worst case — the longest mainnet HRP, `uview`, plus the separator — leaves 14 characters, about 70 bits of jumbled data, below the 2^{125} security target, so 20 is an absolute minimum; and only *initial* characters count, since different displays may otherwise show disjoint substrings of the same string (ZIP-316, *Requirements for both Unified Addresses and Unified Viewing Keys*, and its rationale).

Example 2.10 (A unified address, end to end). We trace the pipeline on the smallest deployed shape — an Orchard-only mainnet UA — using the test vector derived from the 32-byte seed `000102...1f` at account 9, diversifier index $j = 1$ (`zcash_address`, `kind/unified/address/test_vectors.rs`). Every value below is script-computed; the key half runs the deployed `SpendingKey::from_zip32_seed` and `FullViewingKey::address_at`.

Receiver. The account key at `m/32'/133'/9'` yields the full viewing key, whose `PRFexpand` image

under domain byte 0x82 gives dk; FF1 encryption of the index bytes (Definition 2.1) gives d_1 , and the key agreement of §2.2 gives pk_d :

dk	82cc9d79742fe5ae9a142b9336a98677b154fe20401eb18998dbed915b0453ce
I2LEOSP ₈₈ (1)	0100000000000000000000
d_1	3fadf8edb20a3301e8260a
pk_d^*	a311f4cbd54d7d6a76baac88c244b0b121c6dc22a8bcce15898e267829fc1e01

Decryption inverts the permutation: $\text{FF1-AES256.Decrypt}(\text{dk}, "", d_1)$ returns the index bytes above, recovering $j = 1$ — subject to the ownership caveat of §2.2. The neighbouring diversifiers $d_0 = \text{e340636542ece1c81285ed}$ and $d_2 = \text{987fd74a2256c596a66f83}$ share no visible structure with d_1 : the sequence is pseudorandom in the index.

Container. The UA has one item, so the raw encoding of Definition 2.6 is the two header bytes `compactSize(0x03) || compactSize(43) = 032b` followed by the 43-byte receiver $d_1 || \text{pk}_d^*$, 45 bytes in all. Appending the padded HRP, 75 ("u") then fifteen zero bytes, gives the 61-byte F4Jumble input M with $\ell_L = 30$, $\ell_R = 31$. The four rounds of Definition 2.9, letters as there:

$a = M[0..30)$	032b3fadb8edeb20a3301e8260aa311f4cbd54d7d6a76baac88c244b0b121
$b = M[30..61)$	c6dc22a8bcce15898e267829fc1e0175000000000000000000000000000000
$x = b \oplus G_0(a)$	e6f1529b03a0ad0a4209da8e4365ca14a5988bf1d03084a1c326f929c574ca
$y = a \oplus H_0(x)$	37b35eb8784bb2b312f9d35e628cfa9aa8045ef172728dd0cbff84dc81a
$d = x \oplus G_1(y)$	ad26d7fb043e09cbf18e8f4d80bd2423ecb01719170463d644be4c76daab6b
$c = y \oplus H_1(d)$	98979f9a7d2dfce7b1616a8d2b9975fe87878171b6fcf33b303f770b4aa64

The plaintext structure is legible in a and b — the TLV header 032b, the diversifier, the padding — and none of it in $F4Jumble(M) = c \parallel d$.

String. Regrouping the 61 jumbled bytes into 98 five-bit symbols (the last padded with two zero bits) and appending the 6-symbol Bech32m checksum yields, with the HRP and separator, $1 + 1 + 98 + 6 = 106$ characters:

u1nztelxna9h7w0vtpd2xjhxt4lpu8s9cmdl8n8vcr7actf2ny45nd07cy8cyuhuvw3axcp545y0ktq9cezuzx84jyhex8dk4tdvwhu4dl

Decoding reverses each stage per Definition 2.8, the inverse jumble running the same rounds in reverse order. The generating scripts reproduce this string through the deployed encoder (`zcash_address,unified::Address::encode`), re-encode all 60 vectors of the file above, and match all 8 vectors of `f4jumble,src/test_vectors.rs`, in both directions.

2.6 Unified viewing keys

A unified full viewing key (UFVK) and unified incoming viewing key (UIVK) reuse the container wholesale: the same item layout (Definition 2.6), the same containment rules, and the same jumbled Bech32m pipeline under their own HRP’s (Table 3), with viewing-key items in place of receivers at the same typecodes (Table 2; ZIP-316, *Encoding of Unified Full/Incoming Viewing Keys*). The items:

- *Orchard* (0x03): the UFVK item is the 96-byte raw full viewing key of §2.3; the UIVK item is the 64-byte raw incoming viewing key $dk \parallel I2LEOSP_{256}(ivk)$.
- *Sapling* (0x02): the UFVK item is 128 bytes, $EncodeExtFVKParts(ak, nk, ovk, dk) = ak^* \parallel nk^* \parallel ovk \parallel dk$ under Jubjub compression, and SHOULD be derived from the account-level ZIP-32 extended full viewing key (ZIP-316; the encoding function is defined in ZIP-32). The UIVK item is 64 bytes, $dk \parallel I2LEOSP_{256}(ivk)$ with a zero ivk invalid (per protocol specification version 2025.6.0) — deliberately *more* than the protocol specification’s Sapling incoming viewing key, which is ivk alone: carrying dk is what makes UAs derivable from a UIVK.
- *Transparent P2PKH* (0x00): the UFVK item is 65 bytes, the chain code c followed by the 33-byte compressed public key — the last 65 bytes of the BIP-32 extended public key — at the account level $m/44'/coin_type'/account'$; the UIVK item has the same shape one level down, at the external change level $m/44'/coin_type'/account'/0$ (the deployed doc comment pins $m/44'/133'/account'/0$ for mainnet). Items of type P2SH (0x01) MUST NOT appear in Revision 0 UFVKs or UIVKs (ZIP-316, *Encoding of Unified Full/Incoming Viewing Keys*).

The deployed item types mirror the sizes exactly: `unified::Fvk::{Orchard([u8; 96]), Sapling([u8; 128]), P2pkh([u8; 65])}` and the `unified::Ivk` analogues (64/64/65).

Remark 2.11 (Viewing-key items are pruned). The transparent items deliberately omit the BIP-32 extended-key metadata — depth, parent fingerprint, child number — because the child number would reveal the wallet account index; deserialization reconstitutes dummy attributes (depth 3 for the account key, depth 4 for the IVK, parent fingerprint `0xffffffff`). The Sapling item is pruned as well: the deployed `DiversifiableFullViewingKey` is exactly (ak, nk, ovk, dk) with no chain code, so round-tripping a UFVK loses the ability to perform non-hardened Sapling child derivation below the account — consistent with the ZIP, which only says the item *SHOULD* be *derived from* the account-level extended full viewing key.

The priority list of Table 2 does not govern viewing keys the way it governs receivers: a consumer of a UVK normally takes the *union* of the information in all items rather than selecting one, and although the current FVK and IVK types reuse the receiver typecodes, future viewing-key types need not correspond to receiver types (ZIP-316, *Encoding of Unified Full/Incoming Viewing Keys*). The deployed container nevertheless returns `items()` sorted in preference order for UVKs just as for UAs, keeping `items_as_parsed` for round-trips; the one place preference genuinely orders viewing-key material is the selection of outgoing viewing keys, which ZIP-316 specifies separately.

Namely, ZIP-316 refines which ovk a sender uses for its outgoing ciphertexts (the C_{out} construction of the *Ironwood Guide*, “Delivery: the note reaches its owner”): the Sender determines a sending account — preferably from the wallet API, else when all inputs belong to a single account — and a *preferred sending protocol*, the most preferred receiver type among the funds being sent, and *SHOULD* use the external or internal ovk (according to the transfer type) of that protocol from the account-level full viewing key; if no account can be determined, it *SHOULD* pick one source address of the preferred protocol and use its FVK’s ovk (ZIP-316, *Usage of Outgoing Viewing Keys*). Deployed: `UnifiedFullViewingKey::select_ovk(scope, input_sources)` picks the OVK of the highest-preference pool represented among the inputs.

The external/internal split just used comes with a strict capability separation: a full viewing key holder can derive both the external and the internal incoming viewing keys and ovks; the external IVK holder can derive neither the internal IVK nor any ovk; the external ovk holder can derive neither the internal ovk nor any IVK. For the shielded protocols this follows from one-wayness of `PRFexpand` and of $\text{CRH}^{\text{ivk}}/\text{Commit}^{\text{ivk}}$, with the internal-key derivations specified in ZIP-32 (ZIP-316, *Deriving Internal Keys*). The Orchard internal full viewing key replaces only the commitment randomizer — the `0x83` derivation of Definition 1.6, from which the internal dk, ivk, and ovk follow by the standard component derivations. Transparent P2PKH accounts, having no protocol-level ovk, get the synthetic `0xd0` pair of Definition 1.7 (ZIP-316, *Deriving Internal Keys*).

A UIVK derives from a UFVK item by item, preserving typecodes: the Sapling FVK maps to its IVK per protocol specification §4.2.2, the Orchard FVK to its external-scope IVK per §4.2.3, and the transparent P2PKH account key to the external-change key by non-hardened child derivation at index 0; items with unrecognized typecodes *MUST* be dropped (as they must be again when deriving a UA from a UIVK). The deployed method `UnifiedFullViewingKey::to_unified_incoming_viewing_key` maps Sapling via `to_external_ivk`, Orchard via `to_ivk(Scope::External)`, transparent via `derive_external_ivk`, and sets the unknown-item list to empty, matching the ZIP (ZIP-316, *Deriving a UIVK from a UFVK*).

2.7 Deriving a unified address from a UIVK

A UA derives from a UIVK at a *single* diversifier index j , and j must be valid simultaneously for every item: the Sapling component requires j to yield a valid Sapling diversifier (Definition 2.3); the transparent component requires $0 \leq j \leq 2^{31}-1$, since j becomes the BIP-44 non-hardened index-level child, giving the receiver path `m/44'/coin_type'/account'/0/j`; the Orchard component imposes no constraint (ZIP-316, *Deriving a Unified Address from a UIVK*). The deployed index map makes

the transparent constraint concrete: the function `to_transparent_child_index` splits the 11 little-endian index bytes into the low 4 and the high 7, requires the high 7 to be zero, and parses the low 4 as a `NonHardenedChildIndex`, whose maximum is $2^{31} - 1$ — exactly the ZIP range.

Which receivers a generated UA carries is decided per call, by the request type `UnifiedAddressRequest`: either `AllAvailableKeys` or `Custom` over `ReceiverRequirements`, one setting of `Require`, `Allow`, or `Omit` for each of `Orchard`, `Sapling`, and `P2PKH`; the constructor rejects any combination in which both `Orchard` and `Sapling` are `Omit`, since no shielded receiver could result. The preset constants are `ALLOW_ALL = (Allow, Allow, Allow)`, `SHIELDED = (Allow, Allow, Omit)`, and `ORCHARD = (Require, Omit, Omit)`. Intersecting two requirements prefers `Require` and `Omit` over `Allow` and errors on `Require` meeting `Omit`.

Generation at an index, `UnifiedIncomingViewingKey::address(j, request)`, derives each requested receiver at the same j : the `Orchard` receiver via `address_at(j)`, which is total and never fails; the `Sapling` receiver via the diversifiable IVK's `address_at(j)`, which yields no receiver at an invalid diversifier — an error only when `Sapling` is `Required`, a silent omission under `Allow`; the transparent receiver via `to_transparent_child_index` and public derivation, where an out-of-range index errors under `Require` and omits under `Allow`. The result must retain at least one shielded receiver. The search `find_address(j, request)` increments j *only* on `InvalidSaplingDiversifierIndex`, returns `DiversifierSpaceExhausted` on overflow past $2^{88} - 1$, and aborts on any other error; `default_address` is `find_address(0)`. The deployed *default unified address* therefore sits at the smallest index valid for all required receivers.

Remark 2.12 (Whose convention the default address is). ZIP-316 does not specify address-search semantics at all; the smallest-valid-index default above is a `zcash_keys` convention, not a ZIP mandate — only ZIP-32's per-protocol default diversifiers (Definition 2.3) are normative. The search's tolerance is also narrow: it skips an index only for an invalid `Sapling` diversifier, so with a `Required` transparent receiver a search that starts or arrives at an index of 2^{31} or beyond — or a failed BIP-32 child derivation, probability about 2^{-127} — aborts generation rather than advancing (`zcash_keys, keys.rs`).

Remark 2.13 (Index 0 and cross-wallet recovery). Diversifier index 0 fails to yield a valid `Sapling` diversifier with probability approximately $\frac{1}{2}$, so a wallet requiring a `Sapling` receiver often issues its first UA at a higher index — yet some wallets always generate a transparent P2PKH address at index 0. All Zcash wallets **MUST** therefore assume there may be transparent funds at diversifier index 0 of each ZIP-32 account, even if the wallet itself would never generate a UA at that index; reliable cross-wallet fund recovery depends on this (ZIP-316, note in *Deriving a Unified Address from a UIVK*).

2.8 Revision 2 (specified, not deployed)

Revision 2 of ZIP-316 is a Draft as of the zips snapshot c6b7035; everything in this subsection is classified *specified*, and none of it exists in the deployed crates — the reference implementation is `librustzcash` PR #2199, and a search of the local `zcash_address` sources finds none of the new HRP, typecodes, or expiry logic. Revision 2 adds (ZIP-316, *Revisions*; Revision 2 changelog):

- *New HRPs*. Unified addresses split into `zu` (shielded-only; transparent items prohibited) and `tu` (transparent-enabled); UVFKs use `uvf` and UIVKs `uvi`; testnet forms append `test` to each HRP.
- *Metadata items*. Typecodes `0xC0–0xFC` are reserved for metadata, which **MUST NOT** be selected as receivers and **MUST NOT** confer viewing authority. The subrange `0xE0–0xFC` is *MUST-understand*: a consumer that does not recognize such an item **MUST** treat the whole container as unsupported. Typecodes `0xFFFA–0xFFFF` are experimental, and registry additions come via 2xxx-series ZIPs (ZIP-316, *Metadata Items*; *Typecode Registry*; *Adding new types*).
- *Address expiration*. Typecode `0xE0` carries a little-endian u32 Address Expiry Height — the address is expired once the chain height exceeds it — and `0xE1` a little-endian u64 Unix-time Address Expiry Time. A Revision 2 Sender **MUST** set a nonzero `nExpiryHeight` in transactions sent to an address

that defines an Address Expiry Height; MUST NOT send if its normal `nExpiryHeight` would exceed the Address Expiry Height, which would leak the address's expiry into the transaction; and SHOULD arrange expiry within 24 hours when only an expiry time is given. Expiry metadata is retained when deriving a UIVK and must not increase when deriving a UA; because shared expiry metadata can link diversified addresses, per-address variation is RECOMMENDED (ZIP-316, *Address Expiration Metadata*).

- *Viewing keys.* The at-least-one-shielded-item requirement is dropped for UVKs, and P2SH viewing-key items (BIP-388 wallet policies) are added (ZIP-316, Revision 2 changelog).

Remark 2.14 (A retroactive rule the deployed parser does not enforce). Revision 2 also legislates for Revision 0 strings: a Revision 0 container — identified by its `u/uview/uivk` HRP — MUST NOT contain MUST-understand typecodes, and consumers MUST reject violations (ZIP-316, *Metadata Items*). The deployed parser predates this amendment and does not enforce it: every typecode from `0x04` to `0x02000000` parses as ignorable `Typecode: :Unknown`, so a `u`-prefixed address carrying an `0xE0` expiry item parses successfully today (`zcash_address, kind/unified.rs`). This is correct Revision 0 behaviour under the pre-amendment text and a violation of the Revision 2-amended text; we flag the divergence rather than resolve it.

3 Fees (ZIP-317)

For an ordinary wallet-created transaction, any fee is public: it is the transparent value the transaction leaves unclaimed. Consensus also permits a zero fee, and coinbase transactions do not pay one. For a shielded protocol this publicity imposes two requirements that an ordinary fee market does not. The fee must be *computable from public transaction fields alone* — a fee that depended on private data (which notes were spent, what change was returned) would leak that data through the amount paid (ZIP-317, *Requirements*) — and it should be *uniform across wallets*, because a wallet paying an idiosyncratic fee fingerprints every transaction it builds. ZIP-317 resolves both with a single *conventional fee*: a deterministic function of the transaction's public shape that every wallet is expected to pay.

Remark 3.1 (A convention, not a consensus rule). The conventional fee is a wallet convention, not a consensus requirement. ZIP-317 (*Fee calculation*) states: “It is not a consensus requirement that fees follow this formula; however, wallets SHOULD create transactions that pay this fee, in order to reduce information leakage, unless overridden by the user.” Enforcement is economic, not validity-level: nodes are expected to relay, and the recommended block-template algorithm to preferentially mine, transactions paying at least the conventional fee (§3.6).

ZIP-317 is organised into *revisions* (Last-Updated 2026-06-26): Revision 0 is Active and is the deployed fee mechanism; Revision 1 (a fee contribution for Ironwood-pool Actions, NU6.3) and Revision 2 (memo bundles, depending on the undeployed ZIP 248) are both Draft. A contribution defined by a revision is included in all later revisions. This volume documents Revision 0 as the base rule and states Revisions 1–2 below. Although its document status remains Draft, Revision 1 is implemented and is the fee contribution used for active NU6.3 Ironwood bundles (§3.3); Revision 2 remains *specified, not deployed*.

3.1 The conventional fee

The unit of account is the *logical action*: a pool-neutral measure of how much verification work and chain space a transaction demands. Its inputs are public transaction fields defined in the protocol specification §7.1 (*Transaction Encoding and Consensus*): the total byte sizes `tx_in_total_size` and `tx_out_total_size` of the transparent `tx_in` and `tx_out` fields, and the counts `nJoinSplit`, `nSpendsSapling`, `nOutputsSapling`, `nActionsOrchard`, and — from v6 — `nActionsIronwood`.

Definition 3.2 (Logical actions, ZIP-317 Revision 0). The transaction’s pools contribute

$$\begin{aligned} \text{contribution}_{\text{Transparent}} &= \max\left(\left\lceil \frac{\text{tx_in_total_size}}{150} \right\rceil, \left\lceil \frac{\text{tx_out_total_size}}{34} \right\rceil\right), \\ \text{contribution}_{\text{Sprout}} &= 2 \cdot n\text{JoinSplit}, \\ \text{contribution}_{\text{Sapling}} &= \max(n\text{SpendsSapling}, n\text{OutputsSapling}), \\ \text{contribution}_{\text{Orchard}} &= n\text{ActionsOrchard}, \end{aligned}$$

and *logical_actions* is the sum of the contributions of the applicable revision (ZIP-317, *Fee calculation*).

Definition 3.3 (Conventional fee). The conventional fee, in zatoshis, is

$$\text{conventional_fee} = \text{marginal_fee} \cdot \max(\text{grace_actions}, \text{logical_actions}),$$

with *marginal_fee* = 5000 zatoshis per logical action and *grace_actions* = 2 (ZIP-317, *Fee calculation*).

The two byte-size divisors are the standard P2PKH sizes, chosen so that one typical transparent input or output weighs one logical action. The standard input size $150 = 36 + 110 + 4$ decomposes as the outpoint (36: the 32-byte transaction id and 4-byte index naming the output being spent), the script — 1 overall-length byte, 1 signature-length byte, a 72-byte signature, 1 sighash-type byte (the flag selecting which parts of the transaction the signature commits to), 1 pubkey-length byte, a 33-byte pubkey, and 1 byte of margin, totalling 110 — and the sequence field (4). The standard output size $34 = 8 + 26$ decomposes as the value (8) and the script — 1 script-length byte plus the 25-byte P2PKH script (ZIP-317, *Rationale for the chosen parameters*).

Why logical actions, and why these constants. A naïve schedule would charge per input plus per output. But an Orchard Action spends one note *and* creates one (the *Ironwood Guide*, “One Action, one payment: the walkthrough”), and Orchard bundles are padded to a minimum shape (§3.2); charging inputs and outputs separately would penalise exactly that forced padding. Logical actions count an Action once and, via the max in each contribution, do not discriminate between pools (ZIP-317, *Rationale for logical actions*). The transparent contribution is size-based rather than count-based so that large scripts cannot be undercounted: a 2-of-3 P2SH multisig input is roughly 70% larger than a P2PKH input and is charged proportionally (ZIP-317, *Rationale for the chosen parameters*). The grace window *grace_actions* = 2 exists so that padding a one-action transaction to two actions — the standard behaviour of *zcashd* and the mobile SDKs — costs nothing; it also makes a low-value input worth sweeping: within the window, an input worth less than the marginal fee is still economic to include when a second input covers the payment plus fee. Two is also *min_actions*, the smallest economically relevant change-producing transaction (ZIP-317, *Rationale for the chosen parameters*). Finally, *marginal_fee* = 5000 makes the minimal two-action fee 10,000 zatoshis, deliberately restoring the conventional fee that predated ZIP-313 (which had lowered it to 1,000).

Later revisions. Revision 1 (Draft, NU6.3) adds

$$\text{contribution}_{\text{Ironwood}} = n\text{ActionsIronwood};$$

Ironwood-pool Actions use the same Action structure as Orchard, so they are counted identically (ZIP-317, Revision 1). Revision 2 (Draft; ZIP-231 memo bundles, dependent on the Draft, undeployed

ZIP 248 transaction format) adds

$$\begin{aligned} \text{contribution}_{\text{Memos}} &= \max(0, n\text{MemoChunks} - \text{free_memo_chunks}), \\ \text{free_memo_chunks} &= \begin{cases} 2 & \text{if } n\text{OutputsSapling} + n\text{ActionsOrchard} + n\text{ActionsIronwood} > 0, \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

bounding the additional fee for a ZIP-231 memo bundle by 0.0019 ZEC (ZIP-317, Revision 2). Revision 1 is implemented in the deployed crates and applies to NU6.3 Ironwood bundles (§3.3); Revision 2 is not implemented.

3.2 The fee is computed on padded counts

The counts in Definition 3.2 are those of the final, serialised transaction — *after* the builder pads each shielded bundle to its minimum shape. The deployed padding rules are those of the crates `librustzcash` builds against (commit `e30517e4`: `sapling-crypto` 0.7.0 and `orchard` 0.15.0, per its `Cargo.lock`):

- *Sapling*: a Transactional bundle with any spend or output requested (or with `bundle_required` set) uses `num_outputs = max(outputs, 2)`, the constant 2 being `MIN_SHIELDED_OUTPUTS`. Spends are never padded under `BundleType::DEFAULT`: `num_spends` equals `spends` when `spends > 0` and 0 otherwise, so an output-only bundle has zero spend descriptions; the floor `max(spends, 1)` applies only when `bundle_required` is set. An empty, non-required bundle is omitted entirely.
- *Orchard*: a Transactional bundle with `requested = max(num_spends, num_outputs) > 0` (or `bundle_required`) uses `num_actions = max(requested, min)`, the floor `min` being the bundle type’s `pad_to_minimum` count, default `DEFAULT_MIN_ACTIONS = 2`. The `max` form of `requested` applies when the bundle’s flags enable cross-address transfers, as the baseline’s pre-NU6.3 bundles do (`Flags::ENABLED`); Remark 3.4 states the sum rule that replaces it when the flags disable them.

Consequently, under the default bundle types used by the wallet, any nonempty Sapling or Orchard bundle contributes at least two logical actions, and a one-input, one-output payment within a single shielded pool pays exactly the 10,000-zatoshi minimum. The low-level builders also expose unpadded bundle types, for which that two-action conclusion need not hold.

The builder wires the padded counts into the fee call: the method `get_fee` of the transaction builder passes the raw count of Sapling spends added, the Sapling output count via `BundleType::num_outputs` (padding to 2 when the bundle is nonempty), and the Orchard action count via `BundleType::num_actions` (padding to `DEFAULT_MIN_ACTIONS = 2` when nonempty). The proposal-stage balance computation in `zcash_client_backend` does the same — additionally counting the prospective change outputs — using `sapling::builder::BundleType::DEFAULT` and `orchard::builder::BundleType::DEFAULT`, except for the current ZIP-318 canonical-crossing Ironwood bundle’s explicit unpadded choice. Fee estimation and final construction therefore agree on the shielded counts by construction.

Remark 3.4 (The flag-dependent Orchard action count). Release `orchard` 0.15.0 — the version the baseline `Cargo.lock` pins — makes the requested-action count flag-dependent: when a bundle’s flags disable cross-address transfers, a spend and an output can never share an `Action`, so `requested = num_spends + num_outputs` replaces the `max`. Its documentation warns explicitly that ZIP-317 fee estimation must account for the larger count. The deployed builders disable cross-address transfers only for legacy-Orchard-pool bundles under NU6.3, whose bundle version mandates the restriction; every bundle the deployed wallet builds before NU6.3 keeps the flag enabled (`Flags::ENABLED`) and the `max`, so the fee arithmetic above is unchanged.

3.3 The deployed fee rule

Module `transaction::fees::zip317` of `zcash_primitives` implements Definition 3.3 as the type `FeeRule`; the constructor `FeeRule::standard()` instantiates the four ZIP constants,

```
MARGINAL_FEE = 5000,  GRACE_ACTIONS = 2,
P2PKH_STANDARD_INPUT_SIZE = 150,  P2PKH_STANDARD_OUTPUT_SIZE = 34,
```

and the derived `MINIMUM_FEE = 10,000` zatoshis is `MARGINAL_FEE · GRACE_ACTIONS`. Non-standard parameters are gated behind the `non-standard-fees` feature and reject zero input or output sizes. The generic trait method is

```
fee_required(params, target_height,
transparent_input_sizes, transparent_output_sizes,
sapling_input_count, sapling_output_count,
orchard_action_count, ironwood_action_count) -> Zatoshis
```

The ZIP-317 implementation ignores `params` and `target_height` and computes

$$fee = 5000 \cdot \max\left(2, \max\left(\left\lceil \frac{t_{in}}{150} \right\rceil, \left\lceil \frac{t_{out}}{34} \right\rceil\right) + \max(s_{in}, s_{out}) + a + i\right),$$

where t_{in} is the sum of the *known* transparent input sizes, t_{out} the sum of the transparent output sizes, s_{in}, s_{out} the Sapling spend and output counts, a the Orchard action count, and i the Ironwood action count (zero in the V5 transactions the wallet builds before NU6.3); overflow yields a `BalanceError`.

One term of Definition 3.2 is absent, a scope restriction rather than a divergence. There is no Sprout term: the `FeeRule` trait has no `JoinSplit` parameter at all, because the deployed wallet stack cannot construct Sprout spends; the $2 \cdot nJoinSplit$ contribution is spec-only. The Ironwood term of Revision 1 is implemented: the parameter $i = \text{ironwood_action_count}$ above enters *logical_actions*, and is zero in the V5 transactions the wallet builds before NU6.3. The deployed rule is exactly ZIP-317 Revision 1 minus Sprout.

Transparent input sizing. Transparent inputs are sized along two paths. At *proposal* stage, a UTXO (unspent transaction output, the transparent pool’s spendable coin) whose `script_pubkey` is P2PKH is counted as `InputSize::STANDARD_P2PKH = Known(150)`; any unrecognised script yields `InputSize::Unknown`. An `Unknown` input makes the fee incomputable: `fee_required` returns `FeeError::UnknownP2shInputs`. The recognised forms are P2PKH, and multisig-in-P2SH where the redeem script is known. At *build* stage, the method `TransparentInputInfo::serialized_len` computes the P2PKH size exactly: $36 + (1 + 74 + 34) + 4 = 149$ bytes — the outputpoint, a `scriptSig` of one `CompactSize` byte plus a 74-byte push of a 73-byte placeholder signature (`MAX_SIG_SIZE = 73`: a 72-byte DER signature plus the sighash-type byte) plus a 34-byte push of a 33-byte pubkey, and the sequence field — one byte under the ZIP’s 150, which includes a one-byte margin; P2SH multisig inputs are computed from the redeem script.

Remark 3.5 (149 versus 150). The proposal path therefore prices a P2PKH input one byte above its serialised size. The discrepancy is deliberate: the `FeeRule` documentation states that the rule “may slightly overestimate fees in case where the user is attempting to spend a large number of transparent inputs. This is intentional and is relied on for the correctness of transaction construction algorithms in the `zcash_client_backend` crate” (`zcash_primitives`, `transaction/fees/zip317.rs`). An over-estimate can only leave a gratuity, never underfund the transaction.

Transparent output sizing. The trait method `OutputView::serialized_size` returns 8 (value) plus the script bytes plus the `CompactSize` length prefix; a P2PKH output is $8 + 1 + 25 = 34$ bytes, matching the ZIP constant.

The standard rule at the wallet layer. The crate `zcash_client_backend` exposes `StandardFeeRule`, an enum whose sole variant `Zip317` delegates to `FeeRule::standard()`, and the extension trait `Zip317FeeRule`, which exposes `marginal_fee()` and `grace_actions()` to the change strategies. This is the standard, default surface. A fixed-fee rule and its `SingleOutputChangeStrategy` are also available behind the `non-standard-fees` feature.

3.4 Change and dust under the fee rule

The fee rule does not act alone; it is consulted by a *change strategy* that turns a tentative selection of inputs and payments into a balanced transaction. Two ZIP-317 strategies exist: `SingleOutputChangeStrategy`, producing one change output, and `MultiOutputChangeStrategy`, splitting change across several notes according to a `SplitPolicy` so the wallet maintains a target number of spendable notes.

Change-pool selection. Change goes to Orchard if the transaction has any Orchard input or output; otherwise to Ironwood if it has any Ironwood input or output, so change from an Ironwood spend stays in that pool; otherwise to Sapling if it has any Sapling input or output; otherwise — fully transparent flows — to a caller-specified fallback pool. Once NU6.3 is active, the turnstile overrides an Orchard-preferred result to Ironwood unless the transaction spends Orchard notes and strictly less value returns to the pool than those notes remove from it. The source marks the strategy “naive” (TODO) (`zcash_client_backend`, `fees/common.rs`, `select_change_pool`).

The balance computation. The function `single_pool_output_balance` first computes *min_fee*, the fee required with *zero* change outputs. Then:

1. if $total_in < subtotal_out + min_fee$, it fails with `ChangeError::InsufficientFunds`, reporting *available* = *total_in* and *required* = $subtotal_out + min_fee$ — the figure the input-selection loop of §3.5 feeds back;
2. at exact equality, when all flows are transparent and no change memo is requested, it emits no change output and the fee is *min_fee*;
3. otherwise it computes *max_fee* with the target change-output count and derives $change = total_in - (subtotal_out + total_fee)$.

In the shielded case a *zero-valued* change output is deliberately retained even at exact balance: omitting it would let an adversary who knows the recipients’ outputs conclude, from the absence of change, that the inputs sum exactly to payments plus fee; the zero-valued output also carries the change memo.

Dust policy. A change value below a dust threshold is handled by a `DustOutputPolicy`. The default is (`DustAction::Reject, None`), with the unset threshold defaulting to the marginal fee, 5000 zatoshis (`default_dust_threshold = fee_rule.marginal_fee()`); `Reject` still permits exactly-zero change per the previous paragraph. The `AllowDustChange` action emits the dust change output anyway. The `AddDustToFee` action folds the dust into the fee — with a guard: if the resulting fee exceeds the computed fee by more than $10 \cdot \text{MINIMUM_FEE} = 100,000$ zatoshis the fold is not performed and the ordinary change output is kept, protecting the user from a mis-set threshold; the documentation notes that `AddDustToFee` inherently leaks more information.

Uneconomic inputs. An input is classified as dust iff its value is *at most* the marginal fee (`zcash_client_backend`, `fees/common.rs`, `check_for_uneconomic_inputs`). Dust is admitted only where it rides along free. Per pool *p*,

$$allowed_p = \min(\#dust_p, (\#outputs_p + \#change_p) - \#nondust_p) \quad (\text{saturating at } 0),$$

i.e. dust inputs that do not increase the padded action count. Then, if the baseline logical-action count is below *grace_actions*, at most *one* extra dust input may be added free — pools tried in the order transparent, Sapling, Orchard, Ironwood — admitted iff the hypothetical action count with it does not exceed the baseline, where the hypothetical count is $\max(t_{in}, t_{out}) + \max(\text{padded Sapling spends},$

padded Sapling outputs) + padded Orchard actions + padded Ironwood actions with transparent inputs assumed P2PKH. When the presence of a change output is uncertain, the minimum allowance over both manifests is used (`possible_change` construction, same file). Dust beyond the allowance is returned as `ChangeError::DustInputs`, which the input selector converts into an exclusion list (§3.5).

Remark 3.6 (Strictness divergence from the ZIP wording). The ZIP speaks of inputs “with value *below* the marginal fee” (ZIP-317, *Requirements* and *Rationale for the chosen parameters*), while the deployed check classifies value $\leq \text{marginal_fee}$ — a note of exactly 5000 zatoshis included — as dust, and the SQL layer (§3.5) likewise never offers notes with value ≤ 5000 . The `ChangeError::DustInputs` documentation itself acknowledges the determination is “potentially conservative”. We flag this as conservative code behaviour against the ZIP wording, not an error in either (`zcash_client_backend`, `fees/common.rs`; `zcash_client_sqlite`, `wallet/common.rs`).

Splitting change. The strategy `MultiOutputChangeStrategy` fetches account metadata counting the wallet’s existing notes above `min_split_output_value` (default fallback `SplitPolicy::MIN_NOTE_VALUE = 500,000` zatoshis, documented as `MARGINAL_FEE · 100`); the split count shrinks until each split note is at least the minimum split value (or a quarter of the post-transaction average note value when no minimum is set); if fewer splits than targeted result, the fee is recomputed for the actual count. The division remainder is added to the first change output.

Example 3.7 (Fee of a split-change transaction). Fix a five-way split policy with minimum split value 1,000,000 zatoshis and a wallet holding no other notes. Spending a 7,500,000-zatoshi Sapling note to pay 1,000,000 incurs a fee of 30,000 zatoshis — one spend against six outputs, the payment plus five change notes, is six logical actions — and leaves five change notes of 1,294,000 zatoshis each. Spending a 6,000,000-zatoshi note under the *same* policy shrinks the split: at the five-change fee of 30,000 the change totals 4,970,000 — 994,000 per note, below the minimum — so the strategy emits four change notes; the fee, recomputed for the payment plus four change notes (five logical actions), is 25,000, and each change note carries 1,243,750. The figures are recomputed by script from the ZIP-317 formula and the split rule above, and match the strategy’s own tests. The same tests verify that the strategy refuses an exact-balance two-output transaction without change, requiring the change output’s marginal fee, so that a recipient cannot reason about the sender’s input values from the absence of change.

3.5 Input selection: the fee-escalation loop

Fee and input selection are mutually dependent: adding an input can raise the logical-action count, which raises the fee, which can require another input. The deployed selector, `GreedyInputSelector`, resolves the circularity by iteration rather than by solving for a fixed point directly. Starting from an empty selection, each round:

1. chooses source pools by the preference order of §4.2 — the first pool in order whose selected notes cover the requirement is spent alone, and pools otherwise accumulate in preference order until the total covers it;
2. calls `ChangeStrategy::compute_balance` on the current selection (the fee evaluated on padded counts, prospective change included);
3. on `Ok`, emits the `Proposal` with the balanced fee; on `Err(DustInputs)`, adds the offending notes to an exclusion list and retries; on `Err(InsufficientFunds{required})`, sets the new target to *required* — which already embeds the fee recomputed for the enlarged, padded action count — and reselects notes from scratch with `TargetValue::AtLeast(required)`.

The loop terminates when a reselection fails to *strictly* increase the total selected value, returning `InsufficientFunds`. Escalation is thus implicit: each added note can raise the logical-action count by one, raising the required total by one marginal fee, which the next round’s selection must cover.

The note store enforces the dust rule below the selector: both note-selection queries in `zcash_client_sqlite` filter `rn.value > min_value` with `min_value = zip317::MARGINAL_FEE = 5000`, so notes worth at most 5000 satoshis are never even offered to selection. Value-targeted selection scans notes oldest-first under a running-sum window, taking every note while the running sum is below the target plus the single note that crosses it. The candidate notes themselves are the product of the synchronisation described in the *Ironwood Guide* (“Delivery: the note reaches its owner”); this section takes that spendable set as given. Section 10 develops the scanning pipeline that reconstructs it.

ZIP-320 (TEX) transactions. A payment to a transparent-source (TEX) address builds two transactions: the first funds an ephemeral transparent output, which the second spends towards the TEX address. The selector prices the ephemeral input as a standard 150-byte P2PKH input and the ephemeral output as a 34-byte P2PKH output. The value the ephemeral output must carry is obtained by balancing the *second* transaction with a zero-valued ephemeral input (`EphemeralBalance::Input(0)`) and reading the required field of the resulting `InsufficientFunds` error. In `propose_send_max`, the second transaction’s one-transparent-in, one-transparent-out fee is `fee_required([Known(150)], [34], 0, 0, 0, 0) = 10,000` satoshis — the grace minimum.

Shielding. The function `propose_shielding` computes a trial balance over all spendable UTXOs of the source addresses; on `DustInputs` it removes exactly the offending outpoints and recomputes once; the proposal is emitted only if the total reaches the caller’s shielding threshold, else `InsufficientFunds`.

3.6 Relay, mempool, and block template

The conventional fee binds economically through three node-side mechanisms, all specified in ZIP-317 and ZIP-401. None is a consensus rule. The zebra claims below are verified against its sources; the `zcashd` claims rest on the ZIP text alone.

Definition 3.8 (Unpaid actions). For a non-coinbase transaction,

$$\text{unpaid_actions}(tx) = \max(0, \max(\text{grace_actions}, tx.\text{logical_actions}) - \lfloor tx.\text{fee} / \text{marginal_fee} \rfloor),$$

and 0 for coinbase; a block’s unpaid actions are the sum over its transactions, bounded by `block_unpaid_action_limit = 50` in template construction (ZIP-317, *Recommended algorithm for block template construction*).

Relay. Nodes are expected to relay transactions paying at least the conventional fee. A transaction with more than `block_unpaid_action_limit = 50` unpaid actions will never be mined by the recommended template algorithm, and nodes MAY drop such transactions, or those above a configurable limit (ZIP-317, *Transaction relaying*). Concretely: `zcashd` v5.6.0 exposes `-txunpaidactionlimit` (default 50) and `-blockunpaidactionlimit` overriding the block limit, and its relay threshold — a separate, older mechanism — was simplified in v5.5.0 (`zcash/zcash#6542`) (ZIP-317, *Deployment*). The zebra node is stricter than the ZIP suggests: since v1.8.0 (`ZcashFoundation/zebra#8638`) it hard-codes `BLOCK_UNPAID_ACTION_LIMIT = 0`, and `mempool_checks` — run on every candidate for the mempool, the node’s pool of validated transactions awaiting mining, through `VerifiedUnminedTx::new(zebra-consensus, transaction.rs)` — rejects any transaction whose unpaid-action count exceeds that limit (`zebra-chain, transaction/unmined/zip317.rs`). By Definition 3.8, a transaction has zero

unpaid actions iff it pays at least the conventional fee, so zebra admits to its mempool — and therefore relays — only transactions paying the full conventional fee. The same check retains a legacy fee-rate floor of 100 zatoshis per 1000 bytes, clamped to [100, 1000] zatoshis; the source notes it cannot fail once the zero-unpaid-action check passes, the conventional fee being at least 10,000 zatoshis (same file).

Mempool limiting (ZIP 401, as amended). Each mempool transaction has

$$\text{cost} = \max(\text{memory size in bytes}, 10,000), \quad \text{eviction_weight} = \text{cost} + \text{low_fee_penalty},$$

with $\text{low_fee_penalty} = 40,000$ if the transaction pays less than the ZIP-317 conventional fee and 0 otherwise, so low-fee transactions are evicted preferentially under memory pressure (ZIP-401, *Specification*). The threshold was switched from the legacy fee to the ZIP-317 conventional fee in zcashd v5.5.0 and zebra v1.0.0-rc.7 (ZIP-317, *Mempool size limiting*; ZIP-401, *Deployment*). In zebra, method `cost` returns $\max(\text{serialized size}, 10,000)$ and `eviction_weight` adds 40,000 to that eviction weight (not to the fee in zatoshi) when the fee falls below the conventional fee; eviction samples at random by that weight. Because admission already requires the full conventional fee, the penalty never fires in zebra’s own mempool.

Block template construction. The recommended algorithm assigns each candidate transaction a weight

$$\text{tx.weight_ratio} = \min\left(\frac{\max(1, \text{tx.fee})}{\text{conventional_fee}(\text{tx})}, \text{weight_ratio_cap}\right), \quad \text{weight_ratio_cap} = 4,$$

the $\max(1, \cdot)$ avoiding an all-zero-weight candidate set. Phase 1 selects randomly, with probability proportional to weight ratio, among candidates paying at least the conventional fee, subject to the limits on block size and on the number of signature operations a block’s scripts may perform; phase 2 selects among the remaining candidates subject additionally to the block’s unpaid actions staying at most `block_unpaid_action_limit` (Definition 3.8). Whatever an adversary submits, a block built this way carries at most 50 unpaid logical actions (ZIP-317, *Recommended algorithm for block template construction*). In zebra, the two-phase selection (`select_mempool_transactions`; `zebra-rpc, types/get_block_template/zip317.rs`) initialises the phase-2 budget to the hard-coded limit of 0, so a zebra template carries no unpaid actions at all. Overpaying beyond 4× the conventional fee therefore buys no further priority — a deliberate flattening of the fee market that keeps the conventional fee focal.

Deployment. ZIP-317 fees became the wallet default in zcashd v5.5.0; the block-template algorithm shipped in zcashd v5.5.0 and zebra v1.0.0-rc.3 (zebra has no wallet, so it computes no fees for construction) (ZIP-317, *Deployment*).

4 Transaction construction

The synchronisation procedure of the *Ironwood Guide* (§ “Delivery: the note reaches its owner”) leaves a wallet holding a set of unspent notes, each with a value, a position, and an authentication path available on demand. This section documents how the deployed wallet layer turns that state, together with a payment request, into signed transactions. The pipeline, implemented in `zcash_client_backend`, has two phases. Function `propose_transfer` takes a ZIP-321 payment request, runs input selection and fee-and-change computation, and returns a *proposal* — a complete plan, down to the zatoshi, of every transaction to be built. Function `create_proposed_transactions` executes the proposal: one transaction per step, each proved and signed, all stored together via `store_transactions_to_be_sent`; it returns the transaction ids,

and submission to the network remains the caller’s responsibility. The split places every decision that requires judgement — which notes to spend, what fee to pay, where change goes — into an inspectable value computed before any key is touched. Execution follows that plan mechanically, but it is not byte-for-byte deterministic: proofs, signatures, note randomness, and ciphertexts use fresh randomness (including `OsRng`), so repeated execution can produce different transactions and txids.

Current-head extension. Both the volume baseline `e30517e4` and the 2026-09-03 `librustzcash` snapshot `5e770a91` support Ironwood in this pipeline. A proposal carries separate Ironwood inputs, anchor, witnesses, and padding. After NU6.3, an ordinary payment to an Orchard-protocol receiver targets Ironwood; Orchard change from an Orchard spend stays in Orchard through `add_orchard_change_output`, paired with the required fabricated same-receiver spend. Conventional Ironwood bundles retain default padding, while a canonical ZIP-318 crossing uses an unpadding one-Action Ironwood bundle, the shared rolling expiry, and the separately selected anchor bucket. The builder then constructs, proves, signs, and records the Ironwood bundle beside the Orchard bundle using the same Orchard spend authority. Some backend paths still require pool-specific handling, so Ironwood support is stated explicitly below rather than inferred from Orchard support.

4.1 The proposal

The payment request. The input is a ZIP-321 `TransactionRequest`: a `BTreeMap` from payment index to payment, implemented in `librustzcash`’s `components/zip321` crate. A memo attached to a payment addressed to a transparent recipient is rejected — as `Zip321Error::TransparentMemo` at parse time and again as `Error::Payment(PaymentError::TransparentMemo)` when outputs are added to the builder — because a transparent output has no ciphertext to carry one.

Definition 4.1 (Proposal). A *proposal* carries the fee rule under which it was computed, the minimum target height (the next block height), and a non-empty sequence of *steps* (type `Proposal<FeeRuleT, NoteRef>`), each step describing one transaction: the originating transaction request; a map from payment index to the pool that will pay it; the selected transparent inputs; the selected shielded inputs together with the anchor height at which their witnesses will be computed; references to outputs of prior steps to be spent; the *balance* — the proposed change values and the fee required — and a flag marking shielding steps. Validation of a multi-step proposal rejects forward references, spends of non-ephemeral change produced by a prior step, and intra-proposal double spends.

Target and anchor heights. In `zcash_client_sqlite`, the heights are

$$\text{target} := \text{tip} + 1, \quad \text{anchor} := \min_{\text{pools}} \max\{h \leq \text{target} - c : h \text{ checkpointed}\},$$

the “mempool height” and, with c the confirmation depth and “checkpointed” meaning present in the pool’s shard tree, the highest checkpoint at least c blocks below the target, minimised across Sapling and Orchard so a single height anchors both bundles. Anchor semantics and witness derivation are those of the *Ironwood Guide* (§ “Resting: the note in the commitment tree”, in particular “Anchor choice, reorgs, and the anonymity set”, which already develops the ZIP-315 3/10 defaults); here we record only what the pipeline feeds in.

Definition 4.2 (Confirmations policy). The `ConfirmationsPolicy` of ZIP 315 carries a *trusted* depth (default 3), an *untrusted* depth (default 10), and an `allow_zero_conf_shielding` flag

(default true) (ZIP 315 § “Trusted and untrusted TXOs”). The constructor enforces $\text{trusted} \leq \text{untrusted}$. An output is spendable at the trusted depth if its transaction is user-marked trusted or it was received on an internal-scope key, and at the untrusted depth otherwise; outputs of a shielding transaction inherit the mined height and trust of their transparent source inputs; and transparent UTXOs are spendable at zero confirmations for shielding when the flag is set.

Remark 4.3 (Anchor depth versus untrusted notes). Function `propose_transfer` obtains its height pair with `minconf = trusted` — even when the notes ultimately selected are untrusted and must themselves be 10 confirmations deep. The code comment explains the choice: the shallower anchor admits the maximum number of notes, and trusted-versus-untrusted filtering is deferred to note selection, which tests each note’s own depth. The bundle-wide anchor therefore sits at $\text{depth} \geq 3$, not ≥ 10 — which is what ZIP 315 itself recommends. A wallet SHOULD choose the anchor at the trusted confirmation depth: “if the current block is at height H , the anchor SHOULD reflect the final treestate of the block at height $H - 3$ ” (ZIP 315, § “Anchor selection”). The ZIP’s stated rationale: a rollback past the anchor block invalidates the transaction while its revealed nullifiers link it to any future transaction spending the same notes; an anchor too many blocks back prevents recently received notes from being spent; and a *fixed* depth — rather than one conditioned on whether the spent notes are trusted — avoids leaking their trust status (ZIP 315, § “Rationale for anchor selection”). Neither the ZIP nor the code comment claims an anonymity-set benefit for the shallower anchor.

Remark 4.4 (The ZIP-315 floor is advisory in the API). ZIP 315 states that wallets SHOULD NOT permit spending outputs with fewer than 3 confirmations, but the deployed API exposes `ConfirmationsPolicy::MIN` (1 confirmation) and constructors accepting lower values, gated only by a rustdoc warning about transaction distinguishability (`wallet.rs`, `ConfirmationsPolicy`). We present the ZIP recommendation as the norm and the constructors as a deliberate escape hatch.

4.2 Input selection

The deployed selector is `zcash_client_backend`’s `GreedyInputSelector`; it buckets each payment by receiver type: a transparent address becomes a transparent output; a TEX address is deferred to a second ZIP-320 step (below); a Sapling address becomes a Sapling output; a unified address is paid to its Orchard receiver if present — as an Ironwood-pool output once NU6.3 is active at the target height, the turnstile forcing such payments through the Ironwood bundle, and as an Orchard-pool output before then — else its Sapling receiver, else its transparent receiver, and a unified address with no receiver the wallet supports is an error.

The greedy loop. Selection then iterates: starting with no inputs, call the change strategy’s `compute_balance` (§4.3); on `InsufficientFunds{required}`, ask the backend for spendable notes covering at least `required` (`select_spendable_notes` with `TargetValue::AtLeast`, the target height, the confirmations policy, and the running exclusion list); on `DustInputs`, add the identified notes to the exclusion list and retry; on success, emit the proposal. The loop terminates because the selected value must strictly increase on every round, else the selector fails with `InsufficientFunds`. Pool preference inside the loop: the shielded pools are ordered with the family matching the payment’s outputs first — for a payment to an Orchard receiver, Ironwood (once NU6.3 is active) and then Orchard, ahead of Sapling; otherwise Sapling, then Ironwood, then Orchard — and pools the caller’s spend policy forbids are dropped. The first pool in order whose selected notes cover the requirement alone is spent alone; otherwise pools accumulate in preference order until the running total covers it. Spending stays in the pool family matching the outputs where possible, crossing pools only when needed, which keeps single-pool payments free of cross-pool linkage. Within a pool, the `zcash_client_sqlite` backend selects the *oldest* unspent notes first: a window-function running sum over notes ordered oldest-to-newest takes every note below the target plus the one note that crosses it.

ZIP-320 (TEX) payments. A TEX address demands transparent funds whose provenance is a single ephemeral transparent address, so the selector builds *two* steps (ZIP 320 § “Design considerations for Senders”). It sizes the second step first: probing `compute_balance` over the TEX outputs with a zero-valued ephemeral input and catching `InsufficientFunds{required}` yields

$$v_{\text{eph}} = \sum_{\text{TEX payments}} v + \text{fee}(\text{tr}_1),$$

where $\text{fee}(\text{tr}_1)$ is computed for exactly one standard P2PKH input (150 bytes) and the TEX payments as P2PKH outputs (34 bytes each). The change-strategy contract requires the second step to balance exactly — `total = fee_required`, no change — which the selector enforces with a runtime assertion. Step tr_0 then replaces the TEX payments with a single ephemeral transparent “change” output of exactly v_{eph} , and step tr_1 spends it and pays the TEX addresses as P2PKH. Paying a TEX address from shielded inputs in a single step is rejected.

Shielding. Function `propose_shielding` targets the `tip + 1` and collects spendable UTXOs from the given source addresses — at zero confirmations under the default policy (Definition 4.2). Funds may be drawn from at most one ephemeral address at a time and never mixed with other addresses (`ProposalError::EphemeralAddressLinkability`), since co-spending would link the ephemeral address to the wallet’s others. Dust UTXOs reported by `compute_balance` are dropped and the balance recomputed once; the proposal is built only if its total reaches the caller’s shielding threshold. The resulting single step has an empty transaction request, `is_shielding` set, and all value flowing to internal change.

4.3 Fees and change

Fees and change are computed *together*, by a `ChangeStrategy` whose one obligation is `compute_balance`. The deployed strategies are `SingleOutputChangeStrategy` and `MultiOutputChangeStrategy` in the `fees::zip317` module; the `fees::standard` module aliases them over `StandardFeeRule`, whose only variant is `Zip317`. Both delegate to a common routine, `single_pool_output_balance`, described below.

Definition 4.5 (ZIP-317 conventional fee, as deployed). The fee rule (ZIP 317 § “Fee calculation”) is

$$\begin{aligned} \text{fee} &= \text{marginal_fee} \cdot \max(\text{grace_actions}, \text{logical_actions}), \\ \text{logical_actions} &= \max\left(\left\lceil \frac{\text{in_size}}{150} \right\rceil, \left\lceil \frac{\text{out_size}}{34} \right\rceil\right) + \max(n_{\text{spend}}^{\text{Sap}}, n_{\text{out}}^{\text{Sap}}) + n_{\text{act}}^{\text{Orch}} + n_{\text{act}}^{\text{lrw}}, \end{aligned}$$

with `marginal_fee = 5 000` zatoshi, `grace_actions = 2`, the standard P2PKH input and output sizes 150 and 34 bytes, and hence a minimum fee of 10 000 zatoshi (the ZIP 317 parameter table). The Ironwood count $n_{\text{act}}^{\text{lrw}}$ is ZIP 317 Revision 1’s contribution (§3.3), zero in any transaction without an Ironwood bundle. The shielded counts fed in are the *padded* counts of §4.5 — `BundleType::num_outputs` and `num_actions` — both here and in the final builder check.

Two deployment caveats. ZIP 317 Revision 0 additionally defines a Sprout contribution $2 \cdot n_{\text{JoinSplit}}$; the wallet never constructs `JoinSplits`, so the deployed `FeeRule` omits the term (`fees/zip317.rs`) — code and ZIP agree on every transaction the wallet can produce, and we flag the omission rather than restate the ZIP formula as implemented. Second, the rule supports P2PKH transparent inputs plus P2SH inputs of known size only; a coin with an unknown P2SH redeem script fails with `FeeError::UnknownP2shInputs`, and the rule intentionally may slightly *overestimate* the fee for many transparent inputs — an overestimate the construction algorithms in `zcash_client_backend` rely on (`fees/zip317.rs`, `FeeRule` rustdoc and `fee_required`).

Change pool. Change goes to Orchard if the transaction has any Orchard input or output flows; else to Ironwood if it has any Ironwood flows; else to Sapling if it has any Sapling flows; else — fully transparent flows, as in shielding — to the caller-supplied fallback pool, subject after NU6.3 activation to the turnstile override of §3.4. The code marks this as deliberately naive (`fees/common.rs`, `select_change_pool`).

Balance computation. Routine `single_pool_output_balance` proceeds as follows. Let in be the sum of transparent, Sapling, Orchard, and Ironwood inputs (including any ephemeral input) and out the corresponding sum of outputs; let fee_0 be the ZIP-317 fee with zero change outputs. If $in < out + fee_0$, fail with `required = out + fee_0` (this is the value the greedy loop feeds back into note selection). If equality holds *and* the flows are fully transparent *and* no change memo is configured, emit no change with $fee = fee_0$. Otherwise recompute the fee with the target number of change outputs (re-derived if the split policy below yields fewer), set $change = in - (out + fee)$, and divide it across the split count with the remainder added to the first output, each share a shielded change value in the selected pool carrying the configured change memo.

Remark 4.6 (Zero-valued change under the default policy). Under the default change strategy, a transaction with shielded flows produces a change output even when inputs exactly balance outputs plus fee: a zero-valued shielded change output is created so that an observer cannot distinguish exact-balance transactions, and so that a shielded output always exists to carry the change memo. Zero-valued change is allowed even under the `Reject` dust action, and the code notes that zero-valued notes need no witness tracking. Only a fully-transparent-flow transaction without a change memo omits change on the exact-balance branch — the second ZIP-320 transaction is the deployed example. The optional `AddDustToFee` policy below can also omit shielded change when no memo requires it. A configured change memo is attached to every proposed shielded change output, is deliberately discarded for the ZIP-320 step that spends the ephemeral input, and itself forces a change output to exist.

Dust. The default `DustOutputPolicy` is `DustAction::Reject` with no threshold override, delegating the threshold to the strategy, whose default is the marginal fee, 5 000 satoshi. If the total change falls below the threshold: under `Reject`, fail with `InsufficientFunds` and the requirement raised by the shortfall (unless the change is exactly zero, which Remark 4.6 permits); under `AllowDustChange`, keep the dust output; under `AddDustToFee`, burn the change into the fee and omit the output (keeping a zero-valued one if a change memo exists) — *except* when the resulting fee would exceed the computed fee by more than $10 \cdot \text{MINIMUM_FEE} = 100\,000$ satoshi (that is, when the folded dust exceeds 100 000), in which case the change output is kept as a defence against value loss.

Uneconomic inputs. An input is *dust* iff its value is at most the marginal fee; the check runs only when the marginal fee is positive. Some dust rides for free: per pool P , up to

$$\min(|\text{dust}_P|, (\text{outputs}_P + \text{change}_P) - \text{nondust}_P) \quad (\text{saturating})$$

dust inputs do not raise the padded action count, plus at most one extra *grace* input — tried transparent first, then Sapling, then Orchard, then Ironwood — if the baseline logical-action count is below `grace_actions` and adding the candidate does not increase the padded count (the padding rules of §4.5 are invoked directly, with and without it). The allowance is the pointwise minimum over all possible change configurations; dust beyond it is returned as `ChangeError::DustInputs`, which the greedy loop converts into exclusions.

Note management: splitting change. The `MultiOutputChangeStrategy` splits change into up to `target_output_count` equal notes (remainder to the first) whenever the account holds fewer than that many notes of at least `min_split_output_value`; the note-count metadata

comes from `InputSource::get_account_metadata` filtered by that minimum, or by the fallback `MIN_NOTE_VALUE = 100 · marginal_fee = 500 000` zatoshi. The split count is

$$\text{split} = \max(1, \text{target} - \#\{\text{qualifying notes}\}),$$

decremented (floor 1) while `change/split < min_split_output_value`; when the minimum is unset it defaults to $(\text{existing notes total} + \text{change}) / (4 \cdot \text{target})$, a quarter of the projected average note value. If fewer than the target number of outputs survive, the fee is recomputed for the actual count. The `SingleOutputChangeStrategy` is the degenerate `SplitPolicy::single_output()`.

4.4 Executing the proposal

Function `create_proposed_transactions` executes the steps in order. Only *transparent* outputs of prior steps may be spent by later steps: shielded chaining is impossible, because a note’s position in the global commitment tree — and hence its witness (the *Ironwood Guide*, § “Resting: the note in the commitment tree”) — is unknown before the note is mined. Each ephemeral output must be consumed by a later step exactly once, else `EphemeralOutputLeftUnspent`. All transactions are created first and then stored together, so a failure cannot leave a partially transmitted multi-step plan.

Anchor and witnesses per step. For each pool a step involves, the anchor is the shard-tree root at the proposal’s anchor-height checkpoint (`root_at_checkpoint_id`), including for a pool with outputs but no inputs. Using a recent real anchor in that case makes an output-only bundle indistinguishable from one containing real spends. Each selected note’s Merkle path is computed with `witness_at_checkpoint_id_caching` at that same height. The mechanics are those of the *Ironwood Guide* (§ “Resting: the note in the commitment tree”) and are not repeated here.

Builder configuration. The `zcash_primitives` builder is constructed with `BuildConfig::Standard` carrying the per-pool anchors and Orchard/Ironwood padding choices, which normally select `BundleType::DEFAULT`; the Orchard builder exists only if NU5 is active at the target height, and the Ironwood builder only for a format and branch supporting Ironwood. The expiry height is the target plus `DEFAULT_TX_EXPIRY_DELTA = 40` blocks, the post-Blossom default of ZIP 203 (`builder.rs`, `DEFAULT_TX_EXPIRY_DELTA`; ZIP 203 § “Changes for Blossom”). The rustdoc on `Builder::new` still says “20 blocks”; code and ZIP agree, the rustdoc is stale — cite the constant, not the comment. The transaction version is `TxVersion::suggested_for_branch` of the target height’s branch — V5 for the NU5 through NU6.2 branches, V6 for the NU6.3 (Ironwood) branch — and the lock time of built transactions is 0.

Spends, outputs, and keys. Sapling spends are added under the external or internal (change) full viewing key according to each note’s recorded ZIP-32 scope; Orchard spends are added under the account’s Orchard full viewing key, the scope being derived internally (`FullViewingKey::scope_for_address` inside `SpendInfo::new`). Payment outputs are added in payment-index order, then the change outputs from the proposal balance: Sapling change to `ufvk.sapling().change_address()` (internal scope), Orchard change to `ufvk.orchard().address_at(0, Scope::Internal)`, and ephemeral ZIP-320 change values to freshly reserved ephemeral transparent addresses (`reserve_next_n_ephemeral_addresses`; the reservation persists even if construction later fails, so an address is never silently reused). The signing keys marshalled for the build: transparent secret keys derived per input address; *two* Sapling expanded spending keys — `usk.sapling()` and `usk.sapling().derive_internal()` — since external and internal notes sign under different keys; but a *single* Orchard spend-authorising key (`SpendAuthorizingKey`), because Orchard internal and external addresses share `ask` (the *Ironwood Guide*, § “The recipient: keys and addresses”). Ironwood spends use that same Orchard full-viewing and spend-authorizing key, but enter the separate Ironwood builder and commitment tree; current output and change

routing is summarized in the current-head extension above. All randomness for the build is drawn from `OsRng`.

Remark 4.7 (OVK policy: none for change). Under the default `OvkPolicy::Sender`, external outputs are encrypted with the ovk selected by `UnifiedFullViewingKey::select_ovk` over the input sources — the Orchard ovk if any Orchard or *Ironwood* input is spent, else the Sapling ovk, else the external-scope ovk derived from the transparent account key per ZIP-316 — while wallet-internal change outputs get *no* ovk at all (`internal_ovk = None`) (`wallet.rs`; `zcash_client_backend/src/wallet.rs`, `OvkPolicy`; `zcash_keys/src/keys.rs`, `select_ovk`). A change output’s C_{out} is therefore unrecoverable by *any* outgoing viewing key; the wallet finds its change through the internal-scope ivk during scanning (trial decryption per the *Ironwood Guide*, § “Delivery: the note reaches its owner”). This matches the `OvkPolicy::Sender` rustdoc but contradicts descriptions elsewhere of change “encrypted to the internal OVK”; the `None` is deliberate. Policy `OvkPolicy::Custom` supplies both keys explicitly; policy `OvkPolicy::Discard` uses none, so the sender cannot decrypt even its own external outputs.

4.5 Padding, dummies, and shuffling

The deployed bundle builders at the volume baseline are orchard 0.15.0 and sapling-crypto 0.7.0, per `librustzcash’s Cargo.lock`. Their current builder APIs give the padding rules below. In Orchard the requested-action count is $\max(s, o)$ under cross-address-enabled flags and $s + o$ when NU6.3 requires cross-address transfers to be disabled (§5.13).

Orchard. The cross-address-enabled branch uses `BundleType::DEFAULT`: flags `ENABLED` (spends and outputs both enabled) and `bundle_required = false`. With s requested spends and o requested outputs, with cross-address transfers enabled, the action count is

$$n_{\text{act}}(s, o) = \begin{cases} 0 & \text{if } \max(s, o) = 0, \\ \max(\max(s, o), 2) & \text{otherwise,} \end{cases}$$

the floor being `MIN_ACTIONS = 2`. The spend list is extended to n_{act} with dummy spends and the output list with dummy outputs; the two indexed lists are then shuffled *independently* and zipped pairwise into Actions, so the spend–output pairing within an action is random, and `BundleMetadata` records each requested spend’s and output’s post-shuffle action index. For cross-address-disabled bundles the requested count is instead $s + o$, and pairing must preserve the address restriction; the independent-shuffle description does not apply to that branch (§5.13).

Orchard dummies. A dummy spend (protocol specification § 4.8.3, “Dummy Notes (Orchard)”) is built from a freshly random `SpendingKey`, its full viewing key’s address at index 0 (external scope), and a zero-valued note whose ρ is Extract of a random Pallas point (`Rho::from_nf_old(Nullifier::dummy)`), with a dummy Merkle path of a random `u32` position and 32 random $\mathbb{F}_{p_{\text{Pallas}}}$ siblings. Anchor consistency is skipped for zero-valued notes (`has_matching_anchor` returns true), which is exactly what lets a dummy coexist with any real anchor. A dummy output is a zero-valued note to a freshly random address, with no ovk and an all-zero memo.

Per-action randomness. At `ActionInfo` construction, the value-commitment trapdoor `rcv` is sampled uniformly from the Pallas scalar field — one per action, padding actions included. When the action is built, the spend-auth randomiser α is sampled per spend, giving $\text{rk} = \text{ak} + [\alpha] G^{\text{SpendAuth}}$; the output note’s ρ is the action’s own `nf_old`, and its `rseed` is freshly sampled by `Note::new`. The cryptographic roles of `rcv`, α , and `rseed` (with its derived ψ , `rcm`, `esk`) are those of the *Ironwood Guide* (§ “Conservation without disclosure: value commitments and the binding signature” for `cv/rcv`; § “Spend authorisation: RedPallas re-randomisation” for `ak/\alpha/rk`; § “Delivery: the note reaches its owner” for the `rseed` derivations).

Sapling. Under `BundleType::DEFAULT` in `sapling-crypto 0.7.0`, a non-empty bundle pads outputs to $\max(o, 2)$ but never pads spends ($num_spends = s$ when $s > 0$, else 0; the $\max(s, 1)$ floor applies only with `bundle_required` set), so an output-only bundle has zero spend descriptions; the two-output floor (`MIN_SHIELDED_OUTPUTS`) is a rule the source traces to `zcash/zcash` issue 3615. An empty, non-required bundle has zero of both. Spends and outputs are extended with dummies, independently shuffled, and the positions recorded in `SaplingMetadata`. Each spend samples its own rcv and α (uniform in the Jubjub scalar field) and each output its own rcv ; the binding-signature key is $bsk = \sum rcv_{spend} - \sum rcv_{output}$.

4.6 Build order and signatures

Construction 4.8 (`Builder::build`). The build proceeds in a fixed order:

1. check version compatibility;
2. compute the fee over the actual padded counts (Definition 4.5) and require the balance invariant below;
3. build the Sapling bundle and create its zk-proofs (proofs before signatures, because V4 transactions commit to proofs in the digest);
4. build the Orchard bundle and, on current v6 paths, the Ironwood bundle, both unproven;
5. assemble the unauthorised transaction data and compute the digest parts (`TxidDigester`);
6. authorise the transparent inputs;
7. compute the shared shielded sighash;
8. apply the Sapling signatures;
9. create the Orchard proof and apply its signatures, then do the same for Ironwood when present;
10. freeze the result into the final `Transaction`.

Definition 4.9 (Balance invariant). Before building, the builder requires, exactly,

$$\text{value_balance} - \text{fee} = 0,$$

where the value balance sums the transparent, Sapling, Orchard, and — on current v6 paths — Ironwood balances; a negative difference is `Error::InsufficientFunds` and a positive one `Error::ChangeRequired`. The proposal’s change outputs are precisely what make the invariant hold.

One sighash for everything. The message for *all* shielded spend-auth signatures and every present shielded pool’s binding signature is the single 32-byte ZIP-244 sighash,

$$\text{sighash} := \text{signature_hash}(\text{tx}_{\text{unauth}}, \text{SignableInput}::\text{Shielded}, \text{txid_parts});$$

the digest tree itself is constructed in the *Ironwood Guide* (§ “Conservation without disclosure: value commitments and the binding signature”, subsection “What the signatures sign: ZIP-244 digests and their v6 form”) and not repeated. For each Orchard-protocol action (in either the Orchard or Ironwood pool), with $rsk = ask + \alpha$ and $rk = ak + [\alpha] G^{\text{SpendAuth}}$ in the notation of the *Ironwood Guide* (§ “Spend authorisation: RedPallas re-randomisation”),

$$\text{spendAuthSig} := \text{Sign}_{rsk}(\text{sighash}).$$

The binding key is $\text{bsk} = (\sum_{\text{actions}} \text{rcv})$ converted via `into_bsk`, and the builder asserts consistency with $\text{bvk} = \sum \text{cv}^{\text{net}} - \text{ValueCommit}(\text{rcv} = 0, \text{value_balance})$. In `Bundle::prepare`, the binding signature $\text{Sign}_{\text{bsk}}(\text{sighash})$ is created and every dummy spend is signed immediately with its stored random ask randomised by that action’s α ; `Bundle::sign` then signs every action whose ak matches the given authorising key, and `finalize` fails with `MissingSignatures` if any action remains unsigned.

Remark 4.10 (Proving-key cache). The 2026-08-30 `librustzcash` snapshot 91f448b5 derives the Orchard/Ironwood circuit version once and reuses a process-wide cached proving key under `std`; the `no-std` fallback builds it once for that transaction. The Sapling provers remain arguments, typically a `LocalTxProver`.

4.7 After the build

The wallet immediately decrypts its own shielded outputs to record sent-note data: Orchard outputs via `decrypt_output_with_key`, trying the internal-scope `ivk` and then the external-scope `ivk` (same-address post-NU6.3 Orchard change can be external-scoped); Ironwood outputs under the internal Orchard `ivk`; and Sapling via `try_sapling_note_decryption` under its internal `ivk`. It uses the Orchard/Ironwood `BundleMetadata` and `SaplingMetadata` to map each requested output to its post-shuffle position. Transparent outputs are *not* reordered — the source notes “there is no reason to do so because it would not usefully improve privacy” — so `vout` index n is the n -th added transparent output. The fee recorded for each sent transaction is taken from the proposal’s balance, not recomputed from the built transaction.

The PCZT path. PCZT construction parallels the signing path: `build_for_pczt` produces the `PcztParts`, and the `Creator`, `IO-Finalizer`, and `Updater` roles attach ZIP-32 derivation paths (Orchard, Ironwood, and Sapling `m/32'/coin'/account'`; transparent BIP-44 `m/44'/coin'/account'/scope/index`) and the proposal metadata under the proprietary fields `zcash_client_backend:proposal_info` and `:output_info`; only single-step proposals are supported.

Orchard-to-Ironwood pool migration. The `librustzcash` snapshot 91f448b5 also contains the released `zcash_pool_migration` 0.1.0 engine and `zcash_client_sqlite` 0.22.0 store. The engine plans denominations, builds and signs the migration transactions as PCZTs, schedules them by block height, and persists its state through a wallet backend; the consuming application remains responsible for broadcast and for reporting the result. The complete privacy and scheduling rules, and the boundary between these released backend pieces and an end-user wallet flow, are in the *Ironwood Guide*, § “Scheduled Orchard-to-Ironwood migration (ZIP-318)”.

Remark 4.11 (Builder-version currentness). The volume baseline pins orchard 0.15.0. That release includes the NU6.3 bundle versions, cross-address restriction, explicit change-output handling, and version-specific proving keys described in §5.13; those features are deployed, not design-stage. As of 2026-09-05, standalone orchard head be4f6594 remains release 0.15.5, while the `librustzcash` lockfile at 5e770a91 resolves 0.15.3. Neither changes the padding construction derived above.

5 Partially created transactions (PCZT)

The *Ironwood Guide* constructs a transaction as if one machine held everything at once: the notes and their authentication paths, the full viewing key, the spend authorizing key, the proving key, and a source of randomness. Deployed wallets do not enjoy that luxury. A hardware signer holds the spending key but cannot compute a Halo 2 proof and may not fit a whole transaction in memory; a threshold of co-signers must each contribute a share; a proof may be delegated to a machine that must never see spending authority. ZIP-374 resolves this by standardising the intermediate object those parties exchange: the *Partially Created Zcash Transaction* (PCZT), a format that carries the

information each participant in transaction creation needs to perform its step, and that holds the accumulated proofs and signatures while the set is still incomplete — signers can be fully offline (ZIP-374, Abstract). The ZIP is Status Draft, Revision 0, owned by Jack Grigg, created 2024-12-09 (ZIP-374, header); The ZIP fixes the v1 reference encoding to pczt 0.7.0. The 2026-08-30 librustzcash snapshot 91f448b5 ships pczt 0.9.3: release 0.8.0 added the v2 encoding and Ironwood support, and 0.9.0 made the convenience serializer emit the minimum encoding version that can represent the logical PCZT (pczt, CHANGELOG.md; §5.13). Thus the role semantics below describe release 0.9.3; references to 0.7.0 identify only the ZIP’s byte-exact v1 baseline. At current librustzcash head 5e770a91 (2026-09-03), the crate version is still 0.9.3, but unreleased changes make the default transparent Signer and Spend Finalizer accept only SIGHASH_ALL after checking input and signature consistency; explicit policy overrides permit other sighash types (pczt and zcash_transparent CHANGELOG.md, Unreleased). Those changes are not attributed to the released role descriptions below.

The Motivation section of the ZIP names four problems the format solves: (i) offline and hardware signers that cannot compute Halo 2 or Groth16 proofs and may not hold a whole transaction in memory; (ii) proof delegation, where the prover needs the note’s private data and the full viewing key but no spending key material; (iii) multiparty transactions — FROST threshold signing (RFC 9591), transparent multisig (ZIP-48), and inputs contributed by several parties, merged deterministically; and (iv) pre-authorization and deferred proving for v6 transactions, where anchors become authorizing data (ZIP-229) (ZIP-374, Motivation).

The design deliberately reuses the role structure of Bitcoin’s PSBT (BIP 174 / BIP 370) with Zcash-specific additions. PSBT itself cannot be used: it has no representation for shielded inputs and outputs, no notion of proofs as distinct from signatures, and its key-value encoding tolerates unknown fields — undesirable for signers that must understand everything they authorize (ZIP-374, Motivation; pczt, README.md). The ZIP’s Requirements section fixes the resulting contract: the format must not require spending key material to be present (dummy-spend keys are the one exception, and must be removable before Signers see the PCZT); a signer must be able to determine from the PCZT alone exactly what it authorizes; merging must be deterministic and fail loudly on conflict; and the encoding must be compact enough for memory-constrained signing devices and strictly versioned (ZIP-374, Requirements).

Two scope restrictions follow from history. PCZTs do not support Sprout and cannot create v4-or-earlier transactions: the v4 format predates the ZIP-244 transaction digest, and the role separations — in particular proving independent of signing — do not hold for it (ZIP-374, Specification). Two PCZT versions are specified: v1 supports creation of v5 transactions (ZIP-225); v2 supports v5 and v6 (ZIP-229). A PCZT version does not pin a transaction version — a v2 PCZT may carry a v5 or a v6 transaction, a v1 PCZT only v5 (ZIP-374, Versioning). The released pczt 0.9.3 parses both: `Pczt::parse` accepts version numbers 1 and 2, rejecting any other with `ParseError::UnknownVersion`. Its `serialize` emits v1 for a representable pre-v6 PCZT with a canonical-empty Ironwood bundle and no v2-only compact field state, and v2 otherwise; callers that require a fixed wire version use `pczt::v1::Pczt` or `pczt::v2::Pczt` (pczt, src/lib.rs and CHANGELOG.md, 0.9.0).

5.1 Roles and lifecycle

Definition 5.1 (ZIP-374 roles). Transaction creation is divided among the following roles, each entitled to perform a specific step on a PCZT (ZIP-374, Roles):

Creator (a single entity). Initializes the global fields and the empty per-protocol bundles.

Constructor. Adds transparent inputs and outputs, Sapling spends and outputs, and Orchard or Ironwood actions.

IO Finalizer (anyone). Declares the input/output set complete, computes the binding-signature keys, and signs dummy spends (§5.5).

Updater (anyone). Attaches metadata later roles need: derivation paths, full viewing keys, witnesses, anchors.

Prover. Attaches zero-knowledge proofs (§5.8).

Signer. Attaches signatures (§5.7).

Combiner. Merges parallel copies of a PCZT (§5.9).

Redactor. Removes fields that later entities do not need.

Spend Finalizer. Assembles the transparent `script_sigs` (§5.10).

Transaction Extractor. Produces and verifies the final transaction (§5.11).

A *Verifier* is “not a step in the transaction lifecycle, but a way of inspecting a PCZT” and checking its internal consistency on behalf of another entity that lacks the capacity to do so itself — for example a hardware signer (ZIP-374, Roles). The checks such a verifier runs are the per-action verification equations of Construction 5.8. Release pczt 0.9.3 exposes a `public roles::verifier` module whose `Verifier` gives access to the protocol-specific verification surfaces; the underlying equations are implemented by the protocol crates.

The lifecycle is not a fixed pipeline. Proofs and signatures slot in asynchronously: proof creation needs no spending key material, and for v5 transactions signatures commit to the anchor (through the sighash) but not to proofs, while proofs do not depend on signatures — so the Prover and Signer roles can run in either order or in parallel, their outputs merged by a Combiner (ZIP-374, Roles and Prover). From the v6 format onward the two roles become independent even of the anchor choice (§5.13).

5.2 Structure

Definition 5.2 (Field kinds). Every structure in a PCZT contains three kinds of fields (ZIP-374, Specification: Structure):

- *transaction effecting data*: required, part of the final transaction, always present;
- *transaction authorizing data*: initially empty, filled in as the PCZT is processed (proofs, signatures);
- *context data*: committed to by, or relevant to, the effecting data; present so that roles can be performed (openings, keys, paths).

The current `Pczt` struct is `{global, transparent, sapling, orchard, ironwood}` with all four bundles non-optional (pczt 0.9.3, `src/lib.rs`); the historical 0.7.0 v1 implementation carried the first three, and release 0.8.0 added the `ironwood` slot (§5.13). The bundles are not logically optional because a PCZT does not always contain a semantically valid transaction, and there may be phases where protocol-specific metadata must be stored before it is known whether that protocol has any inputs or outputs (ZIP-374, Structure).

Global fields. The `global` structure carries (ZIP-374, Global): `tx_version` (u32; MUST be 5 in PCZT v1) and `version_group_id` (u32; must be consistent with `tx_version`); `consensus_branch_id` (u32; non-optional because it commits to the consensus rules that will apply to the transaction); `fallback_lock_time` (Option<u32>; assumed 0 if omitted); `expiry_height` (u32;

ZIP-203); `coin_type` (u32, SLIP 44 — not part of the transaction and without consensus effect, present so later roles can check network-specific data such as derivation paths); `tx_modifiable` (a u8 bitfield, next paragraph); and `proprietary`, a map from UTF-8 strings to byte strings.

Definition 5.3 (The `tx_modifiable` bitfield). Field `tx_modifiable` tracks which parts of the transaction are still open to change (ZIP-374, Global):

Bit	Mask	Meaning	Merge
0	0b0000_0001	Transparent Inputs Modifiable	AND
1	0b0000_0010	Transparent Outputs Modifiable	AND
2	0b0000_0100	Has SIGHASH_SINGLE	OR
3–6	—	reserved; must be zero (merge fails otherwise)	—
7	0b1000_0000	Shielded Modifiable	AND

The Creator sets bits 0, 1, and 7 and clears bit 2; a Constructor checks the relevant bit before adding an input or output and may clear it; the IO Finalizer clears bits 0, 1, and 7 if there are shielded spends or outputs, and otherwise leaves the bitfield unchanged (§5.5). A Signer clears bit 0 when adding any signature that does not use SIGHASH_ANYONECANPAY — which includes *all* shielded signatures — and clears bit 1 for any signature not using SIGHASH_NONE; a SIGHASH_SINGLE signature also clears bit 1, because removing the paired output would not remove the signature. Bit 2 is set by any Signer using SIGHASH_SINGLE and tells Constructors to preserve the input/output pairing. Every Signer clears bit 7 (ZIP-374, Global).

The split between transparent and shielded modifiable flags is deliberate: Zcash’s sighash pins *all* non-transparent effecting data as soon as any input is signed, so separate flags keep purely-transparent PCZTs semantically close to PSBTs, where Constructor and Signer roles can interleave when inputs are not SIGHASH_ALL (ZIP-374, Rationale). The deep construction of the sighash itself is the ZIP-244 digest covered by the *Ironwood Guide* (the transaction digest); this section only ever cites it.

Transparent inputs and outputs. A `TransparentInput` carries the outputpoint (`prevout_txid` [u8;32], `prevout_index` u32), `sequence` (Option<u32>, default 0xFFFFFFFF), the lock-time constraints `required_time_lock_time` ($\geq$ 500000000) and `required_height_lock_time` ($>$ 0, $<$ 500000000), the `script_sig` (set by the Spend Finalizer), `value` (u64) and `script_pubkey` — both required, because the Transaction Extractor needs every input’s UTXO to derive the shielded sighash for the binding signatures — `redeem_script` (P2SH only), `partial_signatures` (a map from 33-byte public keys to signatures), `sighash_type` (Signers MUST use it; Spend Finalizers MUST fail on signatures that mismatch it), `bip32_derivation`, four preimage maps (RIPEMD160/SHA-256/HASH160/HASH256), and `proprietary` (ZIP-374, `TransparentInput`). A `TransparentOutput` carries `value`, `script_pubkey`, `redeem_script`, `bip32_derivation`, `user_address` (Option<UTF8String>, set by an Updater; Signers MUST parse it and confirm that it contains the recipient, directly or as a receiver within a Unified Address), and `proprietary` (ZIP-374, `TransparentOutput`). Transparent partial signatures are stored as DER-encoded ECDSA — the elliptic-curve digital signature algorithm over Bitcoin’s curve secp256k1, the transparent pool’s signature scheme — with the sighash-type byte appended, `sig_bytes` = `DER(sig) || sighash_type`, matching the Bitcoin/PSBT convention.

The Sapling bundle. A `SaplingBundle` carries lists of spends and outputs, a `value_sum` (i128, net spends minus outputs — wide enough that per-spend and per-output values can be redacted while the net remains), the `anchor` ([u8; 32]; in PCZT v1 set by the Creator and immutable, because the v5 sighash commits to it), and `bsk` (Option; None until the IO Finalizer runs, used by the Transaction Extractor for the binding signature). Each `SaplingSpend` requires `cv`, `nullifier`, and `rk` (effecting),

and optionally carries a per-spend 192-byte Groth16 zkproof (Prover), a 64-byte `spend_auth_sig` (Signer), recipient (43 bytes), value, `rcm` or `rseed` (`rcm` only for pre-ZIP-212 notes; the Prover needs one of them), `rcv` (the IO Finalizer compresses these into `bsk`; the Prover requires it; it opens `cv`, hence is redactable), `proof_generation_key` (`ak`, `nsk`) (Updater sets, Prover requires), witness (a `u32` position and a 32-level path), `alpha`, `zip32_derivation`, `dummy_ask` (a Constructor chooses it; the IO Finalizer uses and clears it; Signers MUST reject PCZTs containing it), and `proprietary`. Each `SaplingOutput` requires `cv`, `cmu`, `ephemeral_key`, `enc_ciphertext`, and `out_ciphertext` (whose length depends on the transaction version), and optionally carries `zkproof`, `recipient`, `value`, `rseed` (the Prover requires it, in preference to disclosing the shared secret), `rcv`, `ock` (lets a Signer verify `out_ciphertext`; None under the OVK policy None), `zip32_derivation`, `user_address`, and `proprietary` (ZIP-374, `SaplingBundle/SaplingSpend/SaplingOutput`).

The Orchard bundle. An `OrchardBundle` carries (`pczt`, `src/orchard.rs`; ZIP-374, `OrchardBundle/OrchardAction`):

actions. A list of `OrchardAction` structures.

flags (u8). Bit 0 `enableSpend`s, bit 1 `enableOutput`s; the remaining bits are reserved zeros in PCZT v1.

value_sum ((u64, bool)). The net value of spends minus outputs. The ZIP never states the boolean's meaning; only the reference implementation defines it (`true` = negative, via the orchard crate's sign mapping) — a spec gap we flag in §5.14.

anchor ([u8; 32]). Set by the Creator; fixed for the life of a v1 PCZT.

zkproof (Option<Vec<u8>>). A *single* Halo 2 proof for the whole bundle, set by the Prover — unlike Sapling's per-spend and per-output proofs.

bsk (Option<[u8; 32]>). The binding signing key, set by the IO Finalizer.

Each `OrchardAction` is $\{cv^{net}, spend, output, rcv\}$: the net value commitment, a spend half, an output half, and *one* optional `rcv` per action, since an action carries a single net value commitment (the *Ironwood Guide*, conservation of value).

An `OrchardSpend` carries: `nullifier` and `rk` (32 bytes each; effecting, required); `spend_auth_sig` (64 bytes; Signer); `recipient` (43 bytes, the raw Orchard payment address encoding; a Constructor sets it, the Prover requires it); `value` (the Prover requires it; a Signer may verify it against `cv_net` and confirm change amounts; redactable); `rho` and `rseed` (32 bytes each); `fvk` (96 bytes, `ak || nk || rivk` — the full viewing key of the note's recipient; an Updater sets it, the Prover requires it); `witness` (a `u32` position plus a 32-level, 1024-byte authentication path; Updater sets, Prover requires); `alpha` (32 bytes; a Constructor chooses it, the Signer requires it; redactable once `zkproof` and `spend_auth_sig` are set); `zip32_derivation`; `dummy_sk` (a Constructor chooses it for dummy notes; the IO Finalizer requires and clears it; Signers MUST reject PCZTs containing it); and `proprietary`. An `OrchardOutput` carries the effecting fields `cmx`, `ephemeral_key`, `enc_ciphertext`, and `out_ciphertext` (all required), plus `recipient`, `value`, `rseed` (the Prover requires it instead of disclosing the shared secret), `ock`, `zip32_derivation`, `user_address` (Signers MUST parse it and confirm that it contains the recipient), and `proprietary` (ZIP-374, `OrchardSpend/OrchardOutput`). The witness, anchor, and the note-commitment machinery these fields feed are exactly those of the *Ironwood Guide* (the commitment tree and authentication paths); we do not re-derive them here.

ZIP-32 derivations. A `Zip32Derivation` pairs a *seed_fingerprint* (`[u8; 32]`, the ZIP-32 seed fingerprint) with a *derivation_path* (`List<u32>`), where each index may carry the hardened flag, bit 31 (*index* | `0x80000000`). Shielded spend paths are generally fully hardened; transparent paths

generally include a non-hardened section matching BIP 44 or the ZIP-320 ephemeral-address path. The orchard crate rejects any non-hardened component at parse time — Orchard does not support non-hardened derivation — and its method `extract_account_index` recognizes exactly the pattern `[32', coin_type', account']` under a matching seed fingerprint.

5.3 Encoding

Construction 5.4 (PCZT encoding). A PCZT is encoded as

$$\underbrace{[0x50, 0x43, 0x5A, 0x54]}_{\text{ASCII PCZT}} \parallel \text{I2LEOSP}_{32}(\text{version}) \parallel \text{version-specific body},$$

an 8-byte header followed by the body. A parser rejects inputs shorter than 8 bytes (`TooShort`), a wrong magic (`NotPczt`), and — in the workspace crate — any version other than 1 or 2 (`UnknownVersion`); a parser encountering an unknown version **MUST NOT** parse the remainder. Files use the `.pczt` extension (ZIP-374, Encoding).

The body is a profile of the Postcard wire format — byte-for-byte the `serde` serialization of the `pczt` Rust crate, with the ZIP text canonical (ZIP-374, Encoding). Its primitives:

- `varint(n)`: emit n seven bits at a time, least-significant group first, one byte per group; every byte except the last has bit 7 set; encoders **MUST** emit the shortest encoding. A k -bit integer occupies at most $\lceil k/7 \rceil$ bytes, and in a maximum-length encoding the final byte contributes only the high $(k \bmod 7)$ bits — the fifth byte of a `u32` is at most `0x0F`; decoders **MUST** reject encodings exceeding either limit.
- Signed `i128` (the Sapling `value_sum`): $\text{encode}(n) = \text{varint}(\text{zigzag}(n))$ with $\text{zigzag}(n) = (n \ll 1) \oplus (n \gg 127)$, the right shift arithmetic and the result reinterpreted as unsigned; a non-negative n encodes as $2n$, a negative n as $2|n| - 1$; at most 19 bytes.
- `Option`: a tag byte `0x00` (`None`) or `0x01` followed by the value.
- `List`: `varint(count)` then the elements.
- `Map`: `varint(count)` then the entries in strictly ascending lexicographic key order, duplicates forbidden.
- `Structs`: field concatenation in the ZIP's field order, with no framing or tags.

The `v1` body is the concatenation

$$\text{Global} \parallel \text{TransparentBundle} \parallel \text{SaplingBundle} \parallel \text{OrchardBundle},$$

with the transparent bundle a pair of input and output lists, the Sapling bundle `{spends, outputs, value_sum, anchor, bsk}`, and the Orchard bundle `{actions, flags, value_sum, anchor, zkproof, bsk}` (ZIP-374, Encoding and Data type encodings).

A minimal PCZT, byte by byte. The smallest PCZT the Creator can emit makes the positional encoding concrete. Instantiating the Creator for NU6.2 (branch `0x5437F330`), expiry height 10,000,000, coin type 133, and building with no spends or outputs then serializing explicitly with `v2::Pczt::serialize` yields the 32-byte version-2 stream below (computed by script against the `pczt` crate; every byte is assertion-checked):

offset	field	bytes
0	magic "PCZT"	50 43 5a 54
4	version 2 (raw LE u32)	02 00 00 00
8	tx_version = 5 (varint)	05
9	version_group_id (varint)	8a ce 9c b5 02
14	consensus_branch_id (varint)	b0 e6 df a1 05
19	fallback_lock_time = None	00
20	expiry_height (varint)	80 ad e2 04
24	coin_type = 133 (varint)	85 01
26	tx_modifiable = 0b1000_0011	83
27	proprietary: empty map	00
28–31	four absent bundles	00 00 00 00

Three features of the Postcard discipline are visible at once. Integers are LEB128 varints, so `coin_type = 133` costs two bytes where 1 would cost one — and re-encoding with coin type 1 indeed shortens the stream to 31 bytes, shifting every later field: the encoding is positional, which is precisely why fields cannot be inserted within a version (cross-checked by script: single-field variants differ exactly at the annotated offsets). Absence is one 0x00 byte: the version-2 format wraps each of the transparent, Sapling, Orchard, and Ironwood bundles in an `Option` that collapses when the bundle is empty, so the entire bundle machinery of Construction 5.4 above costs four bytes until a first spend or output materialises. And the initial `tx_modifiable = 0x83` sets the two PSBT-inherited modifiable bits — transparent inputs (bit 0) and outputs (bit 1) — plus ZIP-374’s Zcash-specific shielded-modifiable bit 7, per Definition 5.3.

The convenience `Pczt::serialize` does not always emit v2: since release 0.9.0 it chooses the minimum encoding version that can represent the logical PCZT. This empty NU6.2 example is representable in v1 and therefore needs the explicit v2 serializer to produce the table above; explicit v1 output uses `v1::Pczt::serialize`.

Extensibility is deliberately narrow. New fields cannot be added within an existing PCZT version, because the Postcard encoding is positional; unlike PSBT there are no non-proprietary arbitrary key-value maps, so a Signer’s view of a PCZT is complete. The proprietary maps are the only within-version extension point and are reserved for private use; format evolution happens only through new versions namespaced by the header’s 4-byte version (ZIP-374, Extensibility and Proprietary Use fields). Serialization is deterministic: the crate’s round-trip test checks `serialize(parse(serialize(p))) = serialize(p)`.

5.4 Creation and construction

The deployed `Creator`, `Creator::new(consensus_branch_id, expiry_height, coin_type, sapling_anchor, orchard_anchor)`, selects the transaction format from the consensus branch: NU5 through NU6.2 use v5 (`version_group_id = 0x26A7270A`), while NU6.3 uses v6. It starts with `tx_modifiable = 0b1000_0011` (bits 0, 1, 7 set), Orchard flags = `0b0000_0011` (spends and outputs enabled), and empty transparent, Sapling, Orchard, and Ironwood bundles, with Sapling `value_sum = 0` and Orchard `value_sum = (0, false)`. The Ironwood anchor starts absent and may be supplied with `with_ironwood_anchor` for v6; its flags enable spends, outputs, and cross-address transfers. Builder options also set fallback lock time and per-pool flags. The Orchard initialization is non-negative zero, matching the v2 draft’s canonical empty `OrchardBundle` and `EMPTY_ORCHARD` (§5.14).

The crate ships no `Constructor` implementation (“The roles currently without an implementation are: `Constructor`”; `pczt, src/roles.rs`). In the deployed stack the `Constructor` step is performed by the `zcash_primitives` transaction builder: `Builder::build_for_pczt(rng, fee_rule)` checks that the value balance minus the fee is exactly zero (failing with `InsufficientFunds` or `ChangeRequired` otherwise), then builds the per-pool PCZT bundles *without proving or signing*, returning a `PcztResult` whose `PcztParts` carry the parameters, version, branch id, `lock_time = 0`, expiry height, and the

four bundles (transparent, Sapling, Orchard, and Ironwood), together with the Sapling, Orchard, and Ironwood bundle metadata. Then `Creator::build_from_parts(parts)` assembles the `Pczt`, setting `tx_modifiable = 0b0000_0000` — nothing modifiable, plus the `Has-SIGHASH_SINGLE` bit if any input uses it — `fallback_lock_time = Some(lock_time)`, and `coin_type` from the network parameters.

The Orchard side of that construction produces, per action (`orchard`, `src/builder.rs`, `SpendInfo::into_pczt`, `OutputInfo::into_pczt`, `build_for_pczt`):

$$cv^{\text{net}} := \text{ValueCommit}^{\text{Orchard}}(v^{\text{net}}, r_{cv}), \quad nf_{\text{old}} := \text{the spent note's nullifier under } fvk,$$

a fresh uniformly random α , and $rk := \text{Randomize}(ak, \alpha)$ — the RedPallas re-randomization of the *Ironwood Guide* (spend authorization) — populating the spend's recipient, value, rho, rseed, fvk, witness, and alpha (all `Some`), plus `dummy_sk` for dummy spends. The output half builds the new note with $\rho := nf_{\text{old}}$ of the paired spend, computes `cmx`, and runs the note encryption of the *Ironwood Guide*, populating recipient, value, and rseed. It currently sets `ock = None` with an in-code TODO (“Extract ock from the encryptor ...”), so the deployed Constructor never fills `ock` even when an outgoing viewing key is used — a divergence from the ZIP's field description (§5.14).

Bundles are padded with dummy actions and their positions randomized, so a consumer must map logical spends and outputs to bundle positions through the builders' metadata: `spend_action_index(n)` and `output_action_index(n)` on the Orchard `BundleMetadata`, and `SaplingMetadata` for Sapling. The wallet backend uses exactly this to attach per-output metadata to the right PCZT actions.

5.5 IO finalization: the binding-signature key

Once the input/output set is complete, the IO Finalizer freezes it. The deployed role fails with `NoSpends/NoOutputs` if the transaction would have no spends or no outputs; if there are shielded spends or outputs it clears `tx_modifiable` bits 0, 1, and 7. It computes the transaction-version-aware *shielded sighash*

$$sighash := \text{signature_hash}(tx_data, \text{SignableInput}::\text{Shielded}, txid_parts),$$

where `tx_data` is the *effects-only* transaction data reconstructed from the PCZT — proofs and signatures do not affect the sighash — and `txid_parts = tx_data.digest(TxidDigerster)`. The helper supports the v5/ZIP-244 and v6 transaction digests. It then calls the Sapling, Orchard, and Ironwood per-pool `finalize_io(sighash, 0sRng)`.

Construction 5.5 (Orchard IO finalization). Every action's `rcv` must be present. The role computes

$$bsk := \sum_{a \in \text{actions}} rcv_a \in \mathbb{F}_{p_{\text{Vesta}}},$$

converted through its 32-byte little-endian representation into a RedPallas binding signing key (`ValueCommitTrapdoor::into_bsk`), and recomputes

$$bvk := \left(\sum_a cv_a^{\text{net}} \right) - \text{ValueCommit}^{\text{Orchard}}(\text{value_sum}, 0),$$

where $\text{ValueCommit}^{\text{Orchard}}(v, r) := [v] V^{\text{Orchard}} + [r] R^{\text{Orchard}}$ with

$$V^{\text{Orchard}} = \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-cv}, v),$$

$$R^{\text{Orchard}} = \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-cv}, r)$$

— the net value commitment of the *Ironwood Guide* (conservation of value). It fails with `ValueCommitMismatch` unless `VerificationKey(bsk) = bvk`: this is the ZIP’s requirement that `value_sum` be consistent with the value commitments and trapdoors being aggregated. Finally, for each action with `dummy_sk` present, it *takes* the key (`Option::take`, guaranteeing that no spend authorizing key material remains after this role), derives `ask = SpendAuthorizingKey(sk)`, and signs the action with the shielded sighash (ZIP-374, IO Finalizer).

The Sapling analogue is $bsk := \sum_{\text{spends}} rcv - \sum_{\text{outputs}} rcv$ as a Jubjub scalar, made into a RedJubjub binding signing key, checked against $bvk = (\sum_{\text{spends}} cv - \sum_{\text{outputs}} cv)$ adjusted by `value_sum` as an i64 (`InvalidValueSum` if it does not fit); dummy spends are then signed with their `dummy_ask`, which is taken and cleared (the workspace pins sapling-crypto version 0.7, same algorithm). Ironwood uses the Orchard analogue on its separate bundle and anchor.

Remark 5.6 (Why a dedicated role). The key `bsk` cannot be computed by the Transaction Extractor without preserving every `rcv` until extraction — revealing the opening of every value commitment to all intermediate participants — and cannot be maintained incrementally by Constructors, because with parallel construction and a Combiner the incremental update would be wrong. A dedicated role that runs once, after IO completion, resolves both. The same role signs dummy spends so that no Signer ever parses spending key material (ZIP-374, Rationale, “Adding an IO Finalizer role”).

5.6 Updating

The deployed Updater wraps the protocol crates’ updater types. Method `update_global_with` sets proprietary fields; the wrapped Orchard `ActionUpdater` exposes, per action, the setters `set_spend_zip32_derivation` and `set_spend_proprietary` on the spend half, and on the output half the setters `set_output_zip32_derivation`, `set_output_user_address`, and `set_output_proprietary` (with Sapling, Ironwood, and transparent analogues). Release 0.9.3 also exposes setters for Sapling, Orchard, and Ironwood anchors and spend witnesses. Anchor updates are restricted to the v6 format under NU6.3, must precede the relevant proof, and may fill an absent anchor or repeat the same one but may not replace a different present anchor. Witnesses likewise must be attached before the corresponding proof.

5.7 Signing

Construction 5.7 (The Signer role). The deployed Signer (pczt feature `signer`), `Signer::new`, parses all bundles, extracts an effects-only `TransactionData`, and computes `txid_parts` and the shielded sighash of §5.5 *once*, cached across signatures. Its methods are `shielded_sighash()` and `transparent_sighash(index)` — the latter

`signature_hash(tx_data, SignableInput::Transparent(input i), txid_parts)`

under the input’s declared `sighash_type` — the signing methods `sign_transparent`, `sign_sapling`, `sign_orchard`, and `sign_ironwood`, and their `append/_apply_` variants for externally produced signatures (a hardware device), and `finish()`. To sign an Orchard spend, `alpha` must be present; with the wallet’s `ask`:

$rsk := ask + \alpha$ (the RedPallas `SpendAuth` randomization), require `VerificationKey(rsk) = rk`

— failing with `WrongSpendAuthorizingKey` otherwise — then

`spend_auth_sig := RedPallas.Sign(rsk, sighash).`

The external path `apply_orchard_signature` accepts a signature iff it verifies under the action's `rk` over the shielded sighash (`InvalidExternalSignature` otherwise); the Ironwood path applies the same RedPallas checks to its separate action bundle. After any shielded signature all three modifiable bits are cleared; after a transparent signature the bits are updated per its sighash type as Definition 5.3 specifies.

All shielded signatures behave as committing to the whole transaction — there is no shielded `SIGHASH_ANYONECANPAY` — which is why adding a shielded signature clears all modification permissions (ZIP-374, Global, `tx_modifiable`). Before signing an Orchard or Ironwood spend, the deployed Signer calls `verify_nullifier` (Construction 5.8) and treats `Missing{Recipient, Value, Rho, RandomSeed}` as acceptable — those fields may legitimately have been redacted for this signer — but hard-fails on any actual inconsistency; likewise for Sapling.

Construction 5.8 (Per-action verification equations). The orchard crate exposes, for Signers and Verifiers:

```

verify_cv_net :  cvnet  $\stackrel{?}{=} \text{ValueCommit}^{\text{Orchard}}(v_{\text{spend}} - v_{\text{output}}, \text{rcv})$ ;
verify_nullifier :  rebuild the note from (recipient,  $v, \rho, \text{rseed}$ ),
                    require fvk owns it (fvk.scope_for_address is Some),
                    and nf  $\stackrel{?}{=}$  the note's nullifier under fvk;
verify_rk :  ak from fvk,  rk  $\stackrel{?}{=} \text{Randomize}(ak, \alpha)$ ;
verify_note_commitment :  rebuild the output note with  $\rho := \text{nf}_{\text{old}}$  of the paired spend,
                           cmx  $\stackrel{?}{=} \text{Extract}(\text{NoteCommit}_{\text{rcm}}(...))$ ,

```

with the nullifier, randomization, and commitment constructions those of the *Ironwood Guide*. For dummy notes ($v = 0$) an `expected_fvk` argument is ignored in favour of the spend's own `fvk` field.

A signer implementation “should only need to implement”: Pedersen commitments using Jubjub/Pallas arithmetic (note and value commitments); BLAKE2b and BLAKE2s and the PRFs/CRHs built on them; the nullifier check; the KDF plus note decryption (`AEAD_CHACHA20_POLY1305`); the `SignatureHash` algorithm; RedJubjub/RedPallas signatures; and a source of randomness — no proving circuitry (`pczt`, `src/roles/signer/mod.rs`, module doc). For dependency-constrained environments a `low_level_signer::Signer` variant (not feature-gated) exposes `sign_ironwood_with/sign_orchard_with/sign_sapling_with/sign_transparent_with` closures over the parsed bundles plus `tx_modifiable`.

5.8 Proving

The Prover attaches proofs; per the ZIP it requires, for each Orchard action, the spend's recipient, value, rho, rseed, fvk, witness, and alpha, plus the action's `rcv` and the bundle anchor; delegating proving therefore reveals note private data, but not spending authority (ZIP-374, Prover). The deployed role (`pczt` feature prover) offers `requires_sapling_proofs` (any Sapling spend or output missing its zkproof), `requires_orchard_proof`, and `requires_ironwood_proof` (actions non-empty and the corresponding bundle zkproof `None`), letting a wallet skip downloading the Sapling parameters when they are unneeded; `create_orchard_proof(pk)` parses the bundle and calls the orchard crate's `Bundle::create_proof`; `create_ironwood_proof(pk)` does the same for Ironwood.

Construction 5.9 (Orchard proof creation from a PCZT). With no actions the call is a no-op. Otherwise, for each action i the prover rebuilds the spend information ($\text{fvk}_i, \text{note}_i, \text{merkle path}_i$) — failing with `WrongFvkForNote` if fvk_i does not own the note — and the output note with $\rho_i := \text{nf}_{\text{old},i}$ (`RhoMismatch` if inconsistent), then forms

$$\begin{aligned} \text{circuit}_i &:= \text{Circuit}(\text{spend}_i, \text{output note}_i, \alpha_i, \text{rcv}_i), \\ \text{instance}_i &:= (\text{anchor}, \text{cv}_i^{\text{net}}, \text{nf}_i, \text{rk}_i, \text{cmx}_i, \text{flags}), \end{aligned}$$

rejecting the identity rk (`IdentityRk`, added in orchard 0.13.0 for the consensus rule introduced in zcashd v6.12.1 / Zebra 4.3.1), and creates a *single* Halo 2 proof over all actions, stored as the bundle’s `zkproof`. The proof commits to the anchor and the flags as public inputs; the statement proved is the Action relation of the *Ironwood Guide*. The Ironwood path applies the same construction with its separate anchor and V3 note-plaintext version.

Because v5 signatures commit to the anchor but not to proofs, and proofs do not depend on signatures, the Prover and Signer can run in either order or in parallel; the `zcash_client_backend` documentation says to use the Combiner “if you create proofs and apply signatures in parallel” (ZIP-374, Prover).

5.9 Combining, redaction, and privacy

Construction 5.10 (Merge algebra). The deployed Combiner — `Combiner::new(Vec<Pczt>)` followed by `combine()` — left-folds pairwise merges; per-protocol bundles are merged before `Global`, “because each is only interpretable in the context of its own global”. The rules:

- Global effecting data (`tx_version`, `version_group_id`, `consensus_branch_id`, `fallback_lock_time`, `expiry_height`, `coin_type`) must be equal; `tx_modifiable` merges bit-by-bit per Definition 5.3.
- Optional fields: `merge( $\ell$ , None) =  $\ell$` ; `merge(None,  $r$ ) =  $r$` ; `merge(Some  $\ell$ , Some  $r$ ) =  $\ell$` iff `$\ell = r$` , else `FAIL`. Maps merge by key union, with values required equal on collision.
- Bundle structure: flags must be equal; if either side’s `bsk` is set (the IO Finalizer has run), action counts and `value_sum` must match; otherwise a strict prefix extension appends the extra actions *only if* the shorter side’s global has `Shielded Modifiable` set — else `FAIL`. Anchors follow the optional-field rule: an absent anchor may merge with a present one, while two present anchors must be equal. The per-action required fields (`cvnet`, `nullifier`, `rk`, `cmx`, `ephemeral_key`, `enc_ciphertext`, `out_ciphertext`) must be equal.

Combining must be functionally order-independent:

$$\text{Combine}(f_A(p), f_B(p)) = f_A(f_B(p)) = f_B(f_A(p)).$$

The equality-or-fail rule intentionally differs from BIP 174, which on conflicting values says the combiner “must choose the value it considers correct” — in practice last-wins — silently discarding data. Because every PCZT field is typed and known up front, requiring equality is always possible and turns every conflict into a loud failure. A Combiner **MUST NOT** combine PCZTs that do not represent the same transaction; once IO is finalized, a PCZT is uniquely identified by the `txid` its effecting data determines (ZIP-374, Combiner and Rationale).

Redaction. The Redactor provides per-bundle, per-field `clear` methods. The Orchard `ActionRedactor` can clear `spend_auth_sig`; the spend’s recipient, value, rho, rseed, fvk,

witness, alpha, zip32_derivation, and dummy_sk; the spend’s proprietary entries (by key or all); the output’s recipient, value, rseed, ock, zip32_derivation, and user_address; the output’s proprietary entries; and rcv. The bundle-level redactor can clear zkproof and bsk. The intended use is multiple independent Signers, each given a PCZT with only the information it needs — for example, without the other signers’ α values.

Privacy. A PCZT contains strictly more information than the transaction extracted from it: potentially counterparty addresses, values, note randomness, rcv (which opens cv), the spending account’s full viewing key, ZIP-32 paths, ock (which opens out_ciphertext), and user-facing address strings. Whoever receives a PCZT at a given stage learns everything present at that stage; participants SHOULD send it only to entities required for the remaining roles and SHOULD use the Redactor to strip fields. Delegating the Prover reveals note private data — not spending authority. The extracted transaction contains none of this extra information (ZIP-374, Privacy Implications).

5.10 Transparent finalization and lock time

Construction 5.11 (Script-sig assembly). The Spend Finalizer constructs each transparent input’s `script_sig`. Two forms are supported:

- *P2PKH*: require exactly one entry (*pubkey*, *sig*) in `partial_signatures` satisfying

$$\text{RIPEMD160}(\text{SHA256}(\text{pubkey})) = \text{the address hash};$$

then

$$\text{script_sig} := \text{PUSH}(\text{sig}) \parallel \text{PUSH}(\text{pubkey}),$$

where the `partial_signatures` value *sig* already consists of the DER signature followed by its sighash-type byte.

- *P2SH-P2MS* (*m*-of-*n* multisig):

$$\text{script_sig} := \text{OP_0} \parallel \text{PUSH}(\text{sig}_1) \parallel \dots \parallel \text{PUSH}(\text{sig}_m) \parallel \text{PUSH}(\text{redeem_script}),$$

where the leading `OP_0` is the dummy for the `OP_CHECKMULTISIG` bug and the signatures are the first *m* available ones *in redeem-script pubkey order* (surplus signatures beyond the threshold are discarded; fewer than *m* is a failure; pubkeys must be compressed 33-byte keys).

After finalization the role clears each input’s required lock times, redeem script, partial signatures, BIP-32 derivations, and all preimage maps, keeping value and `script_pubkey` — the UTXO — so the Transaction Extractor can verify the final transaction.

Construction 5.12 (Lock-time determination). The final `nLockTime` follows BIP 370’s rule (ZIP-374, Determining Lock Time; `pczt, src/common.rs, determine_lock_time`). Let *r* hold if some input has a required time or height lock; let *t_u* (“time unsupported”) hold if some input requires a

height lock (so a time lock cannot serve it), and h_u symmetrically. Then:

$$\text{nLockTime} = \begin{cases} \text{FAIL} & r \wedge t_u \wedge h_u, \\ \text{max required height locks} & r \wedge \neg h_u, \\ \text{max required time locks} & r \wedge h_u \wedge \neg t_u, \\ \text{fallback_lock_time or 0} & \neg r, \end{cases}$$

height winning when both types remain possible; an input specifying neither accepts either type. The crate fails with `IncompatibleLockTimes` when one input requires only time and another only height — and, diverging from ZIP-374/BIP 370, also when a single input specifies both lock types: `determine_lock_time` derives each type’s “unsupported” flag from any input carrying the other type’s field, so a both-specifying input sets $t_u \wedge h_u$ and fails, though the ZIP has it take the height lock.

5.11 Extraction and verification

The Transaction Extractor (`pczt` feature `tx-extractor`) checks transaction-version support and lock-time compatibility, then extracts the per-pool bundles, requiring every `script_sig`, every Sapling zkproof and `spend_auth_sig`, each Orchard and Ironwood bundle’s zkproof of canonical length for its action count, every Orchard-protocol `spend_auth_sig`, and each non-empty bundle’s anchor and `bsk`; it rejects the identity `rk` and an invalid `ephemeral_key`.

On the Orchard side, extraction converts the PCZT bundle into a regular bundle whose authorization is `Unbound = {proof, bsk}`; the canonical proof size is validated already at the `Unbound` stage, so the invariant “an Authorized bundle always has a canonical proof” holds across the binding step (this canonical-size check — `Proof::expected_proof_size` and `NonCanonicalProofSize` — entered orchard in 0.14.0 as the GHSA-2x4w-pxqw-58v9 fix), and `value_balance = i64(value_sum)` with `ValueSumOutOfRange` on overflow. The binding step is:

verify every `spend_auth_sig` under its `rk` over `sighash`, $\text{binding_sig} := \text{Sign}(\text{bsk}, \text{sighash})$,

failing with the `pczt` crate’s `Error::SighashMismatch` if any spend authorization does not verify — the crate maps the `None` that orchard’s `apply_binding_signature` returns on failure; the `bsk/bvk` consistency established by the IO Finalizer (Construction 5.5) guarantees the binding signature verifies under the `bvk` derived from the transaction’s value commitments and value balance.

Finally the extractor *verifies* the completed transaction — proofs and signatures — “so that an invalid transaction is detected before it is broadcast” (ZIP-374, Transaction Extractor). Sapling proof verification requires caller-provided `SpendVerifyingKey/OutputVerifyingKey` (error `SaplingRequired` if absent but needed); the Orchard verifying key, shared by Orchard and Ironwood proof verification, is generated on the fly if not provided via `with_orchard`.

5.12 The deployed wallet pipeline

The wallet layer drives the roles end to end. Function `create_pczt_from_proposal` — taking the wallet database, network parameters, spending account, OVK policy, and proposal (`zcash_client_backend` feature `pczt`) — supports only single-step proposals. It runs the same internal transaction construction as its non-PCZT sibling `create_proposed_transactions`; then chains `Builder::build_for_pczt`, `Creator::build_from_parts`, and `IoFinalizer::finalize_io`; and finally runs an Updater pass that:

1. stores the proposal metadata $\{\text{from_account}, \text{target_height}\}$, postcard-encoded, under the Global proprietary key `zcash_client_backend:proposal_info`;

2. stores a reduced recipient enum — `External`, `EphemeralTransparent`, `InternalShielded`, or `InternalTransparent`, the latter three carrying the receiving account — under the per-output proprietary key `zcash_client_backend:output_info` on each shielded and transparent output;
3. sets `user_address` on external outputs; and
4. attaches derivation paths: to Orchard and Ironwood spends and non-dummy Sapling spends the fully hardened `[ 32', coin_type', account' ]` under the account's seed fingerprint, and to transparent inputs the BIP-44 path `[ 44 | H, coin_type | H, account | H, scope, address_index ]` with `H = 0x80000000` and the last two components non-hardened.

The PCZT is then handed to the caller for the Prover, Signer, and Combiner roles — on other devices if need be.

The return path is `extract_and_store_transaction_from_pczt(wallet_db, pczt, sapling_vk, orchard_vk)`: it runs the Spend Finalizer, reads back the proprietary proposal and output metadata — failing if `proposal_info` is missing, so the PCZT must have come from `create_pczt_from_proposal` — reconstructs Sapling and Orchard output notes from the PCZT's explicit recipient/value/rseed fields (outputs whose recipient is absent are dummies; outputs whose note fields were redacted are treated as deliberately unrecoverable and skipped), runs the Transaction Extractor, recovers memos via `try_output_recovery_with_pkd_esk` with `esk` re-derived from the note (post-ZIP-212; the derivation is the PRFexpand construction of the *Ironwood Guide*, note encryption), computes the fee from the PCZT's transparent input values, and stores the result as a `SentTransaction`, returning the txid. The `sapling_vk` argument is optional so callers can first check `Prover::requires_sapling_proofs` and avoid downloading the Sapling parameters.

The same backend return path does not yet reconstruct or store Ironwood sent-output metadata: the generic Transaction Extractor extracts and verifies the Ironwood bundle, but the subsequent output collection covers only Orchard, Sapling, and transparent outputs. This is a deployed backend gap, not a limitation of the PCZT format or extractor (same source file, `extract_and_store_transaction_from_pczt`).

Feature gating. The `pczt` crate is `no_std` (alloc-based). The Creator, Verifier, Updater, Redactor, Combiner, and low-level Signer are always available; `io-finalizer`, `prover`, `signer`, `spend-finalizer`, and `tx-extractor` are cargo features; per-pool support is gated by the `transparent`, `sapling`, and `orchard` features (Ironwood uses the Orchard feature); and `zcp-builder` enables `Creator::build_from_parts` from the `zcash_primitives` builder output. The crate's end-to-end tests exercise the full pipeline — Creator/Builder → IO Finalizer → Updater → Prover → Signer → Combiner → Spend Finalizer → Transaction Extractor — for the `transparent_to_orchard`, `transparent_p2sh_multisig_to_orchard`, `sapling_to_orchard`, and `orchard_to_orchard` flows.

ZIP-48 multisig. ZIP-48 (transparent multisig) prescribes PCZT as its transaction-construction workflow: one participant constructs a PCZT spending multisig UTXOs, with change to a P2SH address at `m/48'/coin_type'/account'/133000'/1/*` (the `133000'` component reads “Zcash P2SH”); the PCZT is conveyed out-of-band over a private authenticated channel; each participant MUST verify that change outputs are wallet-controlled, verifies that the spend is expected, adds their signature(s), and returns the signed PCZT; one participant combines a threshold of signed PCZTs and extracts the final transaction (ZIP-48, Transaction construction). The crate's `transparent_p2sh_multisig_to_orchard` test exercises exactly this flow.

Tooling. The `zcash-devtool` CLI exposes the full role pipeline as subcommands: `pczt create / create-max / shield / create-manual / pay-manual` (Creator through IO Finalizer via `zcash_client_backend`), `inspect`; `update-with-derivation`; `redact`; `prove`; `sign`; `update-with-signature`, which applies externally created signatures; `combine`; `extract`; `send`; `send-without-storing`; and `to-qr / from-qr` — animated QR rendering and webcam scanning of PCZTs (feature `pczt-qr`), illustrating the constrained-transport hardware-wallet flow. Its `sign` command selects which spends to sign by matching the seed fingerprint and account in the PCZT’s `zip32_derivation` and `bip32_derivation` fields against the wallet’s seed (`extract_account_index`), then calls the Signer’s per-pool methods.

5.13 PCZT v2: deferred anchors and the Ironwood bundle

The draft’s v2 (version number 2 in the header) extends the format for v6 transactions (ZIP-229): it adds an `IronwoodBundle`, adds a `note_version` field (an enum `{V2, V3}`) to Orchard-protocol bundles, makes shielded bundle anchors `Option<[u8; 32]>`, and omits canonically-empty bundles from the encoding (ZIP-374, Versioning and v2 Encoding). Classification per our conventions: *implemented and released* — specified in the Draft ZIP and released in `pczt` 0.8.0 on 2026-07-23 (the v2 module with `Option`-wrapped bundles and empty-bundle omission, the `ironwood` slot, `note_version`, and optional anchors). The current 0.9.3 logical model retains these fields.

For `tx_version` 6, ZIP-374 makes anchors and witnesses deferrable: signatures commit to neither, and the ZIP permits an Updater to set or replace an anchor before proving — replacing an anchor MUST clear the bundle’s proofs, which commit to it as a circuit public input — while witnesses must root to the anchor whenever both are present. In the specified design this enables re-anchoring fully signed transactions with the `txid` unchanged, and spending *unmined* Orchard-protocol notes: an Orchard nullifier depends only on note data, with ρ fixed by the creating action rather than by tree position — impossible for Sapling even under v6, because Sapling nullifiers depend on the note’s position. An all-zeroes anchor sentinel was rejected in favour of `Option` because 0 is a valid Pallas base element (ZIP-374, Anchors and pre-authorization; Rationale, “Optional anchors in PCZT v2”).

Release `pczt` 0.9.3 implements the deferred case but not anchor replacement: `set_sapling_anchor`, `set_orchard_anchor`, and `set_ironwood_anchor` may fill an absent anchor (or repeat the same value) under v6/NU6.3 before proofs, but return `ConflictingAnchor` for a different present anchor and `ProofAlreadyPresent` instead of clearing a proof. Thus the released Updater supports anchoring after signing, but not the ZIP’s fully signed re-anchoring operation (`pczt, src/roles/updater/mod.rs`).

From NU6.3, Orchard-pool actions must be created with cross-address transfers disabled — bundle flag bit 2, `enableCrossAddress`, MUST be 0 in PCZT v1 and in any Orchard bundle mined under NU6.3+ — and wallets pair each real spend with a *fabricated* zero-valued output to the spent note’s own receiver, whose `enc_ciphertext` is random bytes. Signers MUST NOT reject a zero-valued output whose ciphertext does not decrypt on that basis alone, and SHOULD classify it as a tolerable dummy (optionally checking `cmx` against the explicit recipient/value/rseed). Conversely, a fabricated zero-valued spend paired with a real output is a *real* spend — no `dummy_sk`; it is signed via the normal Signer flow with the receiver’s spend authorizing key; only fully-dummy actions carry `dummy_sk` (ZIP-374, Fabricated same-address outputs). The `IronwoodBundle` carries the Ironwood-pool component of a v6 transaction, structurally identical to `OrchardBundle` — ZIP-229 defines the Ironwood pool as a second value pool of the Orchard protocol reusing the Action structure — differing in that `enableCrossAddress` is normally 1 (cross-address transfers are the usual case in the Ironwood pool), the anchor refers to the Ironwood note commitment tree, and `note_version` MUST be V3 (the ZIP-2005 quantum-recoverable note plaintext, lead byte 0x03); a non-empty `IronwoodBundle` in a `tx_version`-5 PCZT MUST be rejected (ZIP-374, `IronwoodBundle` and Rationale).

The orchard 0.15 release line implements the NU6.3-era machinery — bundle-format-aware flag parsing (`Flags::from_byte(flags, format)`, `UnexpectedFlagBitsSet`); the check `verify_cross_address_restriction`, requiring every action’s output addressed to the

same (g_d, pk_d) as its spend when flag bit 2 is 0, enforced by `finalize_io` and `create_proof` and provable only with an Ironwood-version proving key; same-receiver comparison via `Address::same_receiver`; and change outputs paired with a fabricated same-receiver spend (orchard, `src/pczt/verify.rs`, `io_finalizer.rs`, `prover.rs`, `parse.rs`, and `src/builder.rs`; `librustzcash Cargo.toml`). The current standalone head is 0.15.5; the `librustzcash` lockfile resolves 0.15.3.

5.14 Divergences between ZIP-374 and the deployed code

Per our conventions, we flag rather than resolve the following.

- *Dummy-key rejection is not implemented in the reference Signer.* ZIP-374 states that “Signers MUST reject PCZTs that contain `dummy_sk` values” (likewise `dummy_ask`), but no code path in the `pczt` Signer, through the current 0.9.3 source, inspects those fields — only the Redactor offers `clear_spend_dummy_sk/clear_dummy_ask`, and the IO Finalizer clears the keys after use. The normative MUST remains unmet by the crate’s own Signer; an embedding wallet must enforce it separately to conform, as must external signers such as hardware wallets (ZIP-374, `OrchardSpend.dummy_sk`; `pczt, src/roles/signer/`).
- *A v4 PCZT is constructible but dead.* The ZIP forbids creating v4-or-earlier transactions, yet `Creator::build_from_parts` maps `TxVersion::V4` to a PCZT with `tx_version = 4` (Sprout/V3 return `None`). Extraction and therefore roles that parse transaction effects reject that version as `UnsupportedTxVersion`, so the state cannot progress (`pczt, src/roles/creator/mod.rs`; `src/lib.rs, extract_tx_version`).
- *Negative-zero initialization (resolved upstream).* The released `pczt` 0.5.1 had `Creator::new` set the Orchard `value_sum = (0, true)`, differing from the v2 draft’s canonical empty `OrchardBundle (0, false)`; the byte encodings differ, which would have prevented the v2 empty-bundle-omission rule from round-tripping. The workspace head now sets `(0, false)`, matching `EMPTY_ORCHARD`, so the divergence is closed (`pczt, src/roles/creator/mod.rs`; ZIP-374, v2 Encoding).
- *The `ock` field is never populated.* The ZIP describes Sapling/Orchard `ock` as `None` only under the OVK policy `None`, but the deployed Orchard builder always sets `ock = None` (in-code TODO), so the Signer capability of verifying `out_ciphertext` is not currently exercisable for wallet-created PCZTs (orchard, `src/builder.rs`, `OutputInfo::into_pczt`).
- *Unspecified sign convention.* The ZIP declares the Orchard `value_sum` as `(u64, bool)` without stating the boolean’s meaning; only the reference implementation defines it (`true = negative`) — a spec gap (ZIP-374, `OrchardBundle`; orchard, `src/pczt/parse.rs`).
- *Reference-version scope.* The ZIP names `pczt` 0.7.0 specifically as the release that defines the v1 encoding. Current `pczt` 0.9.3 still exposes that fixed encoding through `pczt::v1`, but also ships the ZIP’s v2 encoding and Ironwood bundle. This is no longer an implementation-versus-release skew; 0.7.0 remains a deliberate historical anchor for v1 bytes (ZIP-374, Reference implementation; `pczt, CHANGELOG.md`).

6 Broadcast, expiry, and the transaction lifecycle

The *Ironwood Guide* leaves off with a transaction fully proved, signed, and bound — an artefact whose internal validity is settled. This section follows that artefact through the rest of its life: serialisation and the identifier under which the network knows it, the deployed store-then-broadcast discipline, the expiry mechanism that bounds how long it can remain pending, the bookkeeping by which the wallet tracks it to one of its terminal states, the confirmation policy that governs when its outputs

become spendable, and the truncate-and-rescan procedure that repairs the wallet’s view after a chain reorganisation. Throughout, we develop the shielded path; the transparent-input monitoring machinery (`TransactionsInvolvingAddress` and its filters) is gated behind the `transparent-inputs` feature of `zcash_client_backend` and is mentioned only in passing, so for a shielded-only wallet the monitoring surface is exactly the two status queries of §6.4.

6.1 Serialisation and the transaction identifier

A built transaction is serialised for broadcast by `Transaction::write`, which dispatches on version. Transactions v5 (version-group ID `0x26A7270A`) use `write_v5`, while v6 (version-group ID `0xD884B698`) use `write_v6`, which adds the separately encoded Ironwood bundle; v1–v4 use `write_v4` (ZIP-225; ZIP-229; protocol specification §7.1, *Transaction Encoding and Consensus*). The wallet places the resulting raw bytes in the data field of a `RawTransaction` gRPC message for submission.

The identifier under which the network, the mempool, and the wallet’s own database all track the transaction depends on the version. For v5 and v6, the *txid* is the root of the ZIP-244 digest tree,

$$\text{txid} := \text{BLAKE2b-256}(\text{ZcashTxHash_} \parallel \text{branchid}, d_{\text{hdr}} \parallel d_{\text{trans}} \parallel d_{\text{Sap}} \parallel d_{\text{Orch}}),$$

for v5; v6 appends the separate Ironwood digest d_{Iron} to the same top-level preimage. Every absent shielded pool contributes its protocol-defined empty-bundle digest. Here the personalisation is the 12-byte literal `ZcashTxHash_` (one underscore) followed by `branchid` = $\text{LE}_{32}(\text{CONSENSUS_BRANCH_ID})$, the 4-byte little-endian consensus branch id, and an absent pool hashes the empty string under its own personalisation (ZIP-244, *Txid Digest*). The *Ironwood Guide* assembles this tree in full (“What the signatures sign: ZIP-244 digests and their v6 form”); we do not re-derive it. Two consequences matter for the lifecycle. First, the tree is computed over *effecting* data only: changing the transaction’s consensus effect changes the identifier, while allowed recomputation of excluded attestations such as proofs or signatures leaves both the effect and identifier unchanged. This is ZIP-244’s non-malleability property. For v1–v4 the *txid* is instead the double-SHA-256 of the serialised transaction, which is malleable and therefore reliable only once the transaction is mined. Second, because the personalisation bakes in the branch id, the same transaction has a *different* *txid* on parallel consensus branches — replay protection across forks (ZIP-244, *Txid Digest*). For a purely shielded transaction the sighash coincides with the *txid*, as the *Ironwood Guide* already notes.

The header node of the tree is the one this section leans on:

$$d_{\text{hdr}} := \text{BLAKE2b-256}(\text{ZTxIdHeadersHash}, \text{LE}_{32}(\text{version}) \parallel \text{LE}_{32}(\text{versionGroupId}) \parallel \text{LE}_{32}(\text{branchid}) \parallel \text{LE}_{32}(\text{lockTime}) \parallel \text{LE}_{32}(\text{nExpiryHeight})),$$

with version including the overwinter flag (ZIP-244, node T.1). The field `nExpiryHeight` of §6.3 is therefore *effecting* data, committed by the *txid*: expiry cannot be malleated. Zebra’s mempool relies on exactly this, matching and rejecting expired transactions by their non-malleable *txid*.

One convention trips up every first-time implementer: *txid* bytes are stored and transmitted in protocol order but *displayed* byte-reversed and hex-encoded. The `Display` implementation of `Txid` reverses before encoding, and the `zcash_client_sqlite` view documentation states the same rule for its stored byte vectors.

6.2 Store, then broadcast

The deployed pipeline persists a transaction *before* the network ever sees it. The function `create_proposed_transactions` constructs, proves, and signs each step of a proposal (§6.6 fixes the target height it builds against), then persists the results via `WalletWrite::store_transactions_to_be_sent`, whose contract reads “This must be called before the transactions are sent to the

network”. The ordering is deliberate: a transaction the network might mine but the wallet has forgotten is unrecoverable spend history, whereas a stored transaction the network never saw is merely stale, and stored transactions “should be retransmitted while it is still possible that they could be mined”. In multi-step proposals (e.g. the ZIP-320 pairs of §6.5), storage happens only after *all* steps have been created, so that failure partway through creation cannot leave a transmittable prefix behind.

Persisting a created transaction does two things. First, it records the transaction itself: txid, expiry height read from `tx.expiry_height()`, raw bytes, fee, target height, and `min_observed_height := target height` (Table 4). Second, it marks every spent input as spent — Sapling, Orchard, and Ironwood nullifiers into their respective `*_received_note_spends` tables, transparent outpoints into `transparent_received_output_spends` — which locks those inputs against reselection by later proposals until the transaction is mined, or expires (§6.5).

Submission itself is one gRPC call: the `lightwalletd CompactTxStreamer` method `SendTransaction(RawTransaction)` returns (`SendResponse`), where `errorCode = 0` means accepted and any other value carries an `errorMessage(zcash_client_backend, proto/service.proto)`. Zaino implements the same `CompactTxStreamer` service as a drop-in replacement for `lightwalletd`, so submission is identical against either (`zaino, README.md`).

Remark 6.1 (The rebroadcast responsibility boundary). The `librustzcash` documentation records the obligation to retransmit — stored transactions “should be retransmitted while it is still possible that they could be mined” (`zcash_client_backend, src/data_api.rs`), and no-expiry transactions “should be rebroadcast by the wallet until they are either mined or conflict with another mined transaction” (`zcash_client_sqlite, src/wallet.rs`) — but implements no rebroadcast loop itself. The obligation falls on the embedding wallet application; the deployed CLI reference (`zcash-devtool`) submits exactly once.

6.3 Expiry (ZIP-203)

An unmined transaction cannot be left dangling forever: an indefinitely pending payment is indeterminism for the user and dead weight for every mempool. ZIP-203 (Status Final, Category Consensus) resolves this with a consensus-enforced expiry height.

Definition 6.2 (Transaction expiry, ZIP-203). Every transaction from Overwinter onward carries a field `nExpiryHeight` of type `uint32` — always a block height, never a UNIX timestamp. A transaction t is *expired* at block height H iff

$$(t \text{ is not coinbase}) \wedge (t.nExpiryHeight \neq 0) \wedge (H > t.nExpiryHeight),$$

and the consensus rule states: if a transaction is not a coinbase transaction and its `nExpiryHeight` is nonzero, it **MUST NOT** be mined at a block height greater than its `nExpiryHeight`; a block containing an expired transaction is invalid (ZIP-203, *Specification*; protocol specification §7.1). Expiry at height N thus makes block N the last block that may include the transaction. The sentinel `nExpiryHeight = 0` disables expiry, and for non-coinbase transactions `nExpiryHeight ≤ 499999999` **MUST** hold. From NU5 the bound is removed for coinbase only, and the `nExpiryHeight` of a coinbase transaction **MUST** instead equal its block height — this keeps v5 coinbase txids unique, since the ZIP-244 txid does not commit to the coinbase script height (ZIP-203, *Changes for NU5*; protocol specification §7.1).

The deployed default: exactly 40 blocks. ZIP-203 sets the default expiry delta at 20 blocks before Blossom and, from Blossom activation, 40 blocks from the current height — about 50 minutes at the 75-second target spacing — and permits nodes to override it (`zcashd`’s `-txexpirydelta` option; ZIP-203, *RPC, Config*). The deployed light-wallet stack applies the delta to the target height, normally one

above the chain tip, not to the current tip itself. The constructor `Builder::new` sets

$$n\text{ExpiryHeight} := \begin{cases} \text{targetHeight} & \text{coinbase,} \\ \text{targetHeight} + \text{DEFAULT_TX_EXPIRY_DELTA} & \text{otherwise,} \end{cases}$$

with `DEFAULT_TX_EXPIRY_DELTA = 40`, the coinbase rule applied regardless of network upgrade. A public override, `Builder::with_expiry_height`, replaces this: its documentation permits disabling expiry with height 0 but discourages it as not yet well-tested end-to-end, and coinbase builders reject an override that differs from the target height. A wallet that accepts the default therefore expires its transactions exactly 40 blocks after their target height. The rigidity is a feature: ZIP-315 (Draft; see Remark 6.6) says wallets SHOULD use the default 40-block expiry, and a wallet emitting idiosyncratic expiry deltas would fingerprint its transactions, exactly as an idiosyncratic fee would (§3).

Remark 6.3 (Documentation divergence: the builder’s stale delta). The doc comment on `Builder::new` states that the expiry height is set to the target “plus the default transaction expiry delta (20 blocks)”, while the constant it actually adds is `DEFAULT_TX_EXPIRY_DELTA = 40` (`zcash_primitives, src/transaction/builder.rs`). The code matches ZIP-203 post-Blossom; the comment preserves the pre-Blossom value. We follow the code.

Mempool eviction is node policy, mining is consensus. The consensus rule of Definition 6.2 constrains block *contents*; what nodes do with their mempools is policy layered on top. ZIP-203 specifies that when a transaction expires it is removed from nodes’ mempools, together with *all of its dependent transactions*. Zebra’s `remove_expired_transactions` evicts every mempool transaction once

$$\text{tipHeight} \geq t.n\text{ExpiryHeight},$$

i.e. as soon as the tip reaches the expiry height — at which point no future block can include t — and adds the evicted transaction’s non-malleable txid to its rejection list with `SameEffectsChainRejectionError::Expired`. The two formulations agree operationally with ZIP-203’s “block N or earlier; block $N + 1$ will be too late”.

The expiring-soon door policy. The `zcashd` node additionally refuses (as denial-of-service mitigation, not consensus) to accept into its mempool, or relay, a transaction that is *expiring soon*: one that would be expired within `TX_EXPIRING_SOON_THRESHOLD = 3` blocks of the next block height, where `IsExpiredTx( $t, h$ ) = ( $t.n\text{ExpiryHeight} \neq 0 \wedge t$  not coinbase  $\wedge h > t.n\text{ExpiryHeight}$ )` (`zcashd, src/main.h, src/main.cpp`). The practical consequence for wallets: a transaction submitted with fewer than about four blocks of remaining life is refused at the mempool door. Zebra applies no such buffer: its only admission-time expiry check is the consensus rule itself, evaluated at the next block height. The mempool hands each candidate to the transaction verifier with height `tipHeight + 1` (`zebrad, src/components/mempool/downloads.rs`), and the verifier’s `non_coinbase_expiry_height` rejects only when that height exceeds $t.n\text{ExpiryHeight}$ (`zebra-consensus, src/transaction/check.rs`); no expiring-soon threshold appears anywhere in the mempool component. A transaction with a single block of remaining life is therefore accepted and relayed by Zebra but refused by `zcashd`.

Expiry at a network upgrade. The `zcashd` wallet clamps a new transaction’s `nExpiryHeight` to one less than the next network-upgrade activation height (`zcashd, src/main.cpp`). The general `librustzcash` `Builder` does not: absent an override, it sets expiry to target height plus `DEFAULT_TX_EXPIRY_DELTA = 40`, which can cross an activation. Such a transaction is nevertheless bound to the branch and transaction format for which it was built and cannot be mined under incompatible post-upgrade rules. The SQLite parser’s unmined-transaction path assumes the expiry height lies in the same epoch and uses `BranchId::for_height(nExpiryHeight)` when no mined height is available; callers constructing pre-v5 transactions must therefore avoid a cross-activation

expiry. An unmined v4 transaction with expiry 0 carries no epoch hint at all and cannot be parsed until mined (zcash_primitives, transaction/builder.rs; zcash_client_sqlite, src/wallet.rs, parse_tx).

6.4 Tracking a pending transaction

Between submission and its terminal state, a transaction exists only as a row in the wallet database and an entry in remote mempools. The wallet-side record is the zcash_client_sqlite transactions table (Table 4); note that it may hold data unrecoverable from the chain, e.g. wallet-created transactions that expired unmined.

Column	Semantics
txid	transaction identifier, protocol byte order (UNIQUE; reverse and hex-encode for display)
created	wallet-local creation time
block	height of the containing block, set only once that block has been <i>scanned</i> (foreign key to blocks(height))
mined_height	mined height, settable from a status query without scanning the block; CHECK equal to block when block is non-null
tx_index	index of the transaction within its block
expiry_height	maximum height at which the transaction may be mined, if known
raw	serialised transaction bytes, if retrieved
fee	fee paid, if known
target_height	construction target height; set only for transactions this wallet created
min_observed_height	NOT NULL; minimum of the mempool-observation height and the mined height
confirmed_unmined_at_height	maximum height with positive recorded report of unminedness; CHECK NULL when mined_height is non-null
trust_status	user-assigned trust flag (§6.6)

Table 4: Columns of the transactions table.

Statuses and how they are learned. The backend distinguishes three externally observable states, the enum TransactionStatus: TxidNotRecognized, NotInMainChain (in the mempool, or mined only on a reorged-away fork), and Mined(height). Statuses arrive by evaluating TransactionDataRequest values the wallet emits: GetStatus(txid); Enhancement(txid), which downloads the full raw transaction and subsumes GetStatus; and, behind the transparent-inputs feature, TransactionsInvolvingAddress. These are typically served by lightwalletd’s GetTransaction and GetAddressTxids RPCs and fed back via WalletWrite::set_transaction_status or decrypt_and_store_transaction. Active polling is not optional even for a shielded wallet’s own history: fully transparent transactions, and transactions containing neither shielded inputs nor shielded outputs belonging to the wallet, are invisible to the compact-block trial decryption of the *Ironwood Guide* (“Delivery: the note reaches its owner”), so scanning alone can never confirm their fate.

The wire encoding of a status reply hides a protobuf quirk. The RawTransaction.height field signals: 0 = in the mempool; 0xffffffffffffffff = mined on a fork that is not the main chain; any other value = mined height in the main chain (zcash_client_backend, proto/service.proto). The deployed client maps

gRPC NotFound	↦	TxidNotRecognized,
height $\notin [1, 2^{32} - 1]$	↦	NotInMainChain,
otherwise	↦	Mined(height)

Applying a status. On `TxidNotRecognized` or `NotInMainChain` for an unmined transaction, `set_transaction_status` records a report of unminedness, setting `confirmed_unmined_at_height` to the current chain tip. On `Mined(h)` it sets `mined_height` to h , lowers `min_observed_height` to at most h , clears `confirmed_unmined_at_height`, links block if that block is already scanned, and advances the window of unused transparent address indices the wallet derives and watches past the highest index observed used; in every case the pending `tx_retrieval_queue` entry is deleted. Independently, when chain scanning itself encounters a wallet-relevant transaction in a block, `put_tx_meta` upserts block and `mined_height` to the scanned height along with `tx_index`, takes the minimum for `min_observed_height`, and clears `confirmed_unmined_at_height`.

When to stop asking. A `GetStatus` request is (re)generated for every transaction with `mined_height` `NULL` while any of the following holds (`zcash_client_sqlite`, `src/wallet.rs`, `transaction_data_requests`):

```
confirmed_unmined_at_height IS NULL
∨ (expiry_height > 0 ∧ confirmed_unmined_at_height < expiry_height)
∨ (expiry_height IS NULL
    ∧ confirmed_unmined_at_height < min_observed_height + certaintyDepth),
```

where `certaintyDepth := PRUNING_DEPTH + DEFAULT_TX_EXPIRY_DELTA = 100 + 40 = 140` blocks, and the result is unioned with explicit `tx_retrieval_queue` entries. A transaction with known nonzero expiry is thus polled until unminedness is confirmed at or beyond its expiry height; one with unknown expiry until 140 blocks past first observation; and one with expiry 0 only until unminedness is first confirmed — only the first disjunct can hold for it, so a single `TxidNotRecognized` or `NotInMainChain` reply ends the polling. The in-code comment above the query promises to “continue to query indefinitely” for such transactions and expects them “rebroadcast by the wallet until they are either mined or conflict with another mined transaction” (Remark 6.1); the SQL does not implement that promise — a documentation divergence. For display, the `v_transactions` view exposes the terminal flag

```
expired_unmined := mined_height IS NULL ∧ 1 ≤ expiry_height ≤ max(blocks.height),
```

i.e. nonzero expiry at or below the maximum scanned height.

Remark 6.4 (Documentation divergence: the retrieval-queue codes). The table documentation for `tx_retrieval_queue` says `query_type` is “0 for raw transaction (enhancement) data, 1 for transaction mined-ness information” (`zcash_client_sqlite`, `src/wallet/db.rs`), but the authoritative encoder `TxQueryType::code()` maps `Status` $\mapsto$ 0 and `Enhancement` $\mapsto$ 1 and round-trips through `from_code` (`zcash_client_sqlite`, `src/wallet.rs`). The documentation is inverted relative to the code; the code governs.

6.5 Expiry and fund availability

Storing a transaction locked its inputs (§6.2); expiry is what unlocks them. Note selection excludes a note appearing in `*_received_note_spends` only when the spending transaction is still *unexpired* in the following sense, evaluated at the target height H of the new proposal:

<code>mined_height < H</code>	(mined)	
<code>∨ expiry_height = 0</code>	(never expires)	
<code>∨ expiry_height ≥ H</code>	(unexpired)	(1)
<code>∨ (expiry_height IS NULL ∧ min_observed_height + 40 ≥ H)</code>	(default-delta guess).	

A transaction that has expired unmined (`mined_height` `NULL`, $0 < \text{expiry_height} < H$) fails every disjunct, so its inputs re-enter the spendable set for target heights beyond the expiry height. ZIP-315’s known-spendable definition matches: a TXO must not be “committed to be spent in another unexpired transaction created by this wallet” (ZIP-315, Draft).

Remark 6.5 (The unknown-expiry guess, and an inverted comment). The final disjunct of (1) is a guess that the originating wallet used the default delta. Reading the predicate directly: if the sender in fact used a *larger* delta or disabled expiry, the transaction is treated as *expired* from height `min_observed_height + 41` onward while it is actually still mineable — its inputs unlock prematurely, and a respend can conflict with a late mining of the original. If the sender used a *smaller* delta, the inputs merely stay locked slightly longer than necessary. Either misclassification persists only until the wallet observes the transaction mined or enhances it to learn the true expiry height. The doc comment on `tx_unexpired_condition` describes the first case as “treated as unexpired when it shouldn’t be” (`zcash_client_sqlite, src/wallet/common.rs`), which is inverted relative to its own SQL; we state the direction the predicate implies and flag the comment rather than silently following either.

Nullifiers outlive the lock. Dropping a pending spend’s inputs from the *spendable* set is not the same as forgetting them. The query `get_nullifiers(NullifierQuery::Unspent)` deliberately over-selects: a note spent in an unexpired-but-unmined transaction still contributes its nullifier, so that scanning (the *Ironwood Guide*, “Delivery: the note reaches its owner”) detects the spend if the transaction is later mined; a note’s nullifier leaves the watch set only once the spending transaction is mined or can never expire.

Aftermath of an expiry. ZIP-203 (*Wallet behavior and UI*) draws the user-facing conclusions: an expiring zero-confirmation transaction has even less inclusion guarantee, so UIs and services must never rely on zero-conf transactions in Zcash, and wallets should notify the user of expired transactions that must be re-sent. ZIP-315 (Draft) adds that sent transactions SHOULD be reported with confirmations and, while unmined with an expiry height, an estimate of time until expiry. Re-sending carries a privacy cost the wallet should recognise: ZIP-315 identifies repeated spends of the same TXOs across transactions — the typical aftermath of re-sending an expired payment — as an information leak linking the invalidated transaction to its replacement. For ZIP-320 (TEX) payments the deployed convention is to set the *same* expiry height on the first (shielded-to-ephemeral) and second (ephemeral-to-TEX) transactions, so another wallet on the same seed can infer when the second would have expired without observing it; if the second expires unmined, the ephemeral UTXO should be re-shielded before being re-spent.

6.6 Confirmations, trust, and spendability

Mining is not the end of caution: a freshly mined output can be un-mined by a reorganisation (§6.7), so the wallet imposes a confirmation policy before treating outputs as spendable.

Remark 6.6 (ZIP-315 is a Draft). ZIP-315 (“Best Practices for Wallet Implementations”) has Status Draft. The deployed stack nonetheless implements its central recommendations — the 3/10 confirmation split, zero-conf shielding, the fixed anchor depth, the 40-block expiry — so throughout this section ZIP-315 is cited as the (draft) rationale, not as a finalised standard.

Definition 6.7 (Confirmations policy). A `ConfirmationsPolicy` holds two `NonZeroU32` counts, c_{tr} (*trusted*) and c_{untr} (*untrusted*), with the constructor enforcing $c_{\text{tr}} \leq c_{\text{untr}}$, plus a flag `allow_zero_conf_shielding`. The default is $c_{\text{tr}} = 3$, $c_{\text{untr}} = 10$, with zero-conf shielding allowed, per ZIP-315; the constant `ConfirmationsPolicy::MIN` is `1/1/true`. Confirmations are counted since *and including* the block in which a note was produced.

The trusted/untrusted split is ZIP-315’s: a *trusted* TXO is one received from a party where the wallet trusts it will remain mined in its original transaction — e.g. created by the wallet’s own internal

TXO handling — and everything else is untrusted. The recommended counts rest on two observations: reorganisations are anecdotally shorter than 3 blocks, and value from a trusted TXO is always recoverable after a rollback (the wallet can just re-create the transaction), whereas an untrusted TXO may require asking the sender to re-send (ZIP-315, Draft). Users may also explicitly mark specific transactions trusted via `WalletWrite::set_tx_trust`, persisted in the `trust_status` column — ZIP-315’s “MAY enable users to mark specific external transactions as trusted” provision.

Definition 6.8 (Confirmations until spendable). Fix a target height H and write $x \ominus y$ for saturating subtraction (`BlockHeight` subtraction in `librustzcash` yields a `u32` via `saturating_sub`). Let $h_{\text{tr}} := H \ominus c_{\text{tr}}$ and $h_{\text{untr}} := H \ominus c_{\text{untr}}$, and for an output of a transaction with mined height m (if any) let

$$\text{confs}_{\text{tr}} := \begin{cases} c_{\text{tr}} & m \text{ unknown} \\ m \ominus h_{\text{tr}} & \text{else,} \end{cases} \quad \text{confs}_{\text{untr}} := \begin{cases} c_{\text{untr}} & m \text{ unknown} \\ m \ominus h_{\text{untr}} & \text{else.} \end{cases}$$

Then `confirmations_until_spendable` returns, and the output is spendable iff the result is 0:

- *transparent*: 0 if `allow_zero_conf_shielding`; else confs_{tr} if the transaction is trusted or the receiving key scope is internal; else $\text{confs}_{\text{untr}}$;
- *shielded*: confs_{tr} if the transaction is trusted; else, for internal scope, if the maximum mined height h_s of the transparent inputs to the shielding transaction is known, $h_s \ominus h_{\text{tr}}$ when those inputs are trusted and $h_s \ominus h_{\text{untr}}$ otherwise, and confs_{tr} when h_s is unknown; else $\text{confs}_{\text{untr}}$.

The saturation is what makes the count “decrease to a floor of zero” for long-mined outputs.

The internal-scope shielded branch encodes ZIP-315’s rule that shielded funds produced by zero-conf shielding are unavailable until the *source* UTXOs have sufficient confirmations: spendability of a shielding output is computed from $h_s = \text{max_shielding_input_height}$ and the inputs’ trust status, not from the shielding transaction’s own mined height.

Remark 6.9 (Deployed shielding rule vs. the ZIP-315 text). ZIP-315 is internally inconsistent here. Its formal “Shielded TXOs” bullet requires the source UTXOs to have at least $c_{\text{untr}} - c_{\text{tr}}$ confirmations and the shielding transaction at least c_{tr} ; its later “Transparent UTXOs” paragraph instead requires the source transaction to have the full c_{untr} confirmations, again alongside c_{tr} for the shielding transaction. Write h_s for the maximum source-input height, $h_m \geq h_s$ for the shielding transaction’s mined height, and $d := h_m - h_s$. The two ZIP passages respectively imply

$$H \geq h_s + \max(c_{\text{untr}} - c_{\text{tr}}, c_{\text{tr}} + d) \quad \text{and} \quad H \geq h_s + \max(c_{\text{untr}}, c_{\text{tr}} + d).$$

The deployed rule of Definition 6.8 instead tests only the source-input height, giving $H \geq h_s + c_{\text{untr}}$ for untrusted inputs, with no separate shielding-transaction check. At the defaults and $d = 1$, the formal bullet gives $h_s + 7$, while both the later prose and code give $h_s + 10$; for sufficiently large d , the code becomes laxer than both ZIP readings. The deployed trusted-input branch ($H \geq h_s + c_{\text{tr}}$) and unknown- h_s fallback likewise have no direct ZIP counterpart. There is therefore no single unambiguous ZIP threshold against which to label the implementation stricter or laxer.

The full spendability gate. Deployed note selection admits a shielded note as spendable only when all of the following hold: the note’s transaction is in a *scanned* block (`t.block` non-null); its nullifier and commitment-tree position are known and the containing shard is fully scanned (witness data exists — the *Ironwood Guide*, “Resting: the note in the commitment tree”); its mined height is at or below the anchor height; `confirmations_until_spendable = 0` under the policy; and it is not excluded by the spent-notes clause — spent in a mined or unexpired pending transaction, condition (1).

Target and anchor heights. The heights every predicate above is evaluated against are fixed at proposal time (`zcash_client_sqlite`, `src/wallet.rs` and `src/wallet/commitment_tree.rs`; `zcash_client_backend`, `src/data_api/wallet.rs`):

```
chainTip := max(scan_queue.block_range_end) - 1 (ranges are end-exclusive),
H := chainTip + 1 (the “mempool height”),
anchorHeightpool := max{checkpoint : checkpoint ≤ H - minConf},
anchor := minpools with checkpoints anchorHeightpool (Sapling, Orchard),
```

and `propose_transfer` uses the trusted count, $\text{minConf} = c_{\text{tr}}$ (default 3); the target height that `create_proposed_transactions` hands to `Builder::new` is the proposal’s `min_target_height()`. The anchor is therefore *at least* the trusted confirmation count deep under the wallet’s target-height convention. It equals that depth only when every relevant tree has a checkpoint exactly at $H - c_{\text{tr}}$; checkpoint gaps and the cross-pool minimum can make it older. This fixed upper bound on anchor height avoids leaking whether the notes being spent were trusted, the property ZIP-315’s anchor-selection recommendation secures. ZIP-315 phrases its recommendation relative to the chain head: its nominal $\text{block_chainTip} - c_{\text{tr}}$ is one older than the wallet’s newest permitted bound $H - c_{\text{tr}} = \text{chainTip} - (c_{\text{tr}} - 1)$, but actual checkpoint selection can make the deployed anchor older, so it is not unconditionally one block shallower. The tree and anchor mechanics themselves are the *Ironwood Guide*’s (its commitment-tree section); the wallet-layer content is only this choice of depth.

6.7 Reorganisations: truncate and rescan

A reorganisation invalidates the wallet’s suffix of the chain: mined transactions become unmined, anchors vanish, and witness data beyond the fork point is wrong. The deployed stack detects this during scanning and repairs it by truncation.

Detection. Scanning reports continuity errors: the `ScanError` variants `PrevHashMismatch` (the parent hash of a new block does not match the current tip), `BlockHeightDiscontinuity`, and `TreeSizeMismatch` all satisfy `is_continuity_error() = true`. To surface reorganisations *promptly* rather than on the next unlucky scan, `update_chain_tip` marks the `VERIFY_LOOKAHEAD = 10` blocks above the maximum scanned height with `ScanPriority::Verify` whenever that height is at or below the stable height $\text{tip} - \text{PRUNING_DEPTH}$; ranges marked `Verify` must always be scanned before `ChainTip` ranges. The choice of 10 blocks corresponds to roughly 12.5 minutes, within which a user plausibly received a transaction without their wallet having scanned the block.

Recovery loop. On a continuity error e , the documented algorithm is:

1. choose a rewind height at least one block before the error height (the module docs’ heuristic: `e.at_height() - 10`);
2. call `WalletWrite::truncate_to_height`, which may truncate *lower* than requested (below);
3. delete cached `CompactBlocks` above the returned truncation height, re-download, and rescan — `Verify` ranges first.

The truncation target is constrained by checkpoint availability: `truncate_to_height` truncates to the greatest height at most the requested one that has a note-commitment-tree checkpoint in *every* tree containing data, and errors with `RequestedRewindInvalid` — carrying the safe rewind height, the maximum over trees of their minimum checkpoint heights — if none exists. The constant `PRUNING_DEPTH = 100` controls an inclusive checkpoint window: the code computes its pruning floor as $\text{tip} - (\text{PRUNING_DEPTH} - 1)$, a 99-block height delta when checkpoints are contiguous. It should therefore not be read as an exact 100-block rewind depth. The variant `truncate_to_chain_state` additionally permits precise truncation from provided `ChainState` frontiers even when the target checkpoint was pruned.

What truncation does. Truncating to height h :

- clamps the scan queue so the wallet’s chain-tip view returns to h ;
- resets or nulls transparent outputs’ `max_observed_unspent_height` — a spending transaction may be entirely invalidated by a reorganisation that alters the anchors its shielded spends were built against;
- *un-mines* every transaction with `mined_height > h`: `block`, `mined_height`, `tx_index`, and `confirmed_unmined_at_height` are all set NULL, so the transaction reverts to the pending state of §6.4 and will either be re-observed as mined or expire;
- truncates all populated shielded shard trees — Sapling, Orchard, and Ironwood — to the checkpoint at h (tree and anchor mechanics: the *Ironwood Guide*, its commitment-tree section);
- deletes blocks above h and stale `tx_locator_map` entries.

Sent and received notes are deliberately *not* deleted: they may hold data, such as memos, unrecoverable from the chain. Balance computations must instead simply not count received notes whose transactions are no longer mined — which the spendability gate of §6.6 already guarantees, since an un-mined transaction has no scanned block and no qualifying mined height.

Long-run states. A wallet-created transaction may become *mined* and old enough to fall below the retained checkpoint window; it may become *expired unmined*, with its inputs released (§6.5) and its record retained (Table 4) with the user notified per ZIP-203; or, for the expiry-0 transactions a caller must explicitly request via `with_expiry_height`, it may remain *pending* indefinitely, with polling ending at the first confirmed-unmined reply (§6.4) and any rebroadcast falling on the embedding wallet (Remark 6.1).

7 Payment requests (ZIP-321)

The *Ironwood Guide* constructs everything a wallet needs once the sender knows where to pay: note encryption and detection, authentication paths, anchor selection, and the ZIP-244 transaction digest. What none of that settles is the step before any of it — how the payee communicates to the payer an address, an amount, and a memo, in a form a wallet can consume mechanically. ZIP-321 (“Payment Request URIs”) standardises that step as a URI scheme; it is Status Active, Category Standards/Wallet, created 2020-08-28, owned by Kris Nuttycombe and Daira-Emma Hopwood (ZIP-321, header). This section develops the format, its deployed parser — the `zip321` crate in `librustzcash` — and the pipeline that turns a parsed request into a transaction proposal and finally a transaction.

7.1 Requests, payments, and the single-transaction rule

Definition 7.1 (Payment; payment request). A *payment* is a transfer of funds implemented by a single shielded or transparent output of a Zcash transaction. A *payment request* is a request for a wallet to construct a single Zcash transaction containing one or more payments (ZIP-321, Terminology).

A ZIP-321 URI therefore represents a request for the construction of *one* transaction; implementations SHOULD construct a single transaction paying all the specified addresses (ZIP-321, URI Semantics). The number of payments one transaction can carry is limited only by transaction-construction restrictions: the ZIP notes that if every payment were Sapling, the value-balance limit of the protocol specification, §4.13 (Balance and Binding Signature, Sapling), implies construction may fail above 2109 distinct payments, and that the effective limit may be lower when Orchard or future address

types are involved. Recomputed from the specification, the bound is a transaction-size cap used inside that section’s overflow argument, not a direct value-range consequence: each Sapling output contributes 948 bytes to the transaction — an `OutputDescriptionV4` comprises the 32-byte `cv`, `cmu`, and `ephemeralKey` fields, a 580-byte `encCiphertext`, an 80-byte `outCiphertext`, and a 192-byte proof; an `OutputDescriptionV5` is 756 bytes plus its 192-byte entry in `vOutputProofsSapling` — and a transaction must fit within the specification’s 2 MB maximum transaction size (numerically equal to the 2 000 000-byte block-size limit), so at most $\lfloor 2\,000\,000/948 \rfloor = 2109$ outputs fit (protocol specification, §4.13; field sizes from §7.1, “Transaction Encoding and Consensus”). The specification uses the cap to show that the balance sum cannot overflow the type of committed values; ZIP-321 reuses it as the payment-count ceiling.

7.2 URI syntax

Definition 7.2 (ZIP-321 URI grammar). A ZIP-321 URI is a string derived by `zcashurn` in the following ABNF (RFC 5234), reproduced from ZIP-321 (URI Syntax); the productions `ALPHA`, `unreserved`, and `pct-encoded` are those of RFC 3986, and `base64url` is described below.

```

zcashurn      = "zcash:" ( zcashaddress [ "?" zcashparams ] / "?" zcashparams )
zcashaddress  = 1*( ALPHA / DIGIT )
zcashparams   = zcashparam [ "&" zcashparams ]
zcashparam    = [ addrparam / amountparam / assetparam / memoparam
                  / messageparam / labelparam / otherreqparam / otherparam ]

NONZERO      = %x31-39
DIGIT        = %x30-39
paramindex   = "." NONZERO 0*3DIGIT
addrparam    = "address" [ paramindex ] "=" zcashaddress
amountparam  = "amount" [ paramindex ] "=" 1*DIGIT [ "." 1*8DIGIT ]
assetparam   = "req-asset" [ paramindex ] "=" *base64url
labelparam   = "label" [ paramindex ] "=" *qchar
memoparam    = "memo" [ paramindex ] "=" *base64url
messageparam  = "message" [ paramindex ] "=" *qchar
paramname    = ALPHA *( ALPHA / DIGIT / "+" / "-" )
otherreqparam = "req-" paramname [ paramindex ] [ "=" *qchar ]
otherparam   = paramname [ paramindex ] [ "=" *qchar ]
qchar        = unreserved / pct-encoded / allowed-delims / ":" / "@"
allowed-delims = "!" / "$" / "'" / "(" / ")" / "*" / "+" / "," / ";"

```

Three global properties of the grammar deserve emphasis before the per-parameter semantics.

No authority component. A ZIP-321 URI is not a hierarchical URI: the grammar cannot derive `//` after the scheme, so there is no authority component (ZIP-321, URI Syntax note). For a single payment the recipient address **SHOULD** be placed directly in the hier-part, and a URI `zcash:<address>?... MUST` be treated as equivalent to `zcash:?address=<address>&...` (ZIP-321, URI Semantics); the query-parameter form with `addrparam` and distinct indices serves multiple recipients.

Payment grouping by paramindex. Addresses, amounts, labels, and messages sharing the same `paramindex` — including the empty index — belong to the same payment. An index **MUST NOT** have leading zeros, parameter ordering is insignificant, and index values need not be sequential (ZIP-321, URI Semantics). The grammar bounds indices to `"." NONZERO 0*3DIGIT`, that is $1 \leq i \leq 9999$; the empty index denotes a distinguished payment which the deployed crate stores at map key 0. Rendering and parsing are exact inverses on this range: index 0 renders as the empty suffix, and index $i \geq 1$ renders as `. || dec(i)`; on parse, an absent index maps to key 0, and the leading-zero strings `.0` and `.01` match no production and invalidate the URI (ZIP-321, Invalid Examples).

Percent encoding is confined to qchar. Only the values of `label`, `message`, `otherreqparam`, and `otherparam` — the `qchar` productions — may contain pct-encoded sequences. Addresses, amounts, and parameter *names* must appear literally: the ZIP’s invalid examples include `zcash:taddr?amount=1%30` (encoded amount digit), `zcash:taddr?%61mount=1` (encoded parameter name), and `zcash:%74addr...` (encoded address) (ZIP-321, URI Syntax note and Invalid Examples).

The base64url alphabet. Productions `memoparam` and `assetparam` carry binary content encoded as base64url, RFC 4648 §5, with padding omitted: values 62 and 63 encode as `-` and `_`, and implementations MUST NOT accept the standard-base64-only characters `+`, `/`, `=` (ZIP-321, URI Syntax). The deployed crate decodes with the `BASE64_URL_SAFE_NO_PAD` engine.

7.3 Validity and address semantics

The ZIP imposes structural validity rules beyond the grammar (ZIP-321, URI Syntax and URI Semantics):

- a purported ZIP-321 URI that cannot be parsed according to the grammar MUST NOT be accepted;
- if any non-address parameter carries a given `paramindex`, the URI MUST contain an address parameter with that index; and
- there MUST NOT be more than one occurrence of a given parameter name at a given index.

Implementations SHOULD check that each `zcashaddress` is a valid encoding per the protocol specification, §5.6, other than a Sprout address: a transparent address (Base58Check, §5.6.1.1), a Sapling payment address (Bech32 per ZIP-173, §5.6.3.1), or a Unified Address (Bech32m per BIP 350, §5.6.4.1 and ZIP-316). New address formats SHOULD be supported whether or not ZIP-321 is updated for them; if network context is available, each address SHOULD be checked for the correct network; and every requirement of ZIP-316 applies to payments to Unified Addresses (ZIP-321, URI Semantics).

Sprout addresses MUST NOT be supported in payment requests. The rationale is economic rather than syntactic: since Canopy, ZIP-211 restricts adding value to the Sprout pool, so senders cannot generally satisfy a request to pay a Sprout address (ZIP-321, URI Semantics). Section 7.9 flags where the deployed stack enforces this.

Forward compatibility. A parameter name prefixed `req-` that the parser does not recognise renders the *entire* URI invalid (MUST); unrecognised parameters without the prefix SHOULD be ignored. The prefix is potentially part of a future parameter’s name, not a modifier applicable to arbitrary parameters, and the original parameters `address`, `amount`, `label`, `memo`, `message` never carry it, because every conformant parser is required to understand them (ZIP-321, Forward compatibility).

Backward compatibility. Implementations of the informal pre-ZIP-321 `zcash:` URI scheme generally lacked the `req-` rule, the `paramindex` multi-payment facility, the `address=` query form for single payments, and the base64url memo treatment (ZIP-321, Backward compatibility).

7.4 Amounts

The `amount` parameter is a decimal number of ZEC. If a fraction is present, the separator MUST be a period and both the whole and fractional parts MUST be nonempty; grouping separators are forbidden; leading zeros in the whole part and trailing zeros in the fraction are ignored; the fraction carries at most 8 digits; and the amount MUST NOT exceed 21000000 ZEC (ZIP-321, ZEC Transfer Amount).

Thus 50, 050, and 50.00 all denote 50 ZEC; 0.5 and 00.500 denote 0.5 ZEC; and 50,000.00, 50., .5, and 0.123456789 are invalid.

Construction 7.3 (Amount parsing and rendering). Fix $\text{COIN} := 10^8$ zatoshis per ZEC and $\text{MAX_MONEY} := 21\,000\,000 \cdot \text{COIN} = 2\,100\,000\,000\,000\,000$ zatoshis.

Parse. Given a string matching `1*DIGIT [ "." 1*8DIGIT ]`, let c be the whole part read as an unsigned 64-bit integer, and let $z := \text{int}(\text{rpad}_8(f, 0))$, the fraction f right-padded with 0 to 8 digits ($z := 0$ when the fraction is absent). The value is

$$v := c \cdot \text{COIN} + z$$

with checked (non-wrapping) multiplication and addition, and the parse succeeds iff $0 \leq v \leq \text{MAX_MONEY}$ (the Zatoshis range check).

Render. Let $c := \lfloor v / \text{COIN} \rfloor$ and $z := v \bmod \text{COIN}$. Emit $\text{dec}(c)$; if $z \neq 0$, append `.` followed by $\text{dec}(z)$ left-padded with 0 to 8 digits and with trailing 0 characters trimmed.

7.5 Memos, labels, and messages

The `memo` parameter carries the content of the shielded memo field — the fixed 512-byte field of the note plaintext constructed in the *Ironwood Guide* (note encryption) — as `base64url` without padding. The decoded content **MUST NOT** exceed 512 bytes and, if shorter, is filled with trailing zeros to 512 bytes. A memo whose same-index address does not permit memos — a transparent address — makes the *entire* URI invalid (**MUST**) (ZIP-321, Query Keys).

Construction 7.4 (Memo encoding and decoding). Write `b64u(·)` for RFC 4648 §5 encoding with padding omitted.

Encode. For a 512-byte memo m , the parameter value is `b64u(strip0x00( $m$ ))`, where `strip0x00` removes trailing 0x00 bytes (`MemoBytes::as_slice`).

Decode. Let $b := \text{b64u}^{-1}(\text{value})$, rejecting any input containing `+`, `/`, or `=`; require $|b| \leq 512$; then

$$m := b \parallel 0x00^{512-|b|}$$

(`MemoBytes::from_bytes` zero-pads).

An absent memo parameter yields no memo; at output construction the wallet then substitutes the “no memo” sentinel `MemoBytes::empty() = 0xF6 \parallel 0x00511`.

The `label` parameter names an address (for example, the receiver’s name): a client **SHOULD** display it alongside the address, and it **MUST** be identifiable as distinct from the address itself. The `message` parameter is descriptive text the client may display to describe the payment (ZIP-321, Query Keys). Both are `qchar` values, so arbitrary UTF-8 reaches them only through percent encoding (§7.2).

7.6 Custom assets: req-asset (specified, not deployed)

The `req-asset` parameter extends payment requests to ZSA Custom Assets (ZIP-227), encoding an Asset Identifier and an amount in a single value; absent `req-asset`, the payment is for ZEC. A URI containing both amount and `req-asset` at the same index **MUST** be rejected. Only Orchard receivers are valid for Custom Asset payments, so a Custom-Asset request **MUST** use a Unified Address encoding at least one Orchard receiver (ZIP-321, Custom Assets).

Definition 7.5 (req-asset value encoding). The parameter value is

$$\text{b64u}(0\text{x}00 \parallel \text{I2LEOSP}_{64}(\text{value}) \parallel \text{EncodeAssetId}(\text{AssetId})) :$$

a version byte 0x00, the amount as an unsigned 64-bit little-endian integer of the asset’s minimal units, and the 66-byte encoded Asset Identifier of ZIP-227 — a 75-byte payload, hence 100 base64url characters. Future versions may define different structures; implementations **MUST** dispatch on the version byte (ZIP-321, Custom Assets).

Classification per our conventions: *specified*. The parameter appears in the Active ZIP-321 text, but it depends on ZIP-227, which is Status Draft, and the deployed parser rejects every unrecognised req- parameter — including req-asset — so a Custom-Asset request is invalid to the deployed stack today (zip321, src/lib.rs, to_indexed_param).

7.7 The deployed parser: the zip321 crate

The reference implementation is the zip321 crate in librustzcash (components/zip321, a single file src/lib.rs), published at version 0.9.0; release 0.1.0 (2024-08-20) factored it out of zcash_client_backend 0.13.0, which still re-exports it as a deprecated module zcash_client_backend::zip321 (components/zip321/Cargo.toml and CHANGELOG.md; zcash_client_backend, src/lib.rs). The Guide’s baseline is the local checkout — librustzcash main at commit e30517e4 — as for every other librustzcash citation in this volume, where the crate was 0.8.0. The optional-amount semantics shipped in release 0.7.0 (2026-04-23); current 0.9.0 changes dependencies and fixes the test-dependencies feature, not the parser semantics documented below.

Definition 7.6 (Types Payment and TransactionRequest). Type Payment holds a recipient address (zcash_address::ZcashAddress), an optional amount (Option<Zatoshis>), an optional memo (MemoBytes), an optional label and message, and a list other_params of retained unrecognised parameters. An absent amount means the *sender* chooses the amount — semantics introduced in release 0.7.0 (2026-04-23); releases up to 0.6.0 instead defaulted the field to zero, a behaviour ZIP-321 leaves unspecified (zip321, src/lib.rs; CHANGELOG.md, 0.7.0 entry).

Type TransactionRequest wraps a BTreeMap<usize, Payment>, map key 0 denoting the empty paramindex. Constructor new rejects more than 9999 payments (Zip321Error::TooManyPayments) and validates every non-empty request by round-tripping it through to_uri/from_uri; constructor from_indexed rejects any index key above 9999.

Construction 7.7 (Parsing: TransactionRequest::from_uri). Parsing proceeds in four stages.

- (1) Match the literal zcash: (lowercase; nom tag), then take the characters up to ? as the hier-part address. If nonempty, it must decode as a ZcashAddress and is recorded as an address parameter at payment index 0.
- (2) If input remains, require ? followed by a list, separated by & and consuming the rest of the input, of name[.index]=value pairs, each mapped to a typed parameter (address, amount, label, message, memo, or other); any unrecognised req- name is a hard error (“Required parameter {name} not recognized”).
- (3) Group parameters by payment index, rejecting any duplicate — the same parameter kind, or the same other-parameter name, twice at one index (Zip321Error::DuplicateParameter).

- (4) For each index, build a `Payment`. An address parameter must be present, an explicit amount of 0 must not target a transparent-only address, and a memo requires `can_receive_memo()` on its address; the corresponding errors are `RecipientMissing`, `ZeroValuedTransparentOutput`, and `TransparentMemo`. Labels, messages, and other parameters are collected.

Construction 7.8 (Rendering: `TransactionRequest::to_uri`). An empty request renders as `zcash:.` A request with exactly one payment stored at key 0 renders in hier-part form, `zcash:<address>` followed by `?` and the payment’s parameters if any. Otherwise every payment renders in query-parameter form: `zcash:?` followed by, in ascending key order, an address parameter and then that payment’s amount, memo, label, message, and other parameters, the index suffix omitted for key 0.

Definition 7.9 (Payment validity predicate). Constructor `Payment::new(a, v, m, ...)` returns an error iff

$$(\text{transparentOnly}(a) \wedge v = \text{Some}(0)) \vee (m \neq \text{None} \wedge \neg \text{canReceiveMemo}(a)),$$

with `PaymentError::ZeroValuedTransparentOutput` and `PaymentError::TransparentMemo` for the respective disjuncts, and `Ok` of the payment otherwise. The two predicates live in `zcash_address`: `canReceiveMemo` is true for Sprout, Sapling, and Unified addresses containing a Sapling or Orchard receiver, false for P2PKH, P2SH, and TEX; `transparentOnly` is true for P2PKH, P2SH, and TEX, and for Unified addresses that have a transparent receiver but neither a Sapling nor an Orchard receiver.

Character sets. On parse, a raw `qchar` value may contain alphanumerics plus `- . _ ~ ! $ ' ( ) * + , ; : @ %`, and a parameter name is ALPHA followed by ALPHA/DIGIT/+/-; the value is then percent-decoded and validated as UTF-8. On render, values are UTF-8 percent-encoded over the complement of `qchar`: the ASCII controls together with space and `" # % & / < = > ? [ \ ] ^ _ { | }` (the crate’s `QCHAR_ENCODE` set). Unknown parameters *without* the `req-` prefix are retained: their percent-decoded values land in `Payment::other_params` and are re-rendered, percent-encoded, by `to_uri`.

Totals. Method `TransactionRequest::total()` sums the payment amounts with overflow checking against `MAX_MONEY`, returning `Ok(None)` when any payment lacks an amount (the total is undefined) and `Err(BalanceError::Overflow)` on an out-of-range sum.

Test vectors. The crate’s test module encodes the ZIP’s own valid and invalid examples: `amount=1` parses to 10^8 zatoshis and `amount=123.456` to 12 345 600 000; the boundary 21000000 is valid and 21000000.00000001 invalid; i64-overflow and u64-wraparound amounts are invalid; `paramindex 10000` is invalid; `amount=123.` is invalid; and `req-unknown=x` is invalid.

7.8 From request to proposal to transaction

A parsed `TransactionRequest` enters the wallet through the function `propose_transfer` in module `zcash_client_backend::data_api::wallet`. It takes the wallet database, network parameters, the account to spend from, an input selector, a change strategy, the request, and a confirmations policy, and returns a `Proposal<FeeRule, NoteRef>` by delegating to the input selector’s `propose_transaction`; the proposal is later executed by `create_proposed_transactions`. The

convenience wrapper `propose_standard_transfer_to_address` places a single `Payment` into a one-element request and uses `GreedyInputSelector` with `SingleOutputChangeStrategy`.

The selector requires every payment to carry an amount: a missing amount yields `ProposalError::PaymentAmountMissing(idx)`, so a wallet UI must fill in user-chosen amounts before proposing.

Construction 7.10 (Payment-to-pool assignment). For each $(idx, \text{payment})$ in the request, `GreedyInputSelector` converts the recipient address to the wallet’s network and assigns an output pool:

- a transparent address maps to the transparent pool, emitting a `TxOut`;
- a TEX address maps to the transparent pool of the *second* step of a two-step proposal (below), its payment re-wrapped with no memo;
- a Sapling address maps to the Sapling pool;
- a Unified Address with a supported Orchard receiver maps to Orchard before NU6.3 and to Ironwood from NU6.3 onward (the Ironwood bundle delivers to the same Orchard-protocol receiver); otherwise it maps to Sapling if a supported Sapling receiver is present, then to its transparent receiver, or fails (`GreedyInputSelectorError::UnsupportedAddress`).

The selector then chooses spendable shielded notes to at least the required amount, preferring the pool matching the outputs, computes change under the change strategy, and returns a proposal whose steps embed the originating request and the resulting map `payment_pools : idx ↦ PoolType`.

TEX addresses (ZIP-320). A payment to a TEX address is consumed as a two-step proposal: step tr_0 excludes the TEX outputs and instead creates one ephemeral transparent change output; step tr_1 spends that ephemeral output to the TEX recipients, with the memo forced to `None`. Paying a TEX address from shielded inputs in one step is rejected (`ProposalError::PaysTexFromShielded`).

Step validation. Constructor `Step::from_parts` enforces that the keys of `payment_pools` exactly match the request’s payment indices, and that each selected pool satisfies `recipient_address().can_receive_as(pool)`; an Ironwood assignment is also accepted when the recipient can receive as Orchard, because the pools share the receiver protocol. It further checks that every payment has an amount, and the balance equation

$$v_{\text{transparent}}^{\text{in}} + v_{\text{shielded}}^{\text{in}} + v_{\text{prior steps}}^{\text{in}} = \text{total}(R) + \text{total}(\Delta), \quad (2)$$

where R is the embedded request, Δ the proposed balance — change, including ephemeral outputs, plus fee — and every sum is checked in `Zatoshis`; a violation is a `BalanceError`.

Serialization. A proposal serialises its transaction request as its ZIP-321 URI: the protobuf field `transactionRequest` of message `ProposalStep` (string, field 1, package `cash.z.wallet.sdk.ffi`) is produced by `to_uri` and parsed back with `from_uri`, alongside `PaymentOutputPool{paymentIndex, valuePool}` entries for the pool map (`zcash_client_backend, proto/proposal.proto`). This round-trip is why the crate accepts and renders the empty request `zcash:` — a shielding step has no payments — a deliberate divergence from the ZIP flagged in §7.9.

Execution. At transaction creation, for each (idx, pool) in the step’s pool map the corresponding `Payment` is fetched from the embedded request. A shielded output uses the payment’s

memo, or the sentinel `0xF6 || 0x00`⁵¹¹ when absent (§7.5); a memo on a transparent output raises `Error::Payment(PaymentError::TransparentMemo)`. For a Unified Address, the receiver actually paid is the one recorded in `payment_pools` — the Orchard, Sapling, or transparent receiver extracted from the UA. From here the construction is that of the *Ironwood Guide*: note encryption seals each output’s plaintext, and the spend and binding signatures sign the ZIP-244 sighash.

7.9 Divergences between ZIP-321 and the deployed stack

Our conventions require flagging every point where the deployed code and the ZIP disagree; ZIP-321 accumulates several, all in the parser.

- (D1) *Empty requests.* The crate accepts `zcash:` (and `zcash:?`) as a valid empty Transaction Request and renders empty requests as `zcash: (zip321, src/lib.rs)`, but the grammar of Definition 7.2 requires either an address or `? zcashparams` after the scheme, and the ZIP defines a request as having one or more payments. The divergence is deliberate: proposal serialisation round-trips shielding steps — which have no payments — through the URI form (§7.8).
- (D2) *Sprout recipients.* ZIP-321 says Sprout addresses MUST NOT be supported (§7.3), yet the crate parses a Sprout recipient without error — `ZcashAddress::try_from_encoded` decodes Sprout (`zcash_address, src/encoding.rs`) and `to_payment` performs no Sprout check; indeed `canReceiveMemo` is true for Sprout, so even a memo to one is accepted. Rejection occurs only downstream, at proposal time, when address conversion fails with `ConversionError::Unsupported("Sprout")` because `zcash_keys::address::Address` has no Sprout variant (`zcash_address, src/convert.rs`; `zcash_client_backend, src/data_api/wallet/input_selection.rs`). The deployed stack satisfies the payment-level requirement but not URI-level rejection.
- (D3) *Case sensitivity.* ABNF literal strings (RFC 5234) are case-insensitive, so a strict reading makes `ZCASH:` — and even `AMOUNT=` — grammar-valid; the deployed parser matches only lowercase (`zip321, src/lib.rs, tag("zcash:")` and the literal parameter names). The ZIP text never addresses case, which matters for QR alphanumeric-mode URIs, which are uppercase. We adopt the deployed, lowercase-only behaviour as normative and treat the ABNF case-insensitivity as a defect of the ZIP text: an uppercased URI is unparseable to the deployed stack, so a producer wanting QR alphanumeric mode must not obtain it by uppercasing the URI.
- (D4) *Valueless parameters.* The grammar makes `=` optional for `otherparam` and `otherreqparam` (a bare nonce is derivable), but the crate requires `name=value`, rejecting such grammar-valid URIs (`zip321, src/lib.rs, zcashparam`).
- (D5) *Bare percent signs.* The grammar admits `%` in a value only as pct-encoded (`% HEXDIG HEXDIG`), but the parser accepts `%` as a raw character and its percent decoder passes invalid sequences through unchanged, so `message=100%` is accepted — laxer than the grammar (`zip321, src/lib.rs, qchars` and `percent_decode`). A direct test against the crate confirms it: the values `100%`, `100%zz`, and `100%2` all parse, the malformed sequences surviving percent-decoding verbatim, while `100%25` decodes to `100%`.
- (D6) *Empty list elements.* The production `zcashparam` can derive the empty string, making `a=1&&b=2` arguably grammar-valid, while the crate rejects it (its list elements must be nonempty) — a divergence in the opposite direction of (D4) and (D5).
- (D7) *Zero-valued transparent outputs.* The crate rejects `amount=0` to a transparent-only address (Definition 7.9); ZIP-321 has no such rule. The crate’s changelog asserts such outputs are disallowed by consensus (`zip321, CHANGELOG.md, 0.7.0` entry). The assertion is inaccurate: consen-

sus admits zero-valued transparent outputs — `CheckTransaction` rejects only negative values and values above `MAX_MONEY` (`zcashd`, `src/main.cpp`), and the protocol specification constrains transparent outputs no further, requiring only a nonnegative remaining transparent value pool. What rejects them in practice is *standardness*: a zero-valued spendable output falls below the dust threshold (zero only for unspendable scripts, which no address payment produces), so `zcashd`'s `IsStandardTx` (`src/policy/policy.cpp`) and Zebra's equivalent mempool check (`zebrad`, `src/components/mempool/storage.rs`) refuse to admit or relay it. The crate's rule is sound — such an output is unrelayed — but on standardness, not consensus, grounds.

8 Compact blocks (ZIP-307)

A wallet that keeps payment detection local detects its notes by trial-decrypting every shielded output on the chain (*Ironwood Guide*, §“Delivery: the note reaches its owner”). That obligation costs twice: one key agreement per output for each candidate incoming viewing key, and the bandwidth of downloading every output. ZIP 307 (*Light Client Protocol for Payment Detection*) attacks the bandwidth half. A *compact block* is “a packaging of ONLY the data from a block needed to” (1) detect a payment to one’s shielded address, (2) detect a spend of one’s notes, and (3) update witnesses for new spend proofs (ZIP 307, “Compact Stream Format”). The deployed protocol widens this beyond ZIP 307’s Sapling-only scope to Orchard and Ironwood (not Sprout), and adds a fourth purpose: detecting spends of the coins of the wallet’s transparent addresses (`compact_formats.proto`, header comment, lines 21–26). The computational half survives compaction — every compact output must still be trial-decrypt — and that residual linear scan is one of the costs Project Tachyon’s proposed proof-carrying-wallet design aims to remove (Bowe–Miers, ePrint 2025/2031; tachyon-zcash/tachyon, book/src/proof-tree.md @ 26fdc165). The deployed service surface that carries compact blocks is catalogued in §9.4.

8.1 Normative status and wire-format history

ZIP 307 carries Status: Draft, covers Sapling only, and nowhere mentions Orchard, ChainMetadata, or transparent compact data (`zip-0307.rst`, header and whole file). Its message definitions are stale relative to deployment: the ZIP text defines `CompactBlock{BlockID id = 1; vtx = 3}` and `CompactTx{txIndex, txHash, spends, outputs}`, with element messages named `CompactSpend` and `CompactOutput` (ZIP 307, “Compact Stream Format”), whereas the deployed messages of §8.2 use different names, field numbers, and fields. The renames to `CompactSaplingSpend/CompactSaplingOutput` date to `lightwallet-protocol v0.3.1` (2021-12-09), which also added `CompactOrchardAction` (`lightwallet-protocol, CHANGELOG.md`).

The canonical proto files live in the `zcash/lightwallet-protocol` repository and are vendored — git subtree or symlink — into `lightwalletd`, `librustzcash` (the files under `zcash_client_backend/proto/` are symlinks into `lightwallet-protocol/walletrpc/`), and `Zaino`. The repository’s README states that it “IS not a specification” and defers to ZIP 307 (`lightwalletd, lightwallet-protocol/README.md`) — which is itself stale. No current document therefore fully specifies the deployed wire format. Per our conventions we classify the compact-block format *deployed but underspecified*: we cite `lightwallet-protocol` tagged releases as the de facto definition, and flag each point where ZIP text and deployment disagree (§8.8).

At the pinned revisions, the three codebases vendor three different revisions: `lightwalletd v0.4.1` (`protoVersion` present, no Ironwood); `librustzcash main v0.5.0` (`protoVersion` removed and reserved; Ironwood fields present); `Zaino` an intermediate snapshot carrying *both* the Ironwood fields and `protoVersion` (diff of the three `compact_formats.proto` copies). We flag the skew, and a changelog-date contradiction, in §8.8. Deployed-behaviour claims in this section are verified against `librustzcash e30517e4`, `lightwalletd 61fee32e`, `Zaino 0e057e22`, and `zips 6961098` (checkouts of 2026-07-14).

Since that snapshot, `lightwalletd` has caught up: release 0.5.0 vendored tagged `lightwallet-protocol v0.5.0`, removed `protoVersion`, and added Ironwood parsing and service data; release 0.5.4 now reports `lightwalletProtocolVersion = ``v0.5.0``` (`lightwalletd 09593ed, CHANGELOG.md`). Detailed behaviour comparisons below retain their explicit older pins so that their line citations and divergence findings remain reproducible.

8.2 The deployed messages

The messages live in the protobuf package `cash.z.wallet.sdk.rpc` and declare `proto3` syntax (`compact_formats.proto`, lines 5–11). For ordinary `proto3` scalar, string, bytes, and repeated fields,

Tag	Date	Change
v0.3.1	2021-12-09	CompactOrchardAction added; CompactSpend/CompactOutput renamed CompactSaplingSpend/CompactSaplingOutput
v0.3.2	2021-12-09	CompactOrchardAction.encCiphertext renamed ciphertext
v0.3.5	2023-07-03	ChainMetadata and CompactBlock.chainMetadata added; GetSubtreeRoots/SubtreeRoot added; nul- lifier RPCs added; CompactSaplingOutput.epk re- named ephemeralKey
v0.4.0	2025-12-03	CompactTxIn/TxOut and vin/vout added; BlockRange.poolTypes added; LightdInfo.lightwalletProtocolVersion added; CompactTx.hash renamed txid (serialization- compatible)
v0.4.1	2026-02-20	nullifier RPCs designated deprecated in the proto
v0.5.0	2026-06-30	Ironwood fields added; protoVersion removed, field number 1 and name reserved

Table 5: Wire-format history of lightwallet-protocol (CHANGELOG.md); v0.5.0 is the current tagged release.

absence decodes as the type’s default and an absent scalar is indistinguishable from an explicitly encoded default. Message fields retain presence information. The scanner must defend against default-zero scalar metadata — the tree sizes of §8.5 are the load-bearing case.

```

message CompactBlock {
  uint32 protoVersion = 1;
  uint64 height = 2;
  bytes hash = 3;
  bytes prevHash = 4;
  uint32 time = 5;
  bytes header = 6;
  repeated CompactTx vtx = 7;
  ChainMetadata chainMetadata = 8;
}

message ChainMetadata {
  uint32 saplingCommitmentTreeSize = 1;
  uint32 orchardCommitmentTreeSize = 2;
  uint32 ironwoodCommitmentTreeSize = 3;
}

message CompactTx {
  uint64 index = 1;
  bytes txid = 2;
  uint32 fee = 3;
  repeated CompactSaplingSpend spends = 4;
  repeated CompactSaplingOutput outputs = 5;
  repeated CompactOrchardAction actions = 6;
  repeated CompactTxIn vin = 7;
  repeated TxOut vout = 8;
  repeated CompactOrchardAction ironwoodActions = 9;
}

```

removed in v0.5.0; number and
name reserved
block height
block hash, 32 bytes
parent block hash, 32 bytes
Unix epoch time
“should always be unset (empty)”
compact transactions
end-of-block tree sizes
Sapling tree size at end of block
Orchard tree size at end of block
v0.5.0 only
position within the block
32 bytes, protocol byte order
unpopulated in practice
coinbase input omitted
v0.5.0 only

The container messages, from the deployed protos (lightwalletd vendored copy, walletrpc/compact_formats.proto, lines 14–80; librustzcash copy, lines 14–86):

- Field header “should always be unset (empty)” per the proto comment (lines 27–30). ZIP 307’s block header validation — Equihash (*Consensus Guide*, §“Equihash and the memoryless

model”), bits, difficulty — is therefore unimplementable against deployed servers; the ZIP itself flags the section as “only partially implemented: currently only prevHash is checked” (ZIP 307, “Block header validation”). Client-side, only prevHash and height continuity are enforced (§8.5). ZIP 221’s chain-history commitment was designed to support succinct probabilistic header-chain verification, but the pinned sync client does not consume it (ZIP 221).

- Field `protoVersion` never varies on the wire. `lightwalletd` never sets it, the assignment commented out as a TODO (`parser/block.go`, line 112); deployed Zaino serves 0 on every path (`zaino-state`, `compact_block.rs` line 285, `legacy.rs` line 1113), which for a `proto3` scalar is byte-identical to `lightwalletd`’s absent field. Its lone non-zero constructor (`zaino-fetch`, `src/chain/block.rs`, line 218) is a public helper whose only caller in the pinned repository formats a parse-error Debug string; no serving path calls it. Thus that literal does not reach the audited service’s wire output. Version v0.5.0 removed the field, reserving number 1 and the name (`librustzcash` copy, lines 33–36). The `librustzcash` scanner never reads it. Flagged in §8.8.
- Message `ChainMetadata` states each pool’s note commitment tree size *as of the end of this block*, as `uint32` (`compact_formats.proto`, lines 14–17). A full depth-32 tree has size 2^{32} , which this field cannot represent. Although `ScanError::TreeSizeOverflow` exists, the compact-scanning path at the pinned and current revisions does not emit it; its end-size additions remain `u32` arithmetic. Section 8.5 shows how these sizes turn a compact block into note positions.
- Field `txid` carries the 32-byte transaction identifier in protocol byte order; the proto comment mandates that it “MUST NOT be reversed or hex-encoded” — the reversed hex form is display-only — citing protocol spec §7.1.1, “Transaction Identifiers” (`compact_formats.proto`, lines 52–57).
- The fee field is documented “present if server can provide”, but neither deployed server populates it: `lightwalletd` leaves it commented out with TODO: `calculate fees` (`parser/transaction.go`, line 407) and Zaino hardcodes `fee: 0` (`zaino-fetch`, `src/chain/transaction.rs`, line 369). Clients cannot rely on it. For a transaction with no transparent inputs the proto comment gives the client-side reconstruction (`compact_formats.proto`, lines 59–64):

$$\text{fee} = \text{valueBalanceSapling} + \text{valueBalanceOrchard} + \sum \text{vPubNew} - \sum \text{vPubOld} - \sum \text{tOut},$$

with `valueBalanceIronwood` joining the sum in v0.5.0.¹

- Field `vin` omits the coinbase transaction’s single null-outpoint input; a light client recognises a coinbase `CompactTx` by `index = 0`, coinbase being always first in a block. Server `lightwalletd` implements exactly this, skipping `vin` when `index = 0` (`compact_formats.proto`, lines 70–76; `parser/transaction.go`, lines 398–403). The transparent element messages are `CompactTxIn{prevoutTxid, prevoutIndex}` and `TxOut{value, scriptPubKey}` (`compact_formats.proto`, lines 82–104).

The shielded elements. Each shielded element message keeps only the detection-relevant bytes of its full on-chain description.

A `CompactSaplingSpend` retains only the 32-byte nullifier of the pre-v5, 384-byte `Spend` description counted by ZIP 307 (`cv` 32, `anchor` 32, `nullifier` 32, `rk` 32, `zkproof` 192, `spendAuthSig` 64) — the one field needed to detect a spend of one’s own notes; ZIP 307 quotes this historical comparison as a 90% bandwidth improvement. In v5/v6 the anchor and bundle structure are shared, so 384 bytes is not the marginal size of each additional spend (ZIP 307, “Spend Compression”). A

¹Type `uint32` additionally caps the representable fee at $2^{32} - 1$ zatoshi (≈ 42.9 ZEC); the proto comment does not remark on this.

Message	Fields (bytes)	Payload	Full form	Saving
CompactSaplingSpend	nf (32)	32	384	~90%
CompactSaplingOutput	cmu (32), ephemeralKey (32), ciphertext (52)	116	580 + 32	~80%
CompactOrchardAction	nullifier (32), ephemeralKey (32), ciphertext (52)	148	—	—

Table 6: Compact shielded elements (`compact_formats.proto`, lines 106–130). The Sapling comparison baselines and saving figures are ZIP 307’s historical pre-v5 accounting (“Spend Compression”, “Output Compression”); the 148-byte Orchard total is arithmetic, not stated in the sources. Payload figures count raw bytes; on the wire, a `CompactSaplingSpend` encodes to 34 bytes (36 as a `CompactTx.spends` element), a `CompactSaplingOutput` to 122 (124), and a `CompactOrchardAction` to 156 (159 — its 156-byte payload needs a two-byte length varint). Field headers add 2 bytes per 32-byte field and 2 per 52-byte ciphertext. Computed by script, proto-encoding every compact element of mainnet block 3,412,934 (3 spends, 8 outputs, 10 actions; `GetBlock`, `light-walletd` v0.4.19, 2026-07-15); the sizes are block-independent because all field lengths are fixed.

`CompactSaplingOutput` retains `cmu`, the ephemeral key, and the first 52 bytes of `encCiphertext`, “Total size is 116 bytes” per the proto comment, against 580 bytes of ciphertext plus the 32-byte ephemeral key streamed in full form — ZIP 307’s 80% reduction (ZIP 307, “Output Compression”; `lightwalletd compact_formats.proto`, lines 114–122; full encodings per protocol spec §§7.3–7.4). A `CompactOrchardAction` serves both roles of an Action in one message: its `nullifier` detects spends of the input note, while `cmx`, `ephemeralKey`, and `ciphertext` detect and place the output note (`compact_formats.proto`, lines 124–130; full Action encoding per protocol spec §7.5).

8.3 Omissions and fetch-on-detect

Compact blocks omit the 512-byte memo, the 16-byte AEAD tag, the 80-byte `outCiphertext`, `cv`, `rk`, anchors, zkproofs, and signatures (ZIP 307, “Transaction privacy”). Recovering a memo, or spend information via the outgoing ciphertext, thus requires downloading the full transaction — deployed as the `GetTransaction` RPC. ZIP 307 attaches three requirements to that fetch: the client SHOULD obscure its interest by also downloading numerous uninteresting transactions, SHOULD download all transactions of any block it touches, and MUST convey the privacy reduction to the user (ZIP 307, “Transaction privacy”).

The deployed pipeline expresses the fetch as data: scanning emits `TransactionDataRequest::Enhancement(txid)` — “download of complete raw transaction data” — which the caller satisfies via `GetTransaction` and feeds to `wallet::decrypt_and_store_transaction(zcash_client_backend)`. The crate’s own sync loop documents that it does not yet perform enhancement (`zcash_client_backend, src/sync.rs`, lines 1–10).

8.4 Compact trial decryption

The 52-byte `ciphertext` field is the prefix of `encCiphertext` that encrypts exactly the *compact note plaintext*: lead byte (1) || diversifier d (11) || value v (8) || `rseed` (32), so `COMPACT_NOTE_SIZE` = 52; the full plaintext appends the 512-byte memo, and the full ciphertext a 16-byte tag, giving 580 bytes. ZIP 307’s rationale: these bytes “contain the contents and opening of the note commitment, which is all of the data needed to spend the note and to verify that the note is spendable” (ZIP 307, “Output Compression”). The *Ironwood Guide* constructs compact decryption for Orchard in §“Delivery: the note reaches its owner”; we do not repeat the construction, and record here only the deployed mechanics and the integrity argument.

Function `try_compact_note_decryption` (doc-comment: “Implements the procedure specified

in ZIP 307”) proceeds as follows:

$$\begin{aligned} K &:= \text{KDF}(\text{KA.Agree}(\text{ivk}, \text{epk}), \text{ephemeralKey}), \\ \text{np} &:= \text{ciphertext}[0..52] \oplus \text{ChaCha20}_K(\text{nonce} = 0^{96})[64..116], \end{aligned}$$

a bare ChaCha20 keystream with zero nonce, sought to byte offset 64 — skipping keystream block 0, which the full AEAD reserves for the Poly1305 key. The candidate np is parsed as a compact note plaintext (no memo), then `check_note_validity` accepts iff

- (i) the plaintext parses, and
- (ii) the recomputed commitment matches the on-chain one, $\text{Extract}(\text{NoteCommit}_{\text{rcm}}(\dots)) = \text{cmu}$ (respectively cmx), and
- (iii) for ZIP-212 Sapling and for Orchard/Ironwood, the ephemeral secret esk re-derived from rseed reproduces the published key: the byte encoding of $\text{KA.DerivePublic}(\text{esk}, g_d)$ equals `ephemeralKey`, compared in constant time.

Condition (ii) is the integrity substitute: truncation at 52 bytes discards the AEAD tag, so a compact decryption cannot be authenticated via Poly1305; ZIP 307 instead recomputes the note commitment from the decrypted plaintext and compares against the on-chain cmu, itself authenticated by the binding signature over the transaction (ZIP 307, “Output Compression”). Condition (iii) is the deployed strengthening beyond the tag: it defeats a mauled ephemeral key (*Ironwood Guide*, §“Delivery: the note reaches its owner”). For pre-ZIP-212 Sapling notes the domain cannot re-derive esk, so the implementation accepts without condition (iii).

Lead byte and the ZIP-212 grace period. The lead byte is 0x01 pre-Canopy and 0x02 under ZIP-212; deployed code switches on `zip212_enforcement`: during the grace period of 32,256 blocks after Canopy activation both lead bytes are accepted, and thereafter only 0x02 (`zcash_primitives, src/transaction/components/sapling.rs`, lines 27–40; `zcash_protocol, src/consensus.rs`, line 681). ZIP 307’s pseudocode disagrees with itself here: its two “[Canopy onward]” lines are *both* conditioned on $\text{height} < \text{CanopyActivationHeight} + \text{ZIP212GracePeriod}$, the first rejecting $\text{leadByte} \notin \{0x01, 0x02\}$ and the second rejecting $\text{leadByte} \neq 0x02$ (ZIP 307, “Scanning for relevant transactions”, `zip-0307.rst`, lines 275–276). Read literally, the pair rejects 0x01 *during* the grace period and constrains nothing after it — the inverse of the intent; the second condition should read $\geq$. Deployed code implements the intended semantics. The defect is tracked in `research/upstream-defects.md`, item 1 (ZIP 307: second Canopy-onward lead-byte condition rejects 0x01 during the grace period; verified 2026-07-15 at zips 69610984). We flag the divergence in §8.8.

8.5 Scanning: spends, positions, and tree sizes

Spend detection. The scanner parses every compact nullifier field up front: each `CompactSapling Spend.nf` and each `CompactOrchardAction.nullifier`. A malformed length yields `ScanError::EncodingInvalid` rather than a panic. The parsed nullifiers are compared against the wallet’s known-unspent nullifiers in constant time (`find_spent`); unmatched nullifiers are retained in per-block nullifier maps.

Tree building and positions. Every cmu and cmx in the block — not only the wallet’s — is appended to the wallet’s note commitment tree with `Retention` marks, and each detected note is marked at its position. The position is computed arithmetically, never by replaying the chain:

$$\text{pos}(\text{output } i \text{ of } tx) = \text{start_tree_size} + \text{offset}(tx) + i,$$

where $\text{offset}(tx)$ counts the outputs of earlier transactions in the block (`PositionTracker::sapling, orchard_note_position`). The shardtree that consumes these marks is constructed in the *Ironwood Guide*, §“Resting: the note in the commitment tree”; the observation that a light wallet derives positions from boundary metadata and output order appears in §“Delivery: the note reaches its owner”.

Positions are load-bearing, not merely convenient: the `TreeSizeUnknown` doc-comment records that without the tree size it is “impossible to construct the nullifier for a detected note” — the Sapling nullifier depends on the note’s position, via $\rho = \text{MixingPedersenHash}(\text{cm}, \text{NotePosition})$; Orchard’s does not, but both pools need positions for witnesses (protocol spec, “Computing ρ values and Nullifiers” and “Dummy Notes”).

Where the tree size comes from. The start size for a block is resolved by a ladder (`tree_sizes_around`):

$$\text{start}(\text{pool}) := \begin{cases} s_{\text{prior}} & \text{prior BlockMetadata known;} \\ s_{\text{final}} - n_{\text{block}} & \text{chainMetadata present, by checked subtraction;} \\ 0 & \text{block below the pool’s activation height;} \\ \perp & \text{otherwise (ScanError::TreeSizeUnknown),} \end{cases}$$

where s_{final} is the chainMetadata end-of-block size and n_{block} the block’s own output count. The subtraction is checked_sub: underflow yields `ScanError::TreeSizeInvalid`, which is precisely the guard against proto3’s default-zero metadata (§8.2). After walking all `CompactTx`s the accumulated position must equal the chainMetadata size, else scanning fails with `ScanError::TreeSizeMismatch{given, computed}`.

The error taxonomy routes recovery: `TreeSizeMismatch`, `PrevHashMismatch` (prevHash against the prior block’s hash), and `BlockHeightDiscontinuity` ($\text{height} \neq \text{prev} + 1$) report `is_continuity_error() = true` and trigger the truncate-and-rescan reorganisation path (§6.7); `EncodingInvalid`, `TreeSizeUnknown`, `TreeSizeInvalid`, and `TreeSizeOverflow` do not. The overflow variant is emitted by the full-block scanner, not by the compact path described above. The final tree size also drives checkpointing: the helpers `compact_tx_contains_last_sapling_outputs_in_block`, `..._orchard_actions_in_block`, and `..._ironwood_actions_in_block` compare the running position plus the current transaction’s output or action count against the final size, to decide when to write the block-boundary checkpoint into the commitment tree.

The server side of ChainMetadata. Server `lightwalletd` fills the tree sizes from the node’s `getblock verbosity=1` response fields `trees.sapling.size` and `trees.orchard.size` (`lightwalletd`, `common/common.go`, lines 163–175, 399–400). The deployed backend is `zebrad` (§9.5), whose `getblock` omits each subfield while the pool’s tree is empty — in which case Go’s JSON unmarshal leaves `Size = 0`, which is correct for an empty tree. `TreeSizeInvalid` catches a reported size smaller than the block’s own output count, not zero metadata unconditionally. Historically, `zcashd` (retired; last release v6.10.0, 2025-10-03) computed the sizes by anchor lookup (`GetSaplingAnchorAt/GetOrchardAnchorAt` on `hashFinalSaplingRoot/hashFinalOrchardRoot`) and omitted the subfield when the anchor was unavailable (`zcashd`, `src/rpc/blockchain.cpp`, lines 279–297; the `trees` field dates to `zcashd` commit `cf2b6c796`, “rpc: Add trees field to getblock RPC output”).

8.6 The scan pipeline: chain state, batching, caching

Function `scan_cached_blocks` requires a `ChainState` — the final Sapling, Orchard, and Ironwood tree frontiers as of `from_height - 1` — and asserts `from_height = chain_state.height + 1`; all three trees have depth 32. The `ChainState` comes from the server’s `GetTreeState` RPC

via `TreeState::to_chain_state()`, which hex-decodes and byte-reverses the block hash and parses empty tree fields as empty trees. Trial decryption of compact outputs is batched through `BatchRunners` with batch size 100.

The proto accessors split their failure modes: block-level convenience accessors panic on malformed server data (`CompactBlock::hash` and `prev_hash` if not 32 bytes, `height` if outside `u32`), while per-output accessors (`cmu`, `cmx`, `nf`, ephemeral key) return `CompactFormatError::InvalidLength`, `InvalidValue`; the `From` impls that *build* compact messages from full bundles take `encCiphertext[0..COMPACT_NOTE_SIZE]`. The scanner bounds `CompactTx.index` to `u16` (the expect reads “Cannot fit more than 2^{16} transactions in a block”), and `CompactBlock.time` — `uint32` Unix epoch, “if non-zero” — flows into `ScannedBlock` and thence wallet metadata.

Compact blocks are cached as serialized protos. Crate `zcash_client_sqlite` offers `BlockDb`, a `compactblocks` SQLite table decoded on read (`CompactBlock::decode`), and `FsBlockDb`, one file per block plus a `BlockMeta` row {`height`, `blockhash`, `time`, `sapling_outputs_count`, `orchard_actions_count`} that enables range queries without decoding.

ZIP 307’s own client-operation sketch — append every `cmu` to a depth-32 incremental Merkle tree, keep a per-note incremental witness, and cache roughly 100 recent tree states because “no full Zcash node will roll back the chain by more than 100 blocks” (ZIP 307, “Creating and updating note witnesses”) — describes the pre-shardtree design. The deployed replacement is the position-marked shardtree fed by `GetSubtreeRoots` (RPC added in `lightwallet-protocol` v0.3.5), whose mechanics the *Ironwood Guide* constructs in §“Resting: the note in the commitment tree”.

8.7 Server inclusion semantics and successor surface

Which transactions a compact block contains is itself divergent. ZIP 307 states that “By default, `CompactBlocks` only contain `CompactTxs` for transactions that contain Sapling spends or outputs” (ZIP 307, “Block header validation”). Server `lightwalletd` builds and caches compact representations of *all* transactions, transparent-only ones included, with `vin/vout` populated (`parser/block.go`, lines 120–125), and `GetBlock` serves them unfiltered per the v0.4.1 contract; `GetBlockRange` filters per request — with empty `poolTypes` it prunes each transaction to its Sapling and Orchard components and drops transactions left empty (`common/common.go`, `FilterTxPool/filterBlockPool`, lines 526–653) — so the default sync stream carries shielded-relevant transactions only, as Zaino’s does. The successor Zaino filters out `CompactTxs` whose compact fields are all empty, and supports per-request pool filtering via `PoolTypeFilter` and `BlockRange.poolTypes` (`zaino-proto`, `src/proto/utls.rs`, `compact_block_with_pool_types`, lines 328–380, applied on the serving paths in `zaino-state`, `chain_index.rs`) — a filter added in v0.4.0. At the pinned v0.4.1 contract an empty `poolTypes` means Sapling and Orchard; v0.5.0 extends that legacy default to include Ironwood (`service.proto`, lines 38–41). Both pinned servers therefore default to shielded-relevant transactions on the sync stream; we flag the deployed defaults against the ZIP in §8.8.

Two `lightwalletd` RPCs, `GetBlockNullifiers` and `GetBlockRangeNullifiers`, strip actions to nullifier-only, drop outputs, `vin`, and `vout`, and zero both `ChainMetadata` tree sizes (“these are not needed (we prefer to save bandwidth)”) (`frontend/service.go`, lines 212–302). Blocks so obtained cannot in general feed ordinary scanning: they omit the commitments needed by the position ladder of §8.5, stripped Orchard actions fail compact-output parsing, and zeroed sizes can conflict with preceding tree metadata. These RPCs serve spend-status refresh, not scanning. Both were declared deprecated in the `lightwallet-protocol` v0.4.0 changelog (2025-12-03); v0.4.1 designated them deprecated in the proto itself (`option deprecated = true`, `service.proto`:247, 267), in favour of `GetBlockRange` with `poolTypes` (`service.proto`, lines 244–266; `CHANGELOG.md`, v0.4.0/v0.4.1).

Ironwood. The v0.5.0 fields (`ironwoodCommitmentTreeSize = 3`, `ironwoodActions = 9`) exist only in `lightwallet-protocol` v0.5.0; `librustzcash` gates the corresponding tree accounting on `NetworkUpgrade::Nu6_3`, and the pinned `lightwalletd` revision neither parses nor serves Ironwood data (`zcash_client_backend`, `src/scanning/compact.rs`, lines 687–696;

proto/compact_formats.proto, lines 17, 74). This describes the pinned lightwalletd revision: current lightwalletd 0.5.4 parses and serves Ironwood and vendors the tagged v0.5.0 protocol. The pool is *deployed*, although ZIP 2005 remains Proposed and ZIP 258 remains Draft as a document. NU6.3 is active from Testnet height 4,134,000 and Mainnet height 3,428,143. The pinned Zaino implementation wires both fields through its serving builders (zaino-state, chain_index/types/db/legacy.rs), where they remain empty until NU6.3 data exists on chain.

8.8 Divergences between ZIP-307 and the deployed protocol

Per our conventions, we flag rather than resolve the following.

- *No complete normative specification of the deployed wire format.* ZIP 307 is Draft, Sapling-only, and its message definitions are stale — old names, old field numbers, no Orchard, no ChainMetadata, no vin/vout — while the lightwallet-protocol README disclaims being a specification and defers to ZIP 307 (zip-0307.rst, “Compact Stream Format”; lightwallet-protocol/README.md). The deployed format is specified only by tagged proto releases.
- *The grace-period pseudocode bug.* Both “[Canopy onward]” conditions in ZIP 307’s compact-decryption procedure test height < CanopyActivationHeight + ZIP212GracePeriod; read literally they reject 0x01 during the grace period and constrain nothing after it. Deployed zip212_enforcement accepts both lead bytes during the grace period and requires 0x02 after (zip-0307.rst, lines 275–276; zcash_primitives, src/transaction/components/sapling.rs, lines 27–40; research/upstream-defects.md, item 1, verified 2026-07-15 at zips 69610984).
- *Inclusion semantics.* ZIP 307 says Sapling-relevant transactions only; the deployed default stream includes Orchard data, transparent data is available on request (BlockRange.poolTypes), and GetBlock returns all pools unconditionally (§8.7).
- *A protoVersion that never varies on the wire.* Server lightwalletd never sets it; deployed Zaino serves 0, byte-identical to an unset proto3 scalar; and v0.5.0 removed it with field number and name reserved (issue zcash/lightwallet-protocol#25); librustzcash never reads it (§8.2).
- *Dead header-validation surface.* The proto declares header “should always be unset”; ZIP 307’s header and finalSaplingRoot checks are unimplementable against deployed servers, and only prevHash/height continuity is enforced client-side (§8.2, §8.5).
- *An advertised but unpopulated fee field.* The proto says “present if server can provide”; lightwalletd has a TODO, Zaino hardcodes zero (§8.2).
- *Vendored-revision skew at the pinned revisions.* lightwalletd vendors v0.4.1, librustzcash main v0.5.0, and Zaino an intermediate snapshot with both the Ironwood fields and protoVersion; Zaino’s vendored changelog dates v0.5.0 as 2026-07-03 without the protoVersion removal, while the canonical changelog dates it 2026-06-30 with the removal — so Zaino’s vendored proto still *declares* the field the canonical proto now reserves, though it serves only 0 on the wire (diff of the three compact_formats.proto copies; zaino-proto/lightwallet-protocol/CHANGELOG.md, lines 40–51).

9 The light-client service

A light wallet holds keys and no chain. The *Ironwood Guide* (§“Delivery: the note reaches its owner”) establishes what synchronisation consumes: every compact shielded output on the chain, for trial decryption; every nullifier, for spentness; the note commitment tree’s frontier and subtree roots, for

witness assembly; and, for the few transactions that prove to be the wallet’s own, the full serialised bytes. The deployed layer that delivers these is a single gRPC service, `cash.z.wallet.sdk.rpc.CompactTxStreamer`, spoken between the wallet and an untrusted proxy in front of a full node. This section documents the service as deployed: the authority trail from ZIP 307 through the canonical protobuf definitions (§9.1), the compact data format (§8), the interaction model the ZIP prescribes (§9.3), the twenty RPCs and their semantics (§9.4), the server implementations — `lightwalletd`, the deployed proxy (§9.5); `Zaino`, its successor (§9.6); and `Zebra`’s newer in-process server (§9.7) — and a register of the points where the implementations and the normative texts disagree (§9.8). The linear costs this architecture accepts — trial decryption of every output, a maintained witness per note — are precisely the costs Project Tachyon’s proposed proof-carrying-wallet design targets (Bowe–Miers, ePrint 2025/2031; `tachyon-zcash/tachyon`, `book/src/proof-tree.md` @ 26fdc165).

9.1 Architecture and authority

ZIP 307, “Light Client Protocol for Payment Detection” (Status: Draft), fixes a three-component architecture: a Zcash full node as the root of trust; a *proxy server* that converts chain data into a compact format; and the light client (ZIP 307, § “High-Level Design”). The security model is deliberately narrow: the client obtains *payment detection privacy* against an honest-but-curious proxy, and trusts that proxy to provide a correct view of the best chain; network-level privacy (Tor, Nym) is assumed to be provided separately; spending privacy is out of scope (ZIP 307, § “Security Model”). The chain has carried commitments designed to let a client check that view succinctly since Heartwood (ZIP 221, via `hashLightClientRoot`, renamed `hashBlockCommitments` at NU5, where the field also binds a commitment to the block’s transaction signatures and proofs); no deployed client in the pinned `lightwalletd` or `zcash_client_backend` sources verifies those commitments (ZIP 221; ZIP 244; negative source check at the pinned revisions).

Where the contract lives. The wire contract is *not* in the ZIP. The two protobuf files that define the service — `service.proto` and `compact_formats.proto` — have their canonical home in the `zcash/lightwallet-protocol` repository, which self-describes as the canonical Protobuf definitions but explicitly not a specification: “The Zcash Light Client Protocol is described in ZIP-307” (`lightwallet-protocol`, `README.md`). The authority trail is therefore circular — the ZIP defers detail to the deployed API, and the API repository defers normative force back to the ZIP — and the circle does not close: ZIP 307’s own protobuf sketch (`CompactSpend` and `CompactOutput` messages, a `BlockID` embedded in `CompactBlock`, a `RangeFilter`, a `FullTransaction` return type, a four-RPC `CompactTxStreamer`) differs from the deployed proto (`CompactSaplingSpend`, `CompactSaplingOutput`, `CompactOrchardAction`, flat height and hash fields, `BlockRange`, `RawTransaction`, twenty RPCs), and the ZIP warns that “the details of the API described below may differ from the implementation” (ZIP 307, § “Compact Stream Format”, § “Proxy operation”). We treat the canonical proto files, versioned and change-logged, as the deployed contract, and cite the ZIP for architecture and rationale. Both `lightwalletd` and `Zaino` vendor the proto files by git subtree (`lightwallet-protocol`, `README.md`).

Contract version. The canonical protocol is at v0.5.0 (2026-06-30). Version 0.4.0 (2025-12-03) added transparent data to the compact format — `PoolType`, `CompactTxIn`, `TxOut`, `CompactTx.vin/vout`, `BlockRange.poolTypes` — and the `LightdInfo` fields `upgradeName`, `upgradeHeight`, and `lightwalletProtocolVersion`, and declared `GetBlockNullifiers` and `GetBlockRangeNullifiers` deprecated; v0.4.1 designated the pair deprecated in the proto itself (option `deprecated = true`, `service.proto:247,267`) and documented `GetBlock`’s transparent-data behaviour. Version 0.5.0 added the Ironwood fields and removed and reserved `CompactBlock.protoVersion` (`lightwallet-protocol`, `CHANGELOG.md`).

The proto forks at the pinned revisions. The following is the July snapshot preserved for the behaviour audit. The canonical v0.4.1 has `PoolType = {INVALID, TRANSPARENT, SAPLING, ORCHARD}` and `ShieldedProtocol = {sapling, orchard}`. The fork in `librustzcash (zcash_client_backend/proto/service.proto and compact_formats.proto)` adds `IRONWOOD = 4`, `TreeState.ironwoodTree = 7`, `ShieldedProtocol.ironwood = 2`, `ChainMetadata.ironwoodCommitmentTreeSize = 3`, and `CompactTx.ironwoodActions = 9`. Zaino’s fork (`zaino-proto`) adds `IRONWOOD = 4`, `ironwoodTree = 7`, `ChainMetadata.ironwoodCommitmentTreeSize = 3`, and `CompactTx.ironwoodActions = 9`, but not `ShieldedProtocol.ironwood`, and omits the deprecated options on the two nullifier RPCs (diff of the three proto trees; `zaino, docs/internal_spec.md`, § “Zaino-Proto”). Zaino’s documentation states the local copies exist “for development purposes” with a plan to rely on `librustzcash`’s. The “as deployed” wire surface therefore differs by repository; this volume cites the canonical file and flags fork-only fields where they appear.

Remark 9.1 (Sources and pinning). Deployed-behaviour claims in this section were verified against `lightwalletd` master `61fee32e` (2026-06-25), `zaino 0e057e22` (2026-07-04), and `librustzcash e30517e4` (2026-07-10); Zaino pins `zebra-chain 11.0.0` from `crates.io` (`Cargo.lock`). File-and-line citations below refer to these revisions.

9.2 The compact format

The proxy’s one transformation is compression: it strips from each transaction everything a detecting wallet does not need. Proofs, signatures, and value commitments verify the chain, and the client has delegated chain verification to the proxy; what remains is exactly the trial-decryption input and the nullifier.

Definition 9.2 (Compact Sapling output, ZIP 307). A compact output retains three fields of a Sapling Output description:

```
CompactSaplingOutput = (cmu[32], ephemeralKey[32],
                        ciphertext = encCiphertext[0..52]),
```

116 bytes per output against ZIP 307’s comparison baseline of a 580-byte ciphertext plus its 32-byte ephemeral key — an ~80% reduction (ZIP 307, § “Output Compression”; `compact_formats.proto:114--122`). The 52-byte ciphertext prefix covers the compact note plaintext — the contents and opening of the note commitment. Truncation discards the AEAD tag; integrity of the decrypted prefix is instead checked by recomputing the note commitment, as the *Ironwood Guide* constructs in § “Delivery: the note reaches its owner”.

Definition 9.3 (Compact Orchard action, deployed proto). The deployed format extends the same compression to Orchard, one message per Action:

```
CompactOrchardAction = (nullifier[32], cmx[32], ephemeralKey[32], ciphertext[52])
```

(`compact_formats.proto:124--130`). The Orchard message carries its nullifier inline because an Action both spends and creates; on the Sapling side, spend compression is a separate message containing the 32-byte nullifier. ZIP 307’s ~90% figure compares this with the pre-v5 384-byte Spend encoding, not the marginal v5/v6 spend size (ZIP 307, § “Spend Compression”).

Transparent data. Since v0.4.0 the compact format can carry transparent inputs and outputs (`CompactTxIn`, `TxOut`, `CompactTx.vin/vout`); v0.4.1 fixes the convention that the coinbase

transaction’s single null-outpoint vin is omitted, and that clients identify the coinbase by testing `CompactTx.index = 0` (`service.proto:225--237`; `compact_formats.proto:70--76`).

Remark 9.4 (The fee field is dead on the wire). The proto documents `CompactTx.fee` as “present if server can provide”, with the stateless-server formula for the case of no transparent inputs,

$$\text{fee} = \text{valueBalanceSapling} + \text{valueBalanceOrchard} + \sum v_{\text{PubNew}} - \sum v_{\text{PubOld}} - \sum t_{\text{Out}}$$

(`compact_formats.proto:59--64`); with transparent inputs a stateless server cannot compute the fee at all. Neither deployed server ever sets the field: `lightwalletd`’s parser leaves it at 0 with a “TODO: calculate fees” comment (`lightwalletd, parser/transaction.go:397--432`), and Zaino’s conversions pass `fee = None`, encoded 0 (`zaino, zaino-state/src/chain_index/types/db/legacy.rs:1397--1470`). The deployed contract is therefore that the field carries no information, and a client must treat it as absent; we flag the proto’s “present if” language as unimplemented on both servers.

9.3 The interaction model of ZIP 307

The ZIP prescribes a four-phase client-server interaction (ZIP 307, § “Client-server interaction”):

- (A) *Startup*. The client fetches the latest `BlockID`. A wallet importing a pre-existing seed with no recorded birthday starts at Sapling activation, since no shielded address it can derive predates Sapling (ZIP 307, § “Importing a pre-existing seed”).
- (B) *First sync*. The client requests `GetBlockRange(X, Y)` and, on any subsequent request, verifies the hash of the overlap block — the last block it already holds — against its cached copy, so that a reorganisation is detected as a hash mismatch.
- (C) *Transaction fetching*. Detected notes are completed by fetching full transactions via `GetTransaction`. Each such fetch names a txid to the proxy, so the ZIP mandates cover traffic: clients SHOULD download decoy transactions, or all transactions of a touched block, and MUST warn users of the privacy loss otherwise (ZIP 307, § “Transaction privacy”).
- (D) *Incremental sync*. Later syncs repeat (B) from the cached tip, re-checking the overlap block.

Two hardening measures described by the ZIP are not deployed. Block-header validation by the client is described as only partially implemented — only `prevHash` linkage is checked (ZIP 307, § “Block header validation”). The *bucketing* privacy enhancement — for bucket size N , start each range request at

$$\text{floor}(X) = X - (X \bmod N)$$

instead of at the true resume height X , hash-checking and discarding blocks in $[\text{floor}(X), X]$, so that the request leaks only the bucket and absorbs reorganisations — is proposed and not implemented (ZIP 307, § “Block privacy via bucketing”). Both are ZIP-versus-deployed gaps a reader should not assume closed.

The reference client narrows the surface further: the sync loop in `zcash_client_backend` exercises four shielded-sync RPCs (below) plus, when built with feature `transparent-inputs`, `GetAddressUtxosStream`, issued per account per cycle by `refresh_utxos` (cf. §12.3):

- `get_latest_block`, in `update_chain_tip`;
- `get_subtree_roots`, in `download_subtree_roots`;
- `get_block_range`, in `download_blocks`;
- `get_tree_state`, in `download_chain_state`.

Transaction enhancement via `GetTransaction` is listed as not yet implemented in that module (`src/sync.rs`). The complete reference loop is reconstructed in §10.8.

9.4 The RPC surface

The service defines twenty RPCs (`service.proto:221--326`); Table 7 groups them. Three are deprecated, one is testing-only, and six serve the transparent layer, outside this section’s shielded-sync scope. The four marked ✓ are the ones the reference sync loop consumes for shielded sync (§9.3).

Group	RPC	Sync loop	Notes
Chain	GetLatestBlock	✓	tip height and hash
	GetBlock		one block, all pools
	GetBlockRange	✓	streamed range, pool filter
	GetBlockNullifiers		deprecated (v0.4.0; proto option v0.4.1)
	GetBlockRangeNullifiers		deprecated (v0.4.0; proto option v0.4.1)
Trees	GetTreeState	✓	frontiers at height/hash
	GetLatestTreeState		frontiers at the tip
	GetSubtreeRoots	✓	depth-16 subtree roots
Transactions	GetTransaction		full bytes by txid
	SendTransaction		submit raw bytes
Mempool	GetMempoolTx		compact mempool contents
	GetMempoolStream		full txs until next block
Transparent	GetAddressTxids		deprecated, misnamed
	GetAddressTransactions		
	GetAddressBalance		
	GetAddressBalanceStream		
	GetAddressUtxos		
	GetAddressUtxosStream		
Metadata	GetLightdInfo		server and chain metadata
	Ping		testing-only

Table 7: The twenty RPCs of `CompactTxStreamer` (`service.proto:221--326`, `lightwallet-protocol v0.4.1`). The ✓ column marks the shielded-sync RPCs exercised by the reference sync loop; with feature `transparent-inputs` the loop also issues `GetAddressUtxosStream` per account per cycle.

Remark 9.5 (Hash orders). Two byte orders cross the interface. On the usual block and transaction paths, byte-typed identifiers such as `BlockID.hash` and `CompactBlock.hash` carry raw bytes in little-endian (protocol) order; `lightwalletd` reverses the big-endian hex it receives from the node’s RPCs (`lightwalletd, frontend/service.go:69--90`). Field `TreeState.hash`, by contrast, is big-endian display hex *text* (`service.proto:176--184,307--312`). Requests follow the same split: `TxFilter` takes protocol-order txid bytes that the server reverses into display hex for `getrawtransaction` (`frontend/service.go:403--442`), and the mempool exclude entries are protocol-order byte suffixes the server reverses into display-order prefixes (`frontend/service.go:600--747`). There are two deployed byte-typed exceptions: `GetTreeState` interprets its `BlockID.hash` bytes directly as display-order hex, and `SubtreeRoot.completingBlockHash` is likewise returned in display order (Remark 11.6). String-typed hashes are display-order text.

Chain tip. The RPC `GetLatestBlock(ChainSpec)` returns a `BlockID` — height and hash of the best chain tip; `lightwalletd` proxies `getblockchaininfo` and returns the hash as little-endian raw bytes (`lightwalletd, frontend/service.go:69--90`). This is the chain tip the wallet’s confirmation accounting consumes (§6.6).

Single block. The RPC `GetBlock(BlockID)` returns one `CompactBlock`, and per the v0.4.1 contract includes transaction data for *all* value pools, transparent `vin/vout` included — unlike `GetBlockRange`, which defaults to shielded-only (`service.proto:225--237`). The proxy

lightwalletd serves the request from its block cache first and falls back to the node on a miss; a height above the tip returns gRPC `OutOfRange` (“block %d is newer than the latest block”), a lookup by hash returns `InvalidArgument` (“not yet implemented”, referencing PR #309), and height 0 with no hash is `InvalidArgument` (lightwalletd, `common/common.go:495--624`, `frontend/service.go:166--189`). Zaino accepts lookup by hash — hash takes precedence and is resolved to a height via its chain index — and builds the block with all pools included, matching the v0.4.1 contract; the proto makes hash support optional, so both behaviours are conformant, but they diverge (zaino, `zaino-state/src/backends/state.rs:1954--2058`; `zaino-proto/src/proto/utls.rs:310--323`).

Block ranges. The RPC `GetBlockRange(BlockRange)` streams consecutive `CompactBlocks` over the closed interval $[\min(s, e), \max(s, e)]$, in increasing height order iff $s \leq e$ and decreasing otherwise — lightwalletd computes $j = \text{high} - (i - \text{low})$, Zaino tests $s \leq e$ (`service.proto:250--255`). At the pinned v0.4.1 contract, an empty `poolTypes` field selects the legacy behaviour: prune each block to Sapling and Orchard data only; v0.5.0 and current lightwalletd add Ironwood to that default. Clients MUST verify server capability before setting `poolTypes` non-empty (`service.proto:28--42`). Filtering prunes transaction components and then drops empty transactions, but a block whose transactions were all filtered is still returned, with empty `vtx` (lightwalletd, `common/common.go:574--624`). Zaino validates the request up front — start and end must be present and fit in u32; `POOL_TYPE_INVALID` or unknown pool values are `InvalidArgument` — streams through an mpsc channel of the configured `channel_size`, caps an ascending range at the tip and appends a trailing `OutOfRange` error after the valid blocks, and bounds production by a hard timeout of 4× the service timeout (zaino, `zaino-state/src/backends/state.rs:550--683`; `zaino-proto/src/proto/utls.rs:53--152`).

Nullifier variants. The RPCs `GetBlockNullifiers` and `GetBlockRangeNullifiers` — both deprecated (v0.4.0 changelog; proto option v0.4.1) in favour of `GetBlockRange` with `poolTypes` — return `CompactBlocks` reduced to shielded nullifier data: lightwalletd strips Sapling outputs and transparent vin/vout, reduces each Orchard action to its nullifier, and zeroes both commitment-tree sizes; `GetBlockRangeNullifiers` MUST ignore any `TRANSPARENT` member of `poolTypes`, and lightwalletd filters it out (`service.proto:239--268`; lightwalletd, `frontend/service.go:191--225, 259--305`). Zaino’s `FetchService` backend additionally resolves `GetBlockNullifiers` by hash, as for `GetBlock` (zaino, `zaino-state/src/backends/fetch.rs:929--1041`); its `StateService` backend ignores a supplied hash and reads only the height, silently serving height 0 — genesis — for a hash-identified request (`backends/state.rs:2033--2058`; `research/upstream-defects.md`, item 40).

Tree state. The RPC `GetTreeState(BlockID)` returns the note commitment tree state for a block specified by height or hash: network name, height, block hash as a hex string, block time, and the saplingTree and orchardTree frontiers as hex (`service.proto:176--184`). In lightwalletd the reply derives from the node’s `z_gettreestate` RPC, looping along `Sapling.SkipHash` when a reply carries no `FinalState`; `GetLatestTreeState` calls the same path at `getblockchaininfo`’s tip height (lightwalletd, `frontend/service.go:307--401`). The frontier is what seeds the wallet’s shard tree below its birthday, so that witness assembly never replays the chain’s prefix (*Ironwood Guide*, §“Resting: the note in the commitment tree”). Zaino resolves hash-or-height, reports the BIP70 network name, and returns a seventh field `ironwoodTree` (hex, empty when absent) that exists only in Zaino’s and librustzcash’s proto forks (§9.1); `GetLatestTreeState` reuses `GetTreeState` at the current chain height (zaino, `zaino-state/src/backends/state.rs:2516--2576`; `zaino-proto/proto/service.proto:183`). One stale cross-reference: the proto comment for `GetTreeState` cites “section 3.7 of the Zcash protocol specification”, while the current `protocol.tex` numbers note commitment trees as § 3.8.

Subtree roots. Shielded outputs of a pool are numbered from genesis, and each consecutive group of $2^{16} = 65536$ note commitments forms one depth-16 subtree of the pool’s commitment tree; the stream element is

$$\text{SubtreeRoot}_i = (\text{Merkle root of subtree } i, \\ \text{hash and height of the block containing output } (i + 1) \cdot 2^{16} - 1),$$

so that, for example, Sapling output 65535 — the last of subtree 0 — falls in block 558822 (service.proto:191--200; lightwalletd, common/common.go:178--219). The RPC `GetSubtreeRoots(GetSubtreeRootsArg)` streams these for the chosen shielded protocol from `startIndex`, with `maxEntries = 0` meaning all. The lightwalletd handler calls `z_getsubtreesbyindex` and then, per subtree, fetches the completing block via `GetBlock` at its end height to obtain the hash, reversed to big-endian display-order bytes, the opposite convention from `CompactBlock.hash` (Remark 11.6; lightwalletd, frontend/service.go:832--907). These roots are exactly the retained internal nodes from which the wallet’s shard tree assembles authentication paths without replaying every commitment (*Ironwood Guide*, §“Resting: the note in the commitment tree”). Zaino converts `startIndex` and `maxEntries` to `u16` and returns `InvalidArgument` when either exceeds `u16` range, where lightwalletd forwards the full `uint32` — bounded by design, since a depth-32 tree has at most 2^{16} depth-16 subtrees, but the wire contracts differ; it fills the completing block data via `z_get_block` per subtree, under a 4× service-timeout deadline. One Zebra-specific caveat: during Zebra’s initial subtree-index rebuild the underlying RPC returns an empty list for not-yet-rebuilt indices, where zcashd rebuilds before serving (zaino, zaino-state/src/indexer.rs:429--452, 754--912).

Full transactions. The RPC `GetTransaction(TxFilter)` returns the full serialised transaction. Only lookup by 32-byte txid is supported; a block-plus-index lookup returns `InvalidArgument` (“specify a txid, not a blockhash+num”). The lightwalletd handler converts the little-endian txid bytes to big-endian hex, calls `getrawtransaction` with `verbose=1`, and maps RPC failure to gRPC `NotFound` (service.proto:44--51, 270--271; lightwalletd, frontend/service.go:403--442). The privacy cost of this call — it names a txid of interest to the proxy — is the reason for the ZIP’s decoy mandate (§9.3).

Submission. The RPC `SendTransaction(RawTransaction)` submits via the node’s `sendrawtransaction`. Success returns `SendResponse{errorCode=0, errorMessage=txid hex}`; a node rejection is parsed from the “code: message” error string into a non-zero `errorCode` plus message — node-level rejection travels in-band, not as a gRPC error, so a transport-level success is not an acceptance (service.proto:83--89; lightwalletd, frontend/service.go:454--509). Wallet-side obligations around this call — store-then-broadcast ordering, rebroadcast, expiry — are covered in §6.2.

Remark 9.6 (The `RawTransaction.height` sentinels). An original protobuf-definition error typed the field `uint64` rather than `int64`, forcing sentinel semantics (service.proto:53--81; lightwalletd, common/common.go:626--661):

$$\text{height} = \begin{cases} 0 & \text{transaction in the mempool,} \\ 2^{64} - 1 \text{ (0xffffffffffffffff)} & \text{mined on a non-main-chain fork (zcashd’s } -1 \text{),} \\ h & \text{mined on the main chain at height } h. \end{cases}$$

The proto’s own comments contradict each other: the message-level comment says the field carries “the latest block height” when returned by `GetMempoolStream`, the field-level comment specifies the sentinel mapping above, and a FIXME cites `librustzcash` issue #1484. The deployed servers realise both readings — lightwalletd streams 0, Zaino streams the current best-tip height (§9.8). We flag the mempool-height contract as unresolved (`research/upstream-defects.md`, which tracks this as the fourth known defect, already filed as `zcash/librustzcash#1484`).

Mempool contents. The RPC `GetMempoolTx(GetMempoolTxRequest)` streams `CompactTx` for the current mempool contents, possibly a few seconds stale. The request may carry `exclude_txid_suffixes`: txid byte-suffixes of any length up to 32 bytes, in client-side protocol order; the server reverses and hex-encodes each into a display-order prefix, and a mempool transaction is excluded iff its txid matches a prefix that matches *exactly one* mempool txid — when two or more txids match, none of the matching transactions are excluded, and unmatched entries are ignored (service.proto:157--174,289--301; lightwalletd, frontend/service.go:600--747). Field 2 of the request is reserved for a future `exclude_txid_prefixes`. At the pinned lightwalletd revision, an empty `poolTypes` defaults to Sapling-plus-Orchard pruning; current lightwalletd also includes Ironwood. `POOL_TYPE_INVALID` is rejected. The lightwalletd implementation backs the RPC with a process-global cache — `mempoolMap` from txid to `CompactTx`, plus `mempoolList` — refreshed at most every 2 s from `getrawmempool` plus per-transaction `getrawtransaction verbose=0`, reusing already-fetched entries across refreshes; mempool `CompactTx`s are built with height 0 and no fee (lightwalletd, frontend/service.go:589--708). At the pinned Zaino revision, both backends enforce the same 32-byte suffix bound and exactly-one-match exclusion rule. `StateService` validates and applies `poolTypes`; `FetchService` ignores that field and converts with its default filter, which includes Sapling, Orchard, and Ironwood. Thus requested pruning is backend-dependent (zaino, zaino-state/src/backends/state.rs:2302--2429, backends/fetch.rs:1630--1741).

Mempool stream. The RPC `GetMempoolStream(Empty)` streams full `RawTransactions` of the current mempool and keeps the stream open, closing it when a new block is mined (service.proto:303--305). The lightwalletd implementation is process-global state — `g_txList` in arrival order, a `g_txidSeen` set, `g_lastBlockChainInfo` — polled at most every 2 s via `getBlockchaininfo` and `getrawmempool`; the first stream to observe a tip-hash change resets the shared state and all open streams terminate; each per-client loop sleeps 200 ms between sends; transactions mined since listing (height $\neq$ 0) are skipped; streamed transactions carry height = 0 (lightwalletd, common/mempool.go:14--152, frontend/service.go:581--587). Zaino streams height = current best-tip height, closes the stream when its mempool status flips to `Syncing` on a tip change, and additionally imposes a hard 6× service-timeout deadline — 180 s at defaults — where lightwalletd keeps the stream open until the next block regardless of duration; a long block interval can therefore end a Zaino stream with a deadline error where the proto’s documented close-on-new-block semantics would hold (zaino, zaino-state/src/backends/state.rs:2433--2510, chain_index/mempool.rs:419--480).

Server metadata. The RPC `GetLightdInfo(Empty)` returns server metadata plus chain state assembled from `getinfo` and `getBlockchaininfo`: version, vendor (“ECC LightWalletD”), `taddrSupport` true, chain name, the Sapling activation height — looked up via the Sapling consensus branch ID 0x76b809bb in the upgrades map — the chain tip’s consensus branch ID, block height, build and git fields, `estimatedHeight` (less than the tip while the node itself syncs), donation address, and the next pending upgrade’s name and height (service.proto:97--118; lightwalletd, common/common.go:261--319). At the pinned revision the proxy does *not* populate the proto’s `lightwalletProtocolVersion` field 18; current lightwalletd 0.5.4 reports “v0.5.0”. Zaino reports vendor “ZingoLabs ZainoD”, hard-codes `lightwallet_protocol_version` = “v0.5.0” — a version string whose fields did not match the canonical tagged protocol exactly (research/upstream-defects.md, item 3; zaino-state/src/backends/fetch.rs:2097, state.rs:2776) — defaults the Sapling activation height to 1 on regtest, and carries the connected Zebra node’s build information in the `zcashdBuild/zcashdSubversion` fields (zaino, zaino-state/src/backends/state.rs:2724--2793). Both sides are flagged in §9.8.

Ping. The RPC `Ping` is testing-only; lightwalletd disables it unless started with the flag `--ping-very-insecure`, because the call lets a client create arbitrarily many concurrent server

threads — a resource-exhaustion risk (service.proto:324--325; lightwalletd, frontend/service.go:920--936). Zaino returns gRPC Unimplemented.

Transparent-address RPCs. The six remaining RPCs (GetAddressTxids — deprecated and misnamed, it returns transactions — GetAddressTransactions, GetAddressBalance, GetAddressBalanceStream, GetAddressUtxos, GetAddressUtxosStream) proxy the node’s insight-explorer-derived RPCs getaddressesxids, getaddressbalance, and getaddressutxos; they serve the transparent layer and fall outside this section’s shielded-sync scope. For these, lightwalletd validates t-addresses against the regex `\At[a-zA-Z0-9]{34}\z` and caps each GetAddressTransactions inner fetch loop with a 30-second context timeout (lightwalletd, frontend/service.go:55--164, 511--579, 749--830). Zaino caps the request’s address list at `UTXO_MAX_ADDRESSES = 1000` (InvalidArgument beyond) — a server-side fan-out bound lightwalletd lacks — and honours `start_height` and `max_entries` (0 meaning unlimited) as response-size bounds (zaino, zaino-state/src/backends.rs:29--45, backends/state.rs:2591--2719).

9.5 lightwalletd: the deployed proxy

The deployed server is a Go proxy: each gRPC handler in frontend/service.go translates its request into node JSON-RPC calls through the indirect function common.RawRequest. It supports both zcashd and zebrad backends, detecting the backend from the node’s subversion string (/Zebra: versus /MagicBean:); a zcashd backend must have the lightwalletd or insightexplorer experimental feature enabled or lightwalletd exits; the default assumed node is zebrad (lightwalletd, cmd/root.go:200--263, common/common.go:27--35, 61--64).

Ingestion. A single BlockIngestor goroutine polls getbestblockhash every 2 s; when the tip differs it fetches the next block via *two* getblock calls — first verbose=1 for the txids and hash, a workaround because the built-in parser miscomputes v5 transaction IDs (issue #392), then the raw block by that hash, so a reorganisation between the calls cannot mix two blocks. A prevHash mismatch against the cache tip is treated as a reorganisation and exactly one block is dropped (Reorg(height-1)) per iteration; a getblock failure retries after 8 s (lightwalletd, common/common.go:321--493). The 120 s sleep (common/common.go:483--489) runs only when the cache is empty (height = GetFirstHeight()) and the node returned no block at that height; with the cache now starting at height 0 (cmd/root.go:296--299) the branch is reachable only against a node that cannot yet serve genesis, and its log line (“Waiting for ... Sapling activation height”) is stale wording, not stale behaviour.

The block cache. The only persistent chain-serving state lightwalletd maintains is the compact-block cache: two append-only files per chain, `<data-dir>/db/<chain>/lengths` and `.../blocks`. With $d = \text{protobuf}(\text{CompactBlock})$ the entry for height h is

$$\text{FNV1a}_{64}(\text{LE}_{64}(h) \parallel d) \parallel d \quad \text{in blocks}, \quad \text{LE}_{32}(|d|) \quad \text{in lengths},$$

with the offset table reconstructed on startup and valid lengths constrained to $74 \leq |d| \leq 4,000,000$ bytes; corruption — a bad checksum, an impossible length, a partial entry — triggers automatic cache clearing and redownload (lightwalletd, common/cache.go:21--230). The cache now starts at height 0, previously Sapling activation, because compact blocks carry transparent data (lightwalletd, cmd/root.go:265--300). All other responses are proxied per-request; the mempool caches of §9.4 are in-memory only; no per-client state is kept.

Operational defaults. gRPC binds 127.0.0.1:9067 and HTTP 127.0.0.1:9068 (Prometheus /metrics); TLS is required — the server exits if the certificate or key is unloadable — unless

`--no-tls-very-insecure` (plaintext) or `--gen-cert-very-insecure` (self-signed); the data directory defaults to `/var/lib/lightwalletd`; `--nocache` disables the disk cache; `--sync-from-height` and `--redownload` control cache rebuild; the darkside mock mode (`--darkside-very-insecure`) auto-terminates after 30 minutes (`lightwalletd, cmd/root.go:139--180,375--424,498--499`).

9.6 Zaino: the successor indexer

Zaino is the successor indexer, in Rust: `zainod` is the daemon, `zaino-serve` implements `CompactTxStreamer` over tonic, `zaino-state` holds the chain index and backends, `zaino-fetch` is the JSON-RPC client, and `zaino-proto` vendors the protos. The stated motivations are the `zcashd` deprecation, a Rust stack alongside `librustzcash` and Zebra, and a clean indexer/validator trust boundary: indexing functionality moves out of the validator into Zaino, with Zebra exposing its `ReadStateService` for direct read access (`zaino, README.md, § “Motivations”, § “Project Structure”; docs/internal_spec.md`).

Backends. Configuration `backend = 'fetch' | 'state'` selects one of two chain-fetch backends. Backend `StateService` is backed by Zebra’s `ReadStateService` plus a `TrustedChainSync` (`init_read_state_with_syncer`): it reads Zebra’s state store directly, and at startup blocks until the syncer’s tip height *and* hash match the validator’s JSON-RPC tip — hash compared because height alone is ambiguous during a reorganisation. Backend `FetchService` is backed by the node’s JSON-RPC interface (`zcashd` back-compatibility, or a remote Zebra) and is marked “`#[deprecated]: Will be eventually replaced by BlockchainSource`”. Both feed the same `NodeBackedChainIndex` (`zaino, zaino-state/src/backends/state.rs:1,117--260, backends/fetch.rs:1,83--100`). The shipped example configuration nonetheless selects `backend = 'fetch'` — the deprecated backend — and the repository documentation does not say which backend is the recommended production deployment (`zaino, docs/example_configs/zainod.toml`); we flag the question as open.

The chain index. Server-side state is the `ChainIndex`, with three components (`zaino, docs/internal_spec.md, § “Zaino-State”`). A `Mempool` — a dashmap broadcast map keyed by txid, whose sync task polls `get_best_block_hash` and on a tip change clears the map and notifies subscribers. A `NonFinalisedState` — an in-memory `ArcSwap` snapshot of the top blocks of all chains. A `FinalisedState` — all blocks below the seam — served either by a versioned LMDB-backed persistent database (`v1`) or, with `ephemeral_finalised_state`, by a stateless passthrough to the validator; large syncs and version migrations run in the background while serving continues. The seam is

$$\text{floor}(t) = t - D_{\text{op}} \quad (\text{saturating at genesis}),$$

where t is the tip height and $D_{\text{op}} = \text{OPERATIONAL_NFS_DEPTH} = \text{MAX_BLOCK_REORG_HEIGHT} + 1 = 1001$ — Zebra’s reorganisation limit is local node policy, not consensus; the finalised database serves heights $\leq \text{floor}(t)$, the in-memory state serves heights above it, and in-memory retention is hard-capped at $\text{MAX_NFS_DEPTH} = D_{\text{op}} + 10 = 1011$ blocks below the tip.

Serving. Each gRPC handler delegates to a `LightWalletIndexer` trait implemented by both backend subscribers; streaming responses are tokio mpsc channels wrapped in typed streams. The gRPC server may bind a public address only with TLS configured — rejected at startup otherwise — and the default listen address is `127.0.0.1:8137`; Zaino additionally serves a `zcashd`-style JSON-RPC server, unencrypted and restricted to loopback or private bind addresses by default (`zaino, README.md, § “Server network exposure”`). Service defaults: timeout 30 s, `channel_size` 32; stream deadlines are 4× the timeout (120 s), and 6× for `GetMempoolStream` (180 s). Until the non-finalised snapshot exists, `GetLatestBlock` and `GetMempoolStream` return an

UnavailableNotSyncedEnough error where lightwalletd would proxy the node’s answer immediately; a code TODO says this “probably shouldn’t be an error”, so the startup behaviour may change (zaino, zaino-state/src/backends/state.rs:1940--1951).

Compatibility. Zaino’s integration test suite includes a `client_rpcs` partition that compares its LightWallet gRPC outputs against lightwalletd’s, and the project has committed to serving all Zcash JSON-RPC services required by non-validator clients, superseding zcashd in that role (zaino, docs/internal_spec.md, § “Zaino-Testutils and Integration-Tests”; docs/rpc_api.md).

Current-head update. The detailed account above deliberately describes the pinned July revision. At zaino 78eec726 (2026-09-02; after the latest tag, 0.9.0 of 2026-08-29), validator support is Zebra-only and the old StateService/FetchService split has been replaced by one ZebraValidator composite. JSON-RPC is always present; direct ZebraReadStateAdapter database access is an optional accelerator. Configuration now names the modes `direct` and `rpc`, while retaining `state` and `fetch` only as compatibility aliases (zaino-source-zebra/src/lib.rs; zainod/src/config.rs). Consequently, the behaviour-by-backend claims and divergence table below are a reproducible snapshot, not a claim about current Zaino internals.

9.7 Zebra’s in-process server

Zebra main d1fd7ad (2026-09-04, v6.3.0-188-gd1fd7ad) contains a third implementation, added at 85ce9a3 on 2026-08-18 after the zebra 6.3.0 release. It runs CompactTxStreamer inside zebra: most handlers call Zebra’s JSON-RPC methods in process, while compact blocks and mempool streams read the state and mempool services directly. The surface defines handlers for all twenty RPC names; nineteen are implemented, while the testing-only Ping deliberately returns Unimplemented (zebra-rpc, src/lightwalletd/methods.rs and server.rs).

This server remains unreleased at the cited head, experimental, and disabled unless `rpc.lightwalletd_listen_addr` is configured. It serves plaintext HTTP/2 without TLS or authentication; Zebra therefore directs operators to bind it to loopback or a private address and terminate TLS in a reverse proxy (zebra-rpc, src/config/rpc.rs). Its existence invalidates the former statement that lightwalletd and Zaino were the only maintained server implementations, but it does not retroactively alter the pinned two-server comparison below.

9.8 Divergence register

Table 8 collects the contract-level points at which lightwalletd and Zaino at the pinned revisions differ from each other or from the normative texts; each row is developed, with citations, in the paragraph of §9.4–§9.6 that covers the RPC. The ZIP-versus-deployment gaps — the message sketch, partial header validation, unimplemented bucketing — are flagged in §9.1 and §9.3; the proto forks in §9.1.

Two rows deserve emphasis. The mempool-height row is a live specification bug: the proto’s two comments cannot both be satisfied, the deployed servers realise one reading each, and the FIXME in the canonical file (librustzcash issue #1484) records the conflict without resolving it; a client must therefore treat the height of a streamed mempool transaction as carrying no portable information beyond “not yet mined”. The fee row is a silent one: nothing on the wire distinguishes “fee is zero” from “fee not provided”, so the proto’s conditional language is, in deployed terms, a constant.

Contract point	lightwalletd	Zaino
GetBlock / GetBlockNullifiers by hash (optional in proto)	hash lookup is InvalidArgument, “not yet implemented” (PR #309)	GetBlock supports it; pinned StateService ignores the hash in GetBlockNullifiers
Stream deadlines (proto: GetMempoolStream closes on new block)	none; open until next block	hard 4×/6× timeout (120 s/180 s at defaults)
RawTransaction.height in 0 GetMempoolStream (proto comments contradict; issue #1484)		current best-tip height
GetSubtreeRoots argument range (proto: uint32)	forwards full range to node	InvalidArgument above u16::MAX
CompactTx.fee (proto: “present if server can provide”)	always 0 (parser TODO)	always 0 (fee = None)
lightwalletProtocolVersion (LightdInfo field 18, v0.4.0)	never populated at 61fee32	hard-coded “v0.5.0” despite the vendored-field mismatch
Behaviour before server is synced	proxies the node immediately	UnavailableNotSyncedEnough on GetLatestBlock, GetMempoolStream
Ping	disabled unless --ping-very-insecure	Unimplemented
Transparent-address fan-out	unbounded address lists	capped at 1000 addresses

Table 8: Pinned-revision divergences across the CompactTxStreamer contract. Citations for each row appear in the covering paragraphs of §9.4–§9.6.

10 The scanning pipeline

A light wallet holds no chain. Everything it knows — which notes it owns, which of them are spent, and where each sits in the note commitment tree — it reconstructs by scanning a stream of *compact blocks* supplied by an indexing server. This section develops the deployed reconstruction pipeline of `librustzcash`’s `zcash_client_backend` and `zcash_client_sqlite` crates: the compact input format (§8), the account birthday that bounds the scan (§10.2), the priority queue that orders it (§10.3), chain-tip tracking (§10.4), the per-range scanning machinery (§10.5), persistence (§10.6), reorganisation recovery (§10.7), and the reference loop that assembles the parts (§10.8).

The cost shape of the pipeline deserves stating first. For every block in a suggested range, every compact output and action of every transaction is trial-decrypted against every prepared incoming viewing key of every tracked account — two ZIP-32 scopes per pool per account (§10.5) — so synchronisation work scales with global chain traffic multiplied by tracked keys. This trial-decryption component is independent of the wallet’s own activity; other synchronisation work, such as note enhancement and retained-witness updates, is not. Restore-from-seed replays the trial-decryption computation from the birthday. This is precisely the deployed cost centre that Project Tachyon’s proposed proof-carrying wallet is designed to remove (Bowe–Miers, ePrint 2025/2031; `tachyon-zcash/tachyon`, `book/src/proof-tree.md` @ 26fdc165).

10.1 The compact stream

ZIP 307 (Status: Draft) defines the light-client protocol for payment detection. Its *compact output* retains, of a full shielded output, the 32-byte note commitment, the 32-byte ephemeral key, and the first 52 bytes of the note ciphertext:

$$\underbrace{32}_{\text{commitment}} + \underbrace{32}_{\text{epk}} + \underbrace{52}_{\text{ciphertext prefix}} = 116 \text{ bytes,}$$

an ~80% reduction against ZIP 307’s 580-byte-ciphertext plus 32-byte-ephemeral-key baseline; its *compact spend* is the 32-byte nullifier alone. The quoted ~90% spend reduction uses the pre-v5 384-byte Sapling Spend encoding; v5/v6 share anchor and bundle data, so that number is not the marginal size of every current spend (ZIP 307 §§“Output Compression”, “Spend Compression”). Truncating at 52 bytes discards the AEAD tag, so a compact decryption authenticates differently from a full one: the wallet recomputes the note commitment from the decrypted compact plaintext and compares it with the transmitted commitment. This authenticates the plaintext against that commitment, not the chain: the light client still trusts its data source to supply the correct accepted transaction (ZIP 307 §“Compact Stream Format”). The cryptography of trial decryption — key agreement, KDF, the 52-byte compact note plaintext, the commitment recomputation — is the *Ironwood Guide*’s material (§“Delivery: the note reaches its owner”); this volume treats it as a black box mapping a (compact output, incoming viewing key) pair to a note or $\perp$.

The deployed wire format. The format in production has evolved well beyond ZIP 307’s message definitions. A block is

```
CompactBlock { reserved 1; height=2; hash=3; prevHash=4; time=5;
               header=6; vtx=7; chainMetadata=8 }
```

where field 1 (`protoVersion`) has been removed and `ChainMetadata` carries the running per-pool note commitment tree sizes `saplingCommitmentTreeSize=1`, `orchardCommitmentTreeSize=2`, and `ironwoodCommitmentTreeSize=3` — the field on which position tracking (§10.5) depends. Each `CompactTx` carries its index, txid, fee, Sapling spends and outputs, Orchard actions, Ironwood actions (`ironwoodActions`, field 9, reusing the `CompactOrchardAction` message), and transparent vin/vout (`zcash_client_backend`, `lightwallet-protocol/walletrpc/compact_formats.proto`, vendored at `librustzcash` e30517e4).

Remark 10.1 (Normative anchoring). ZIP 307’s message layout — a CompactBlock with an embedded BlockID, messages CompactSpend and CompactOutput — does not match the deployed format, and the ZIP remains Draft and Sapling-only. No ZIP fully specifies the deployed compact format or service: ChainMetadata tree sizes, CompactOrchardAction, ironwoodActions, GetTreeState, and GetSubtreeRoots exist only in the zcash/lightwallet-protocol proto files, which we therefore cite as the specification of record. Ironwood’s compact-format extensions are likewise absent from ZIP 2005, which covers quantum recoverability, not light-client synchronisation (compact_formats.proto; ZIP 307; ZIP 2005).

The fee field. The fee of a CompactTx is calculable by a stateless server only for transactions without transparent inputs, as

$$\text{fee} = \text{valueBalanceSapling} + \text{valueBalanceOrchard} + \text{valueBalanceIronwood} \\ + \sum v_{\text{PubNew}} - \sum v_{\text{PubOld}} - \sum t_{\text{Out}};$$

with transparent inputs the fee depends on prior-transaction outputs a stateless server cannot provide (compact_formats.proto:63--69).

Service surface. The reference synchroniser drives four RPCs of the CompactTxStreamer service: GetLatestBlock(ChainSpec) → BlockID; GetBlockRange(BlockRange) → stream CompactBlock, addressed by heights only, with an optional poolTypes pruning filter that clients MUST verify the server supports; GetTreeState(BlockID) → TreeState, returning network, height, hash, time, and the hex-serialised Sapling, Orchard, and Ironwood commitment trees; and GetSubtreeRoots(GetSubtreeRootsArg) → stream SubtreeRoot, returning complete-subtree roots with their completing block hash and height, selected by enum ShieldedProtocol { sapling=0; orchard=1; ironwood=2 } (lightwallet-protocol/walletrpc/service.proto:24--321).

Remark 10.2 (Pinned-revision divergence at NU6.3). The pinned lightwalletd tree (commit 61fee32e, 2026-06-25) contains no Ironwood support: its walletrpc protos and Go code lack ironwoodActions, ironwoodCommitmentTreeSize, ironwoodTree, and ShieldedProtocol.ironwood. The protos vendored by librustzcash (commit e30517e4) define all four; Zaino (commit 0e057e22) defines the first three but not ShieldedProtocol.ironwood (§9.1), so its GetSubtreeRoots rejects an Ironwood request with InvalidArgument (zaino-state/src/indexer.rs:761--768). The reference synchroniser unconditionally requests Ironwood subtree roots (zcash_client_backend, src/sync.rs:279--296); an NU6.3-aware wallet therefore requires an updated server — at these pins neither lightwalletd nor Zaino serves the full surface.

10.2 The account birthday

Definition 10.3 (Account birthday). Type AccountBirthday pairs a prior_chain_state: ChainState with a recover_until: Option<BlockHeight>. The *birthday height* is

$$\text{birthday} := \text{priorChainState.height} + 1,$$

the height of the first block to be scanned. A ChainState holds a block height, a block hash, and the final note commitment tree frontier of each pool: Sapling (depth sapling::NOTE_COMMITMENT_TREE_DEPTH = 32), Orchard, and Ironwood — the Ironwood tree is Orchard-shaped (MerkleHashOrchard, the same depth) but the pool is distinct and keeps its own tree.

Constructor AccountBirthday::from_treestate builds the birthday from a lightwalletd TreeState for the last block *prior* to the birthday height. The conversion (TreeState::

to_chain_state) hex-decodes the block hash — reversing the bytes, since zcashd renders block hashes byte-reversed in hex — parses the height, and converts the hex-serialised Sapling, Orchard, and Ironwood trees into per-pool frontiers; an empty tree field yields an empty tree.

Choosing a birthday. For a new wallet the crate documentation directs constructing the birthday from the treestate at tip – 100. Under the library’s 100-block reorganisation assumption, placing the birthday below that depth prevents a reorganisation from mining a transaction intended for the wallet *below* its birthday and causing funds to be missed. This is not a consensus guarantee; Zebra’s local rollback policy permits a deeper window (§10.7) (zcash_client_backend, src/data_api.rs:3199--3211). For a restored wallet the birthday is historical, and the recover_until height marks the chain tip at restoration time: the wallet is *in recovery* until no unscanned ranges remain between the birthday height and recover_until (exclusive). The field exists to avoid confusing shifts of balance and spendability while restore-from-seed is in flight — spendability policy itself is the ConfirmationsPolicy of §4 and §6, which this section does not restate. Progress reporting follows the same split: type Progress carries two scanned-notes/notes-on-chain ratios, one over the recovery window (birthday to recovery height) and one over the scan window (recovery height to tip); either denominator may be zero.

The wallet birthday. The birthday of the *wallet* is the minimum birthday over its accounts — literally SELECT MIN(birthday_height) FROM accounts in the SQLite backend — and adding an account whose birthday lies below the current chain tip triggers a rescan of all blocks at and above that height. Function add_account persists the birthday height together with the Sapling and Orchard frontier tree sizes, calls rewind_to_chain_state targeting the birthday’s prior chain state — installing a Historic-priority rescan range above birthday – 1 up to the wallet’s known chain tip while preserving higher-priority queue entries — and finally inserts an Ignored-priority range covering [saplingActivation, birthday).

10.3 The scan queue

The wallet does not scan the chain linearly; it maintains a priority queue of height ranges and always works on the most urgent.

Definition 10.4 (Scan range). A ScanRange is a half-open range [start,end) of block heights paired with a ScanPriority, with helpers truncate_start, truncate_end, and split_at. The SQLite backend stores the queue in table scan_queue as rows (block_range_start, block_range_end, priority) with both bounds UNIQUE and a CHECK constraint start < end.

Priority	Code	Meaning (doc comment)
Ignored	0	Lowest priority; never scanned.
Scanned	10	Already scanned; not re-scanned.
Historic	20	Advance the fully-scanned height.
OpenAdjacent	30	Ranges adjacent to heights at which the user opened the wallet (vestigial; Remark 10.5).
FoundNote	40	Complete tree shards adjacent to found notes.
ChainTip	50	Complete the latest shard.
Verify	60	Previously scanned range that must be re-verified as still on the main chain (highest).

Table 9: The ScanPriority ladder, ascending, with its SQLite codes.

Remark 10.5 (Six live priorities plus one vestigial). We present the ladder as six live priorities and one vestige. Priority `OpenAdjacent` is declared and mapped to SQL code 30, but no code path in the current `librustzcash` tree ever assigns it: a search at commit `e30517e4` finds only the enum declaration, the code mapping, and two doc-comment mentions (`zcash_client_backend`, `src/data_api/scanning.rs:20--21`). It still participates in the ordering, so we retain it in Table 9.

Suggesting work. Method `WalletRead::suggest_scan_ranges` returns queue entries in descending priority order. Its documented contract has two clauses: `Verify` ranges must be scanned before all others, to avoid continuity errors under reorganisation; and the compact-block source must supply note commitment tree size metadata. The SQLite implementation orders by priority `DESC`, `block_range_end DESC` — nearest the tip first within a priority — and filters to priorities at or above `Historic`.

Remark 10.6 (Contract, not type). The guarantee that a `Verify` range arrives first is a documented contract of the trait realised by the SQLite `ORDER BY`; nothing in the type system enforces it, and a third-party `WalletRead` implementation could violate it. We state it as a trait obligation and cite the one deployed implementation that discharges it (`zcash_client_backend`, `src/data_api.rs:1995--2010`; `zcash_client_sqlite`, `src/wallet/scanning.rs:61--90`).

Queue maintenance. Function `replace_queue_entries` performs every insertion as a spanning-tree merge: existing entries overlapping or adjacent to the query range are deleted and re-inserted together with the new entries, and overlaps are resolved by a dominance rule —

- an inserted `Verify` or `Scanned` range always replaces what it overlaps;
- otherwise an existing `Scanned` range wins (unless a forced rescan is requested);
- otherwise the higher priority wins, and equal priorities coalesce into one range.

Gaps between disjoint ranges are filled at `Historic` priority, so the queue always tiles a contiguous span.

10.4 Tracking the chain tip

Function `update_chain_tip` feeds the queue whenever the wallet learns a new tip height tip. Two derived heights recur, with range upper bounds exclusive throughout:

$$\text{chainEnd} := \text{tip} + 1, \quad \text{stable} := \text{tip} - \text{PRUNING_DEPTH},$$

where `PRUNING_DEPTH` = 100 blocks and heights at or below `stable` are treated by the wallet as outside its assumed reorganisation window. The SQLite implementation proceeds as follows (`zcash_client_sqlite`, `src/wallet/scanning.rs:488--660`):

- (a) it is a no-op if the tip is below `Sapling` activation;
- (b) it is a no-op if the new tip is below the prior maximum scanned height — the caller has caught the chain mid-reorganisation, and the discontinuity error path of §10.7 will resolve it;
- (c) it computes a *tip shard entry* at `ChainTip` priority covering every pool’s last incomplete shard,

$$\left[\max(\text{birthday}, \min_{\text{pools}} \max(\text{subtreeEndHeight})), \text{chainEnd} \right),$$

taking the minimum over the `Sapling`, `Orchard`, and `Ironwood` shards tables and emitting the entry only if that minimum lies below `chainEnd`;

- (d) it computes a *tip entry*: with no scanned blocks, a `Historic` range from the wallet birthday to `chainEnd` (or `Ignored` from Sapling activation when no accounts exist); with no shard meta-data (linear scanning), `Historic` from `maxScanned + 1`; if `maxScanned > stable` (steady state), `ChainTip` from `maxScanned + 1` to `chainEnd`; otherwise a `Verify` range

$$[\text{minUnscanned}, \text{min}(\text{stable} + 1, \text{minUnscanned} + \text{VERIFY_LOOKAHEAD})],$$

where `minUnscanned := maxScanned + 1`.

The `Verify` discipline is documented at the same site: because `Verify` outranks everything, the connectivity check — and any rewind it forces — runs before any `ChainTip` range from this or an earlier tip update is scanned. Constant `VERIFY_LOOKAHEAD = 10` blocks is chosen as roughly 12.5 minutes at 75 s per block, a window within which a user may plausibly have received a transaction without leaving the wallet open; and the `Verify` range is clamped to the stable region so that it cannot itself be reorganised away within the wallet’s assumed depth window and interfere with later `Verify` ranges.

Remark 10.7 (Two deliberate asymmetries). First, when `maxScanned = stable` the `Verify` range above is empty: after a long offline period no `Verify` entry need exist, and connectivity is then checked only implicitly, by the previous-hash comparison of §10.7 when the adjacent range is scanned. Second, case (b) above means `update_chain_tip` deliberately does *nothing* when the tip moves backwards; reorganisation detection lives in the scanner, not in tip updates, and recovery in that case waits for a fetch failure or continuity error.

10.5 Scanning a range

Function `scan_cached_blocks` scans one contiguous range; it takes the network parameters, the block source, the wallet database, a starting height `from_height` with its prior chain state `from_state`, and a block-count limit. It panics unless `fromHeight = fromState.height + 1` — the caller must supply the chain state immediately prior to the range.

Keys. The scanner loads all tracked unified full viewing keys and builds a `ScanningKeys` set (`ScanningKeys::from_account_ufvks`). Per account it registers: the Sapling diversifiable full viewing key’s (External, Internal) pairs of incoming viewing key and nullifier-deriving key; the Orchard full viewing key’s (External, Internal) incoming viewing keys; and the *same* Orchard keys a second time under the Ironwood note-encryption domain — Ironwood notes are Orchard-shaped and decrypt under Orchard viewing keys, but carry version-3 note plaintexts, so `IronwoodDomain` is distinct from `OrchardDomain`, which accepts only version-2 plaintexts. Each incoming viewing key is tagged (`AccountId`, `zip32::Scope`) (Ironwood’s note-encryption and recoverability changes are specified by ZIP 2005 (Proposed), while ZIP 258 activates the pool at NU6.3). A scanning key derives a note’s nullifier only when it carries nullifier-deriving capability: trait method `ScanningKeyOps::nf(note, position)` returns `None` for bare incoming viewing keys, so notes detected with an IVK alone never enter the tracked-nullifier set and their spends cannot be detected.

Batched trial decryption. The scanner makes two passes over the downloaded blocks. The first pass feeds every compact output into a set of `BatchRunners` — one per note-encryption domain: Sapling, Orchard, Ironwood — and flushes. A runner accumulates (domain, output) pairs across transactions and blocks into a `Batch` shared by all prepared incoming viewing keys of its domain; when the accumulated output count reaches the batch-size threshold (100 outputs in `scan_cached_blocks`) the batch is spawned on the global rayon thread pool (`rayon::spawn_fifo`) and a fresh batch begins. Results are delivered per (block hash, txid) over flume channels, and `collect_results` blocks until the relevant batch has run. Within a batch, `zcash_note_encryption::batch::try_compact_note_decryption` batches the cryptography:

`D::batch_epk` parses and prepares all ephemeral keys at once, shared secrets are computed for every (ivk, epk) pair — at the pinned `zcash_note_encryption` 0.4.1 revision, each scalar multiplication is still performed separately — and `D::batch_kdf` derives all symmetric keys in one pass before per-output trial decryption of the 52-byte compact ciphertext; the underlying primitives are again the *Ironwood Guide*’s (§“Delivery: the note reaches its owner”).

The second pass. The scanner then reads the prior block metadata and the wallet’s unspent-nullifier set (`Nullifiers::unspent`, per pool), and calls `scan_block_with_runners` per block, accumulating summary counts, updating the in-memory nullifier set after each block (`Nullifiers::update_with`: nullifiers spent in the block are removed, nullifiers of newly received full-viewing-key-decrypted notes are added), and chaining each block’s metadata into the next — so a spend of a note received earlier in the same batch is detected before anything is persisted.

Within a block, spend detection (`find_spent`) compares each revealed nullifier against the tracked unspent set in constant time — a subtle `CtOption` fold over the whole set; an in-code TODO notes the resulting $O(|\text{nullifiers}| \times |\text{spends}|)$ cost and questions whether constant-time operation is warranted. Nullifiers matching no tracked note are recorded as *unlinked* in a per-transaction nullifier map, in case they match a note found later when scanning an earlier range (`zcash_client_backend`, `src/scanning.rs:826--870`). Note detection (`find_received`) assigns each decrypted note its absolute tree position, $\text{position}(o) = \text{treeSize}_{\text{tx start}} + \text{idx}(o)$, and computes a shardtree retention for every commitment in the block:

$$\text{retention}(c) = \begin{cases} \text{Checkpoint}\{\text{Marked}\} & \text{last in block, wallet's} \\ \text{Checkpoint}\{\text{None}\} & \text{last in block only} \\ \text{Marked} & \text{wallet's only} \\ \text{Ephemeral} & \text{otherwise,} \end{cases}$$

and a note is flagged `is_change` when the receiving account also spent notes in the same transaction.

Position tracking. Positions require tree sizes, which the compact stream must supply. Per pool, the tracker (`PositionTracker::for_compact_block`) reconstructs the block’s starting tree size: from the prior block’s metadata when available; else from the block’s `ChainMetadata` final size minus the block’s output count (error `TreeSizeInvalid` on underflow — the proto-default zero is caught here); else 0 below the pool’s activation height (Sapling, NU5, and NU6.3 for Ironwood respectively); else `TreeSizeUnknown`. The tracker advances by each transaction’s per-pool output count, and at end of block its computed size is checked against `ChainMetadata` — a mismatch is the continuity error `TreeSizeMismatch`. Malformed input from the untrusted server is likewise caught up front: wrong-length nullifier bytes and malformed compact outputs or actions yield `ScanError::EncodingInvalid` — naming pool, txid, and index — rather than a panic.

Output. Each block yields a `ScannedBlock`: height, hash, block time, the wallet-relevant `WalletTx` list — a transaction is retained only if it has at least one wallet spend or output in any pool — and per-pool `ScannedBundles` holding the final tree size, the (commitment, retention) list, and the nullifier map. Transparent data in a `CompactTx` is not yet scanned (librustzcash issue #2187) (`zcash_client_backend`, `src/scanning/compact.rs:240--575`; `src/data_api.rs:2520--2692`). The per-call `ScanSummary` reports the scanned range and counts of spent and received Sapling and Orchard notes; received counts may include notes already spent in later blocks (a scanning-order effect), and spent counts cover only previously detected notes.

Remark 10.8 (A gap in `ScanSummary`). No Ironwood counters exist in `ScanSummary`, although `ScannedBlock` and `WalletTx` carry Ironwood spends and outputs; Ironwood activity is silently excluded from the summary. We read this as an oversight in the deployed code rather than a design decision (`zcash_client_backend`, `src/data_api/chain.rs:437--502,677--685`;

research/upstream-defects.md, item 2: “ScanSummary lacks spent/received counters for the Ironwood pool”, observed at librustzcash e30517e433).

The Ironwood pool otherwise traverses the pipeline as a full peer of Sapling and Orchard: its actions are trial-decrypted under IronwoodDomain with the account’s Orchard viewing keys, its nullifiers form a separate set — method `get_ironwood_nullifiers` is a required trait method, deliberately without a panicking default because it sits on the scan path — and its commitment tree and scan-queue shard extensions are its own.

10.6 Persistence

The scanned range is committed by a single call, `WalletWrite::put_blocks(from_state, blocks)`, which requires sequential blocks in increasing height order with `from_state` the chain state prior to the first block. The backend implementation validates, for the first block b_0 and every pool,

$\text{fromState.height}+1 = \text{height}(b_0)$ and $\text{fromState.treeSize}+|\text{commitments}(b_0)| = \text{finalTreeSize}(b_0)$,

each subsequent height being the predecessor plus one; violation is `NonSequentialBlocks`. In `zcash_client_sqlite` the whole call runs in one database transaction.

Stage 1: rows. Per block, `put_blocks` persists block metadata (heights, hash, time, per-pool final tree sizes and commitment counts); per transaction it persists transaction metadata, queues the txid for full-transaction retrieval (*enhancement*), marks spent notes, and persists received notes — checking each received note’s nullifier against the *persisted* nullifier map (`detect*_spend`) to catch notes spent in a later block range that was scanned earlier, the out-of-order counterpart of the in-memory check of §10.5. It then persists each block’s unlinked-nullifier map and prunes the map beyond `PRUNING_DEPTH`.

Stage 2: trees. Note commitments are assembled into shard subtrees in parallel — chunks of 1024 commitments, built at the pool’s shard height, which is 16 for all three pools (subtrees of 2^{16} commitments; the Ironwood tree is Orchard-shaped) — and each shard tree is updated. The same checkpoint set is enforced across all three trees: each tree gains a checkpoint at every height checkpointed in any other. Checkpoints at or above the NU6.3 activation height that fall on the `ANCHOR_RETENTION_INTERVAL = 288`-block grid are retained as durable anchors exempt from checkpoint pruning. This is the boundary of the present volume: shard-tree structure, witness derivation, and anchor selection are the *Ironwood Guide*’s material (§“Resting: the note in the commitment tree”).

Completing the found notes’ shards. After the tree update, `scan_complete` marks the scanned range `Scanned` in the queue and, if wallet notes were found, extends the queue with `FoundNote`-priority ranges covering the blocks needed to complete each found note’s subtree — the blocks that make the note witnessable. Per pool, the shard of a note at position p is

$$\text{subtreeIndex}(p) = \lfloor p/2^{16} \rfloor \quad (\text{Address}::\text{above_position}(\text{SHARD_HEIGHT}, p)),$$

and the extension spans from the previous shard’s end height — bounded below by the wallet birthday and the pool’s activation height — to the containing shard’s end height plus one. The same pass marks notes `witness_stabilized` once their shard’s block extent is fully `Scanned` and the shard’s end height is at most the pruning floor,

$$\text{floor} := \text{maxScanned} - (\text{PRUNING_DEPTH} - 1).$$

Two notions of scanned height. Out-of-order scanning leaves gaps, so the backend distinguishes `block_max_scanned` — simply `max(blocks.height)` — from `block_fully_scanned`, the height to which this and all preceding blocks above the wallet birthday have been trial-decrypted. The SQLite implementation computes the latter from the queue: it is `end - 1` of the *first* Scanned range (by ascending start), defined only when that range starts at or below the birthday.

10.7 Reorganisation detection and recovery

ZIP 307 prescribes the primitive check: on each sync, re-fetch the block at the previous sync height and compare its hash with the cached copy; a mismatch means the block was orphaned (ZIP 307 §“Client-server interaction”, Phases B and D). The deployed scanner generalises this into a per-block continuity check. Function `scan_block_with_runners` first runs `check_hash_continuity`, returning `ScanError::BlockHeightDiscontinuity` if `height(b) ≠ height(prev) + 1` and `ScanError::PrevHashMismatch` if `prevHash(b) ≠ hash(prev)`; for the first block of a `scan_cached_blocks` call the prior metadata is read from the wallet database (`block_metadata(from_height - 1)`), and for subsequent blocks it is the just-scanned block’s own metadata (`zcash_client_backend, src/scanning/compact.rs:257--289; src/data_api/chain.rs:649--693`). Of ZIP 307’s block-header validation list only this previous-hash check is implemented, and the ZIP’s “block privacy via bucketing” enhancement — rounding the sync start down to a bucket multiple — is explicitly unimplemented (ZIP 307 §§“Block header validation”, “Block privacy via bucketing”).

Classification is explicit: method `is_continuity_error` on `ScanError` returns true for `PrevHashMismatch`, `BlockHeightDiscontinuity`, and `TreeSizeMismatch`; the encoding and tree-size-availability errors of §10.5 (`EncodingInvalid`, `TreeSizeUnknown`, `TreeSizeInvalid`, `TreeSizeOverflow`) are not continuity errors.

Recovery. On a continuity error the reference loop rewinds:

```
rewind := errHeight - 10 (saturating_sub),
```

where any height at least one block before the error is valid and the 10-block offset is a heuristic. It then calls `truncate_to_height(rewind)` on the wallet database, `truncate(rewind)` on the block cache, and returns control so the caller restarts from `update_chain_tip` and a fresh `suggest_scan_ranges`. Truncation semantics — resolution to a shared shard-tree checkpoint at or below the requested height (error `RequestedRewindInvalid`, carrying a safe rewind height, otherwise), un-mining of transactions above it, and the deliberate retention of note rows — are covered in §6.7. This section adds two consequences: the scan queue is trimmed to the truncation height, rolling back the wallet’s view of the chain tip so that the next `update_chain_tip` re-installs `Verify` and `ChainTip` ranges above the new maximum scanned height; and at the pinned commit truncation covers all three shard trees and the nullifier-map locators.

The depth constant. Constant `PRUNING_DEPTH = 100` blocks is `librustzcash`’s reorganisation-safety depth (`zcash_client_sqlite, src/lib.rs:169`), the nullifier-map pruning depth (`zcash_client_backend, src/data_api/11/wallet.rs:55`), and ZIP 307’s suggested witness-cache size — the ZIP asserts 100 blocks is reasonable “as no full Zcash node will roll back the chain by more than 100 blocks” (ZIP 307 §“Client operation”). The claim is grounded in `zcashd: MAX_REORG_LENGTH = COINBASE_MATURITY - 1 = 99 (src/main.h:66)`, and a node asked to reorganise deeper aborts (`src/main.cpp:4332--4337`). Zebra’s rollback window is `MAX_BLOCK_REORG_HEIGHT = 1000`, documented as local-only node policy, not consensus (`zebra-chain/src/parameters/constants.rs:20--30`), so the ZIP’s “no full Zcash node” claim no longer holds for Zebra-backed deployments; `PRUNING_DEPTH = 100` matches the `zcashd`

bound, as its doc comment says (`zcash_client_sqlite`, `src/lib.rs:166--169`). The full spec-versus-deployment divergence register for these windows is the *Consensus Guide*’s (§“Reorg rails, maturity, and the finality floor”).

10.8 The reference sync loop

Module `sync` of `zcash_client_backend` assembles the pieces into the reference flow, whose contract the chain module documents (`src/sync.rs:58--226`, `393--467`; `src/data_api/chain.rs:1--152`):

- (1–2) `update_subtree_roots`: download and store complete subtree roots for Sapling, Orchard, and Ironwood via `GetSubtreeRoots` (cf. Remark 10.2);
- (3–4) `update_chain_tip` with the height from `GetLatestBlock`;
- (5) `suggest_scan_ranges`;
- (6) scan every `Verify` range first: download it via `GetBlockRange`, fetch the prior `ChainState` via `GetTreeState` at `start - 1`, scan, delete the cached blocks;
- (7) scan the remaining ranges, split into chunks of the caller-supplied batch size, restarting from step (3) whenever scanning materially changed the suggestions — a new first range outranking the range just scanned, or a truncation after a continuity error (§10.7).

Blocks pass through a `BlockCache` (an extension of `BlockSource`): `get_tip_height`, `read` — short reads allowed but contiguous from the range start — `insert` — non-contiguity permitted — `delete` and `truncate`. The loop deletes each range from the cache after scanning it, the module observing that “keeping the entire chain in `CompactBlock` files on disk is horrendous for the filesystem”; deletions are spawned concurrently and awaited before the loop returns.

The module documents its own limitations: block batches are not downloaded in parallel with scanning; detected transactions are not enhanced (the full transaction, hence the memo, is not fetched); there are no progress notifications; and there is no interruption mechanism short of process exit (`zcash_client_backend`, `src/sync.rs:1--10`). The chain-module documentation additionally recommends scanning `Historic` ranges in reverse height order — or from both ends inward — so that recent unspent notes are discovered sooner.

11 Commitment-tree synchronisation

Scanning tells a wallet which notes it owns (§10); spending one additionally requires the note’s authentication path against a recent anchor. The construction of that path — the depth-32 tree cut at half depth into height-16 shards under a height-16 cap, the wallet hashing its own shard from scanned leaves and taking every other shard as a single downloaded 32-byte root — is the *Ironwood Guide*’s material (§“Resting: the note in the commitment tree”, subsection “The authentication path from public data”). This section supplies what that construction presumes: the RPCs that deliver tree states and shard roots, their derivation inside the validator, and the client discipline — frontier-seeded scanning, reference-retained roots, shard-completion scan ranges — by which `zcash_client_backend` and `zcash_client_sqlite` turn a compact-block stream plus two RPCs into spendable witnesses.

11.1 The linear protocol and its limit

ZIP 307 (Status: Draft) specifies only the original, linear protocol: as compact blocks arrive in height order, the client appends every note commitment to a local depth-32 incremental Merkle tree and updates one incremental witness per unspent note; because incremental trees cannot be rewound, the ZIP advises caching roughly 100 recent tree and witness states, on the stated ground that no full

node rolls back the chain by more than 100 blocks (`zips/zip-0307.rst:290--320`; the same figure resurfaces as `PRUNING_DEPTH`, §10.7). Under this design a wallet cannot spend a note until it has scanned every block from the note's block to the anchor block — each witness update consumes every intervening commitment — so restore-from-seed implies a complete linear scan before the first spend. Removing that spend-before-sync impossibility is the purpose of the machinery below.

Remark 11.1 (Normative status). No ZIP specifies `GetTreeState`, `GetSubtreeRoots`, or the subtree-root chain index. ZIP 307's Phase A leaves tree-state acquisition an explicit unresolved TODO — “Or, it has to set `X` to the block height at which Sapling activated, so as to be sent the entire commitment tree. [TODO: Decide which to specify.]” (`zips/zip-0307.rst:433--439`) — and the repository holding the canonical protobufs disclaims normativity: “This repository IS not a specification of the Zcash Light Client protocol” (`lightwallet-protocol/README.md, v0.4.1`). As in Remark 10.1, we therefore cite the proto files and the deployed implementations as the specification of record, and classify the subtree-root protocol *deployed but unspecified*.

11.2 The tree-state service surface

Definition 11.2 (Tree-state and subtree-root RPCs). The deployed light-wallet service defines three tree RPCs (`lightwallet-protocol/walletrpc/service.proto:307--316`):

- `GetTreeState(BlockID)` returns `(TreeState)` — the tree state at a block specified by height or by hash;
- `GetLatestTreeState(Empty)` returns `(TreeState)` — the tree state at the server's chain tip;
- `GetSubtreeRoots(GetSubtreeRootsArg)` returns `(stream SubtreeRoot)` — a stream of information about roots of subtrees of the note commitment tree of the requested shielded protocol — Sapling or Orchard in the pinned v0.4.1 protocol; v0.5.0 adds Ironwood.

^a

^aThe proto comment refers the reader to “section 3.7 of the Zcash protocol specification” (`service.proto:308`). The reference is stale: at the pinned spec revision, §3.4 is “Transactions and Treestates”, §3.8 “Note Commitment Trees”, and §3.7 “Action Transfers and their Descriptions” (`protocol/protocol.tex:4048,4306,4248`), likely a pointer into an older revision's numbering. We cite the current numbering.

Definition 11.3 (`TreeState` message). Message `TreeState`, documented in the proto as “derived from the Zcash `z_gettreestate rpc`”, carries

```
network : string = 1 (documented as "main" or "test");  height : uint64 = 2;
hash : string = 3 (big-endian display-order hex);  time : uint32 = 4;
saplingTree : string = 5;  orchardTree : string = 6,
```

where the two tree fields are protobuf *strings* — hex encodings of the legacy-serialised commitment trees of Definition 11.4 — not bytes (`lightwallet-protocol/walletrpc/service.proto:176--184`). The comment gives the two production network names, not a wire-level enum: for example, `lightwalletd`'s test-only darkside mode returns “darkside”. At the pinned `librustzcash` revision the vendored proto adds `ironwoodTree : string = 7`, absent from the deployed v0.4.1 surface (Remark 10.2).

Definition 11.4 (Legacy commitment-tree encoding). The tree state of one pool at one block is serialised in the legacy `CommitmentTree` wire format,

$$\text{ser}(T) = \text{Opt}(\text{left}) \parallel \text{Opt}(\text{right}) \parallel \text{Vec}(\text{Opt}(\text{parents}_1), \text{Opt}(\text{parents}_2), \dots),$$

where `Opt(·)` writes `0x00` for an absent node and `0x01` followed by the 32-byte node otherwise, `Vec(·)` is `CompactSize`-prefixed (Bitcoin’s variable-length integer: counts below 253 in one byte, larger ones as a marker byte then 2, 4, or 8 little-endian bytes), and each node is the 32-byte encoding of a base-field element — Jubjub base for Sapling, Pallas base for Orchard — with non-canonical encodings rejected at parse time. The parser `read_commitment_tree` bounds the parents vector at `DEPTH - 1` entries, and the parsed tree converts losslessly to a frontier via `to_frontier()`. The format’s originator is `zcashd` (retired): its Sapling final state streamed the legacy `SaplingMerkleTree` directly and its Orchard final state streamed through the `OrchardMerkleFrontierLegacySer` wrapper, so both pools use the same wire format, hex-encoded (`zcashd`, `src/rpc/blockchain.cpp:1402--1406, 1432--1436`).

Definition 11.5 (Subtree-root messages). Message `GetSubtreeRootsArg` carries `startIndex : uint32 = 1`, the index of the first subtree to return; `shieldedProtocol = 2` with enum `ShieldedProtocol { sapling = 0; orchard = 1 }` in v0.4.1; v0.5.0 adds `ironwood = 2`. Field `maxEntries : uint32 = 3`, where 0 means “return all entries”. Results stream in index — equivalently height — order. Message `SubtreeRoot` carries `rootHash : bytes = 2`, the 32-byte Merkle root of the completed subtree; `completingBlockHash : bytes = 3`; and `completingBlockHeight : uint64 = 4`. Field number 1 is absent, and no reserved statement documents the gap (`lightwallet-protocol/walletrpc/service.proto:186--200`).

Remark 11.6 (Byte-order warts). Two byte-order pitfalls attach to these messages. First, `SubtreeRoot.completingBlockHash` is sent in big-endian display order — `lightwalletd` sets it to `hash32.ReverseSlice(block.Hash)` — which is the *opposite* convention from `CompactBlock.hash`, parsed into little-endian wire order; the proto documents neither convention (`lightwalletd`, `frontend/service.go:896--900`; `parser/block.go:59--61, 115`). The reference client sidesteps the trap by ignoring `completingBlockHash` entirely, reading only `rootHash` and `completingBlockHeight` (`zcash_client_backend`, `src/sync.rs:247--253`). Zaino constructs the field as the block hash’s `bytes_in_display_order()` (`zaino-state/src/indexer.rs:854--861`), agreeing with `lightwalletd`’s big-endian display order; whether any deployed client consumes `completingBlockHash` remains an open question (the reference client ignores it). Second, a `zcashd` release note records that the serialised `finalRoot` field of `z_gettreestate` was byte-flipped relative to `getrawtransaction` output in releases up to and including v5.2.0, later rectified (`zcashd`, `src/rpc/blockchain.cpp:1326--1330`, help text).

11.3 Server-side derivation

Tree states. The RPC originates in `zcashd`’s `z_gettreestate`: it accepts a block hash or height (negative heights allowed, `-1` denoting the last known valid block), requires the block to be on the main chain, and returns the block’s hash, height, and time together with one object per pool, `{skipHash, commitments : {finalRoot, finalState}}`. The `finalState` — the serialised tree of Definition 11.4 — is present only when the block’s anchor is retrievable from the coins view (`GetSaplingAnchorAt / GetOrchardAnchorAt`); otherwise `skipHash` names the most recent prior block that has one, stopping at the pool’s activation height (`zcashd`, `src/rpc/blockchain.cpp:1303--1455`).

The proxy wraps this in a retry loop. Handler `GetTreeState` of `lightwalletd` rejects a request specifying neither height nor hash, marshals a height as a decimal string and a hash

as big-endian hex, and calls `z_gettreestate`; if the reply's Sapling `finalState` is empty and its Sapling `skipHash` is nonempty, it re-issues the RPC at the `skipHash`, iterating until a final state is found or no skip hash remains, and failing with “`z_gettreestate did not return treestate`” otherwise (`lightwalletd`, `frontend/service.go:311--386`). The walk terminates in practice because `zcashd` hard-stops it at the Sapling activation height (the `saplingActive` lambda; `zcashd`, `src/rpc/blockchain.cpp:1408--1418`). Handler `GetLatestTreeState` calls `getBlockchaininfo`, takes its `Blocks` value as the latest height, and delegates to `GetTreeState` at that height (`lightwalletd`, `frontend/service.go:388--401`).

Remark 11.7 (The skip walk keys on Sapling only). Only the Sapling `skipHash` is consulted: when the walk lands on an earlier block, the returned `TreeState` carries that earlier block's height, hash, and Orchard tree — not the Orchard state of the requested block (`lightwalletd`, `frontend/service.go:311--386`).

Subtree roots. The originating RPC is `z_getsubtreesbyindex` “pool” `start_index` (`limit`), added in `zcashd` v5.6.0 and gated behind `-experimentalfeatures + -lightwalletd`: it returns roots of complete 2^{16} -leaf subtrees of the given pool's note commitment tree, each with `end_height`, the height of the block containing the note commitment that completed the subtree, reading `SubtreeData` records from the coins view (`pcoinsTip->GetSubtreeData(pool, index)`) until data is absent or the limit is reached (`zcashd`, `src/rpc/blockchain.cpp:1457--1539`; `doc/release-notes/release-notes-5.6.0.md:28`). The records are maintained by the validator's incremental tree: `zcashd` stores `SubtreeData{leadbyte, root, nHeight}` and `LatestSubtree{leadbyte, index, root, nHeight}` records under a `uint64_t` subtree index, fixes `TRACKED_SUBTREE_HEIGHT = 16`, and produces a completed subtree root exactly when the tree size crosses a 2^{16} boundary (`zcashd`, `src/zcash/IncrementalMerkleTree.hpp:20--70, 409--411`; `IncrementalMerkleTree.cpp:913--925`):

```
current_subtree_index() = size() >> TRACKED_SUBTREE_HEIGHT.
```

Handler `GetSubtreeRoots` of `lightwalletd` accepts only Sapling or Orchard and forwards to `z_getsubtreesbyindex(pool, startIndex[, maxEntries when > 0])`; for each returned subtree it fetches the block at `end_height`, populates the two completing-block fields from it, and streams the assembled `SubtreeRoot` (`lightwalletd`, `frontend/service.go:832--907`).

Example 11.8 (Mainnet Sapling subtrees 0 and 1). A worked datum recorded in `lightwalletd` source (a quoted `z_getsubtreesbyindex` transcript, `common/common.go:178--210`): Sapling subtree 0 — outputs 0 through 65535 — has root

```
754bb593ea42d231a7ddf367640f09bbf59dc00f2c1d2003cc340e0c016b5b13
```

with `end_height = 558822`, and subtree 1 has root

```
03654c3eacbb9b93e122cf6d77b606eae29610f4f38a477985368197fd68e02d
```

with `end_height = 670209`: measured from Sapling activation at height 419,200, the first 2^{17} Sapling outputs span roughly the chain's first 250,000 post-activation blocks.

Deployed validator. Zebra implements both `z_gettreestate` and `z_getsubtreesbyindex` in its RPC surface, so a Zebra-backed indexer retains tree synchronisation after `zcashd`'s retirement. The successor indexer serves the same gRPC API: Zaino's `get_tree_state` proxies `z_get_treestate` to the backing validator and hex-encodes the final states (at HEAD it also populates an `ironwood_tree` field), its `get_latest_tree_state` reads the chain height and delegates, and its `get_subtree_roots` fetches each completing block via `z_get_block(end_height)`. One behavioural divergence: Zaino maps the proto's `uint32` `startIndex/maxEntries` down to

u16 and returns `InvalidArgument` when either exceeds 65535 — semantically safe, since a depth-32 tree has at most 2^{16} shards, but stricter than `lightwalletd/zcashd`, whose u64 subtree index yields an empty stream instead (`src/chain_index.rs:509--514` versus `zcashd src/zcash/IncrementalMerkleTree.hpp:20--70`). Both backends delegate to the validator: `NodeBackedChainIndex::get_subtree_roots` calls the blockchain source; `Validator Connector::State` issues `ReadRequest::`{`Sapling`,`Orchard`,`Ironwood`}`Subtrees` to Zebra’s `ReadStateService`, and `::Fetch` proxies `z_getsubtreesbyindex`; only the mockchain test source self-computes.

11.4 Client state: shards, cap, and retention

The client-side container is `shardtree`: “a sparse binary Merkle tree of the specified depth, represented as an ordered collection of subtrees (shards) of a given maximum height”, plus a *cap* — the tree above the shard roots — and a checkpoint set enabling truncation. The root sits at address $(\text{DEPTH}, 0)$, shards at level `SHARD_HEIGHT`, and `max_subtree_index() = 2^{\text{DEPTH}-\text{SHARD_HEIGHT}} - 1. The wallet API instantiates one such tree per pool as ShardTree(Store, DEPTH = 32, SHARD_HEIGHT = 16) with block heights as checkpoint identifiers, via with_sapling_tree_mut / with_orchard_tree_mut / with_ironwood_tree_mut. The shard height is documented as chosen to conform “to the structure of subtree data returned by lightwalletd when using the GetSubtreeRoots GRPC call”: SAPLING_SHARD_HEIGHT = ORCHARD_SHARD_HEIGHT = 16, half the tree depth of each pool. The geometry — height-16 shards of 2^{16} leaves under the 16-level cap, at most 2^{16} shards — is the Ironwood Guide’s construction (§“Resting: the note in the commitment tree”); the deployed constants realise it exactly. The SQLite backend constructs each tree as ShardTree::new(store, PRUNING_DEPTH) with PRUNING_DEPTH = 100: at most 100 ordinary recent checkpoints are retained before pruning. Checkpoints occur at represented block boundaries; a block containing no note commitment in any pool need not create a distinct checkpoint. From NU6.3 activation onward, established checkpoints at heights divisible by 288 are additionally retained as durable anchors and are exempt from that automatic pruning. The 100-block depth is the assumed maximum reorganisation depth and ZIP 307’s cache-size guidance (§11.1).`

What survives pruning is governed by per-node *retention*:

- Ephemeral — prunable together with its siblings;
- Checkpoint{id, marking} — the position is retained (its value prunable) while the checkpoint lives, decaying to `Marked`, `Reference`, or `Ephemeral` per the marking when the checkpoint is removed;
- Marked — retained until explicitly removed; the wallet’s own note positions;
- Reference — retained until overwritten by a node with `Ephemeral`, `Checkpoint`, or `Marked` retention; downloaded subtree roots and seed frontiers.

11.5 Ingesting subtree roots

The reference sync driver begins *every* run by refreshing the shard roots: function `update_subtree_roots` downloads all subtree roots — the default `GetSubtreeRootsArg`, i.e. `startIndex = 0`, `maxEntries = 0` — for `Sapling` and, with the orchard feature, `Orchard` and `Ironwood`, converting each streamed message by

```
CommitmentTreeRoot::from_parts(completingBlockHeight, H::read(rootHash))
```

and passing the batches, each with start index 0, to `put_sapling_subtree_roots`, `put_orchard_subtree_roots`, and `put_ironwood_subtree_roots`. Type `CommitmentTreeRoot` pairs the subtree end height with the root hash, and the trait documentation fixes the meaning: each value “should be the Merkle root of a subtree ... containing $2^{\text{SHARD_HEIGHT}}$ note commitments”.

The SQLite backend implements all three puts via one generic procedure, `put_shard_roots` parameterised by the hash H , the tree depth, and the shard height:

- (1) no-op on empty input;
- (2) load the cap and treat it as a standalone tree of $\text{DEPTH} - \text{SHARD_HEIGHT}$ levels whose level-0 leaves are shard roots: the `LevelShifter` wrapper offsets the hash,

$$\begin{aligned} \text{emptyLeaf} &= H.\text{emptyRoot}(\text{SHARD_HEIGHT}), \\ \text{combine}(\ell, a, b) &= H.\text{combine}(\ell + \text{SHARD_HEIGHT}, a, b), \end{aligned}$$

and the downloaded roots are batch_inserted at `Position::from(startIndex)` with `Retention::Reference`;

- (3) write the cap back;
- (4) check contiguity over $[\text{startIndex}, \text{startIndex} + n)$: `check_shard_discontinuity` rejects a call whose shard-index range is neither overlapping nor immediately adjacent to the stored $[\text{min}(\text{shardIndex}), \text{max}(\text{shardIndex}) + 1)$ range, raising `Error::SubtreeDiscontinuity` — subtree roots must arrive without gaps;
- (5) upsert each root into `{pool}_tree_shards(shard_index, subtree_end_height, root_hash, shard_data)`; on conflict only the end height and root hash are updated, so existing scanned shard data is never clobbered, and the fresh `shard_data` written for new rows is a single root leaf with `RetentionFlags::EPHEMERAL`.

The downloaded root is cached beside any persisted subtree rather than merged eagerly — “we want to avoid deserializing the subtree just to annotate its root node, so we simply cache the downloaded root alongside of any already-persisted subtree” — and the read path resolves the pair: `get_shard` deserialises `shard_data` and applies `reannotate_root(Some(rootHash))`.

Remark 11.9 (Idempotent full re-download). The driver fixes `startIndex = 0` on every run although `put_shard_roots` supports arbitrary start indices; the upsert path makes the repetition idempotent, at the cost of re-transferring one 32-byte root (plus completing-block metadata) per completed shard per pool on every sync (`zcash_client_backend, src/sync.rs:80--86, 228--299`). As of 2026-07-15 (mainnet tip 3,413,101): 1,128 complete Sapling subtrees (last completing at block 3,390,009) plus 765 Orchard (3,388,160), so each sync re-downloads 1,893 roots — 60,576 bytes of root data, 136,378 bytes of `SubtreeRoot` messages, 145,843 bytes with the five-byte gRPC frame per stream element (computed by script; counts cross-check against `ChainMetadata` at block 3,413,000: $\lceil 73,932,454/2^{16} \rceil = 1,128$, $\lceil 50,191,264/2^{16} \rceil = 765$). No source documents the fixed `startIndex = 0` as intentional.

11.6 Frontiers as scan seeds

A `TreeState` becomes client state through `TreeState::to_chain_state()`: it hex-decodes the block hash and reverses its bytes (“Zcashd hex strings for block hashes are byte-reversed”), and parses each pool’s legacy tree (Definition 11.4) into a frontier; an *empty* tree-state string parses as the empty tree, the correct state before a pool’s activation. The result is

```
ChainState{ blockHeight; blockHash;
            finalSaplingTree : Frontier<32>; finalOrchardTree : Frontier<32>;
            finalIronwoodTree : Frontier<32>; }
```

“the final note commitment tree state for each shielded pool, as of a particular block height” — the Orchard and Ironwood frontiers sit behind the orchard feature.

Every scanned batch is seeded with the chain state of the block immediately preceding it, and the seed is verified against the server’s own metadata before any tree is touched:

$\text{fromHeight} = \text{seed.blockHeight} + 1$ (asserted), $\text{seed.treeSize}_P + |\text{cmp}_P(b_0)| = \text{finalTreeSize}_P(b_0)$

per pool P , where b_0 is the batch’s first block and the final tree size is read from the block’s chainMetadata; violation yields error `NonSequentialBlocks`. The metadata sizes are what give the scanner absolute leaf positions when no prior metadata is available. Without positions it cannot derive Sapling nullifiers or build any pool’s witnesses (`TreeSizeUnknown`, a hard error); Orchard and Ironwood nullifier derivation itself is position-independent, and a size contradicted by block contents is `TreeSizeMismatch`, a continuity error triggering the rewind of §10.7 (`lightwallet-protocol/walletrpc/compact_formats.proto:14--17,31--40`).

The batch’s note commitments are then assembled into located subtrees in parallel — rayon, chunks of 1024 commitments, `LocatedTree::from_iter` at the shard level — starting at `Position::from(seed.treeSize_P)`, and each pool’s tree is updated by `update_tree`:

- (1) insert the seed frontier (`tree.insert_frontier`) with retention `Checkpoint{id = frontierHeight, marking = Reference}` — the in-source comment: “we insert the frontier with Checkpoint retention because we need to be able to truncate the tree back to this point”;
- (2) `insert_tree(subtree, checkpoints)` for each built subtree;
- (3) add any cross-pool checkpoints missing above the minimum retained checkpoint (the cross-pool checkpoint-unification rule itself, §10.6).

Inside `shardtree`, `insert_frontier` splits a non-empty frontier at the shard boundary: the leaf and the ommers (the sibling hashes a frontier stores, one per filled subtree along the leaf’s path to the root) below the shard level enter the shard containing the frontier’s position, the ommers at levels $\geq \text{SHARD_HEIGHT}$ enter the cap, checkpoint retention adds a checkpoint at the leaf position, and checkpoints beyond `max_checkpoints` are pruned. An empty frontier with checkpoint retention records a checkpoint for the empty tree (`Retention::Marked` is invalid there).

Frontier seeding is also how a wallet is born and how it rewinds. An `AccountBirthday` wraps “the chain state prior to the birthday height” plus an optional `recover_until`; `AccountBirthday::from_treestate` builds it directly from a `TreeState` for the last block *prior* to the birthday, with `height() = priorChainState.blockHeight + 1`, and account creation stores the birthday tree sizes and rewinds the wallet to the birthday chain state, so no block before the birthday needs to be replayed: the prior frontier summarises the required prefix (§10.2). Symmetrically, `truncate_to_chain_state` inserts each pool’s frontier from the target chain state with `Checkpoint{id = targetHeight, marking = None}`, creating the checkpoint on which the subsequent tree truncation lands (recovery flow, §10.7).

11.7 Shard completion and spendability

Two of the scan-queue priorities of §10.3 exist purely for this layer: `ChainTip` is documented as “blocks that must be scanned to complete the latest note commitment tree shard” and `FoundNote` as “blocks that must be scanned to complete note commitment tree shards adjacent to found notes”.

The tip shard. On each tip update the backend computes per pool the maximum `subtree_end_height` over the pool’s shards table — the end of the last complete shard known from `GetSubtreeRoots` — takes the minimum across pools, and enqueues

$$\lceil \max(\text{walletBirthday}, \text{minShardTip}), \text{tip} + 1 \rceil$$

at `ChainTip` priority, so the incomplete tip shard of every pool is scanned even before historic ranges (the full tip-update case analysis, §10.4).

Found notes. After a range is scanned, `scan_complete` maps each detected note position to its shard, $s = \lfloor p/2^{16} \rfloor$ (`Address::above_position(SHARD_HEIGHT, position)`), and extends the scanned range to that shard’s block extent — from the previous shard’s `subtree_end_height` (the pool activation height for shard 0, clamped to the wallet birthday) to the containing shard’s `subtree_end_height + 1` — enqueueing the extensions beyond the just-scanned range at `FoundNote` priority. Completing the shard’s block range is exactly what makes the note’s witness computable.

Witness derivation. The query `witness_at_checkpoint_id(position, id)` requires `position ≤ checkpoint.position()` (else `QueryError::NotContained`) and folds, from the note’s shard upward to the root, the root of each sibling address truncated at leaf count `positionas of + 1` — each sibling root drawn from the note’s own shard (scanned leaves), another shard’s cached root (`GetSubtreeRoots`, Definition 11.5), the cap, or a per-level empty-root constant beyond the tree size. A pruned node that cannot be replaced by an empty-root constant yields `QueryError::TreeIncomplete(addresses)` — the precise failure mode of an unready witness. The mathematics of the fold is the *Ironwood Guide*’s (§“Resting: the note in the commitment tree”); what this volume adds is the deployed readiness condition under which it cannot fail.

Spendability. That condition is enforced in SQL. Writing U_p for the rows of view `v_{pool}_shard_unscanned_ranges`, note selection excludes any note at position p for which some $(s, e, h_0, h_1) \in U_p$ satisfies

$$s \leq p < e \wedge h_0 \leq \text{anchorHeight} \wedge h_1 > \text{walletBirthday},$$

i.e. the note’s shard has an unscanned block range at or below the anchor; the same view drives `unscanned_tip_exists`, which invalidates anchors whose containing shard is not fully scanned. The views compute `startPosition = shardIndex << 16` and `endPositionExclusive = (shardIndex + 1) << 16`, set the shard’s block extent to begin at the previous shard’s end height (the pool activation height for the first shard), join the scan-queue rows whose block ranges overlap that extent, and filter to priority above `Scanned`. Unwinding the predicate recovers the tree-readiness condition: at a given anchor, the note’s own shard must be scanned up to the anchor, every earlier shard must be covered by a downloaded root, and the blocks from the last complete shard boundary to the anchor must have been scanned (the `ChainTip` range above). Whether the note is otherwise eligible to spend, which anchor a transaction selects, and at what confirmation depth, is covered in §4.1 and §6.6.

At the pinned `librustzcash` commit — part of the NU6.3 work of Remark 10.2, not of any release deployed with `lightwalletd v0.4.1` — notes additionally gain a `witness_stabilized` flag once their shard’s block extent is fully `Scanned` and the shard’s end height has at least `PRUNING_DEPTH` confirmations (`floor = maxScanned - (PRUNING_DEPTH - 1)`); stabilised notes bypass the confirmations check at spend time, and the tip shard by definition can never stabilise (`zcash_client_sqlite`, `src/wallet/scanning.rs:416--473`; `src/wallet/common.rs:700--724`).

What this buys, and what remains. The linear protocol’s impossibility is gone: a freshly restored wallet can seed its trees from the birthday frontiers instead of replaying the chain prefix, and can use one 32-byte root per completed shard. A note found during scanning becomes tree-ready once its own shard’s block extent and the tip range are scanned — not after the entire chain history. What remains is the trial decryption of every block from the birthday forward (§10) and the shard-completion scanning of this section; both are cost centres the proposed Tachyon proof-carrying wallet is designed to remove (Bowe–Miers, ePrint 2025/2031; `tachyon-zcash/tachyon`, `book/src/proof-tree.md @ 26fdc165`).

12 What the server learns

A light client outsources block storage and filtering to a server it does not trust with its keys, then reconstructs its own state by trial decryption on the device (*Ironwood Guide*, §“Delivery: the note

reaches its owner”). The keys never travel; the *requests* do. This section inventories what the deployed server-side observer learns from those requests: the stated threat model, the baseline that genuinely holds, the leaks in the deployed request pattern, the operator’s instruments for recording them, and the mitigations that actually ship. Implementation references use the commits pinned above; where the code diverges from ZIP text, we flag the divergence in place.

12.1 The stated security model

ZIP 307 names one primary goal for the light-client protocol.

Definition 12.1 (Payment detection privacy). The serving proxy must not learn which transactions are addressed to a given light wallet. Under the additional assumption of network privacy (“via Tor or similar”), the proxy must also be unable to link different connections or queries as deriving from the same wallet (ZIP 307 §“Security Model”).

The adversary of ZIP 307 is the node/proxy combination, modelled as *honest but curious*: it is trusted for chain-state correctness — the client accepts the block data it serves — but assumed to record and analyse everything it sees. The ZIP explicitly does not address spending notes privately (ZIP 307 §“Security Model”); transaction submission nonetheless flows through the same server, and we treat it in §12.5.

Remark 12.2 (The de facto wire specification). ZIP 307 has Status Draft — it was never finalised — even though the protocol it sketches is the deployed synchronisation layer. The deployed wire format has moved past the sketch: the ZIP’s CompactBlock carries vtx at field 3 and a BlockID, while the deployed message carries vtx at field 7 plus (in the v0.4.1 proto `lightwalletd` vendors) `protoVersion` — removed in v0.5.0 (§8.2) — `prevHash`, `time`, an (always unset) header, and `ChainMetadata`; the deployed CompactTx adds `txid`, `fee`, Orchard actions, and transparent vin/vout entries (CompactTxIn outpoints and full TxOuts) (ZIP 307 §“Compact Stream Format” vs. `lightwalletd`, `lightwallet-protocol/walletrpc/compact_formats.proto`). The `lightwallet-protocol` proto repository, not the ZIP, is the de facto wire specification. The planned in-protocol successor is emptier still: ZIP 314, “Privacy upgrades to the Zcash light client protocol”, has Status Reserved — a title with no content (`zips/zip-0314.rst`). The server implementation delegates likewise: the `lightwalletd` README directs developers to the external wallet app threat model for “important information about the security and privacy limitations of light wallets that use `lightwalletd`”; the repository itself contains no privacy analysis (`lightwalletd`, `README.md`).

12.2 What never leaves the device

The baseline holds. No RPC in the deployed service carries key material: across the full CompactTxStreamer surface (twenty methods), requests carry only public selectors or data such as heights, hashes, txids, transparent addresses, raw transaction bytes, pool or subtree selectors, and — for the testing-only Ping RPC — a duration (`lightwalletd`, `lightwallet-protocol/walletrpc/service.proto`). Incoming viewing keys remain on the device: trial decryption of compact outputs is performed client-side by `scan_block` over locally cached CompactBlocks, by the mechanism of the *Ironwood Guide*, §“Delivery: the note reaches its owner”; witness maintenance over the resulting positions is that of §“Resting: the note in the commitment tree”. Nullifier matching against the CompactSaplingSpend and CompactOrchardAction nullifier fields is likewise local.

The compact form is the reason a server mediates at all: the per-output and per-spend reductions, and the recomputed-commitment integrity substitute for the discarded AEAD tag, are those of §8 (Table 6; §8.4).

Remark 12.3 (The transparent boundary). The baseline above is a shielded-layer property. The same service carries transparent-address RPCs whose requests contain cleartext addresses

— `GetAddressTxids` and `GetAddressTransactions` (an address plus a block range), `GetAddressBalance(Stream)` (an address list), `GetAddressUtxos(Stream)` (an address list plus a start height) — so for the transparent layer the server learns the wallet’s addresses outright, by declaration rather than inference (`lightwalletd`, `walletrpc/service.proto`). We treat that request family as out of scope except where the shielded sync driver itself uses it (§12.3) and where deployed mitigations attach to it (§12.7).

12.3 The observable request schedule

What the server sees is a request schedule. ZIP 307 describes it in four phases: at first start, $A = \text{GetLatestBlock}$ plus `GetBlock` at the tip; on each subsequent synchronisation, $B = \text{GetLatestBlock}$ followed by `GetBlockRange(X,Y)` from the last-synced height X ; $C = \text{GetTransaction(txid)}$ for each transaction of interest after detection; and $D =$ the next `GetLatestBlock` plus `GetBlockRange(Y,Z)` (ZIP 307 §“Client-server interaction”).

The deployed driver refines this but does not change its shape. Per cycle, function run of `zcash_client_backend` (module `sync`, feature `sync`) issues: `GetSubtreeRoots` per shielded pool, `GetLatestBlock`, `GetAddressUtxosStream` over *all* transparent receivers of every account (clear-text, Remark 12.3), `GetTreeState` for the block before each scan range, and `GetBlockRange` for each suggested scan range in `batch_size` chunks. At the pinned commit the subtree-root fetch covers Sapling, Orchard, and Ironwood. Ironwood appears in no light-client ZIP — ZIP 2005 covers quantum recoverability and ZIP 258 only NU6.3 deployment — the request exists solely because the `librustzcash` proto fork defines `ShieldedProtocol.ironwood = 2` and the driver requests it (`src/sync.rs:289--296`; cf. Remark 10.1). The module’s own documentation notes that this simple implementation does not itself enhance transactions; enhancement is part of the documented flow (the crate-root diagram’s “Enhance transactions” step) and is driven by wallets via `transaction_data_requests` (`zcash_client_backend`, `src/lib.rs`, `src/sync.rs`).

The *order* of range requests is itself data. Scan ranges carry a priority (Table 9), and `suggest_scan_ranges` returns them in descending priority, then descending end height (§10.3), so `Verify` and `ChainTip` ranges near the tip are fetched first and `FoundNote` ranges pre-empt Historic backfill. The privacy consequence of that pre-emption is Remark 12.4.

12.4 Leaks in the deployed pattern

ZIP 307 asserts that compact-block synchronisation itself “leaks no information about which transactions the light client is interested in” because the compact stream is a broadcast medium (ZIP 307 §“Transaction privacy”). The same document contradicts the claim in the immediately preceding section, for the request pattern actually deployed: “The above interaction reveals to the server at the start of each synchronisation phase (B and D) the block height which the light client had previously synchronised to. This is an information leak under our security model” (ZIP 307 §“Block privacy via bucketing”). Compact range contents alone do not reveal which outputs trial-decrypt successfully, but the requests’ bounds and timing — and, when an integrating wallet performs them, transaction-specific follow-up requests — provide additional observables. We take them in turn.

Historic backfill reveals an approximate birthday. On account creation, the scan queue marks the range from Sapling activation up to the account birthday as `Ignored` — never downloaded — and queues scanning from the birthday itself. The minimum historic start height visible across the request schedule therefore approximates the wallet’s earliest account birthday. Priority scheduling means this need not be the first range requested, and importing an older account can later lower it.

Range boundaries reveal scan state and progress. In ZIP 307’s linear schedule, a range start is the prior sync height, which is the leak the ZIP concedes above. The deployed priority queue can instead request a verification, shard-completion, or historic range, so its bounds expose scan state and

progress but are not invariably the preceding sync height. ZIP 307’s proposed mitigation is bucketing: for a bucket size N , phases B and D would request

`GetBlockRange( $X - (X \bmod N)$ ,  $Y$ )` instead of `GetBlockRange( $X$ ,  $Y$ )`,

hiding the exact previously-synced height within an N -block bucket (ZIP 307 §“Block privacy via bucketing”). The ZIP marks this “a proposed enhancement that has not been implemented”, and the deployed code confirms: `download_blocks` issues `GetBlockRange` with start and end exactly equal to the suggested scan-range bounds, with no rounding (`zcash_client_backend`, `src/sync.rs`).

Wallet-driven enhancement can follow detection with a fetch. During compact-block scanning, function `put_blocks` records, per scanned block, exactly the subset of transactions relevant to this wallet, and queues a retrieval request for each. An integrating wallet that consumes `transaction_data_requests` later receives `Enhancement(txid)` or `GetStatus(txid)`, and may resolve it via the `GetTransaction` RPC. The simple sync module does not itself perform enhancement, and the API imposes no requirement that a fetch occur immediately after its containing range. When a wallet does issue these requests without cover, however, the selected txids reveal transactions it is enhancing or status-checking; that set is sensitive but can also include sent transactions and transparent-graph dependencies.

Here the code diverges from the ZIP, in the unusual direction: the ZIP is stricter than the deployment. ZIP 307’s full-transaction mitigation is normative — the client “SHOULD obscure the exact transactions of interest by downloading numerous uninteresting transactions as well”, “SHOULD download all transactions in any block from which a single full transaction is fetched”, and “MUST convey to the user that fetching full transactions will reduce their privacy” (ZIP 307 §“Transaction privacy”). No decoy, chaff, or cover-traffic code exists anywhere in `librustzcash` or `lightwalletd` (negative grep over both repositories at the pinned commits), and the deployed RPC surface offers no bulk full-transaction fetch to implement the second SHOULD with — `GetBlock` returns compact data (`lightwalletd`, `lightwallet-protocol/walletrpc/service.proto`). The protocol does not prevent an individual wallet from issuing extra `GetTransaction` requests as cover; the point is that neither pinned reference library supplies that policy automatically.

Enhancement fetches carry no timing decorrelation. The deployed request taxonomy is

`TransactionDataRequest` $\in$ { `GetStatus(txid)`, `Enhancement(txid)`,
`TransactionsInvolvingAddress{..., request_at, ...}`,
`GetSpendingTx(.)` }

(the last two variants are feature-gated). Only `TransactionsInvolvingAddress` carries a `request_at` field, documented as the instant chosen so as to “decorrelate this request to a light wallet server from other requests made by the same caller”, ignorable when the supplier of chain data is “private, trusted-for-privacy”. The shielded-note variants `Enhancement` and `GetStatus` have no analogous jitter: the SQLite backend emits them with no `request_at`. The one deployed timing defence (§12.7) protects transparent address checks, not the shielded fetch-on-detect path.

Remark 12.4 (FoundNote clustering — an inferred leak). When a note is found, `scan_complete` inserts scan ranges at priority `FoundNote` extending the scanned range so as to complete the note-commitment-tree shard(s) containing the found note, and `suggest_scan_ranges` orders by priority, so the server observes out-of-order, high-priority `GetBlockRange` fetches that can narrow the blocks around detected notes (`zcash_client_sqlite`, `src/wallet/scanning.rs`; `zcash_client_backend`, `src/data_api/scanning.rs`). No source documents this as a leak — neither ZIP 307 nor the code documentation — and the statement above is this volume’s own inference from the scan-queue code. It requires adversarial review — in particular, independent confirmation that `FoundNote` ranges surface to the server as distinct requests rather than being absorbed into contiguous suggested ranges — before it may be cited as fact.

Mempool polling can fingerprint the session. If a client calls `GetMempoolTx`, field `exclude_txid_suffixes` of the `GetMempoolTx` request carries suffixes of txids the client already holds, so a mempool-polling client additionally reveals which mempool transactions it has already seen, a session-linkable fingerprint (`lightwalletd`, `walletrpc/service.proto`). No source documents this as a leak; we record it for completeness.

12.5 The submission channel

Spending is out of ZIP 307’s stated scope (§12.1), but the deployed submission path runs through the same server when a wallet chooses that path. The `SendTransaction` handler hex-encodes the received raw transaction and forwards it to the backing node’s `sendrawtransaction` JSON-RPC (`lightwalletd`, `frontend/service.go`), so the operator holds the complete transaction *before* network broadcast and can link it to the submitting IP address, the TLS session, its timing, and — absent circuit isolation (§12.7) — to the same session’s sync and enhancement traffic. The backing full node also sees the transaction and the `GetTransaction` txid pattern, but without client IPs: `lightwalletd` terminates the client connections. The lifecycle the transaction then follows — store-then-broadcast, expiry, confirmation counting under the `ConfirmationsPolicy` — is described in §6.

12.6 The operator’s instruments

What the operator *can* record is a structural property, independent of any logging switch. The gRPC endpoint is operator-terminated TLS, so every connection exposes the peer IP and its open and close times — the peer may be a Tor exit or another proxy, not the wallet’s own IP. A Prometheus gauge, `grpc_server_connections_current`, tracks the live connection count, and `go-grpc-prometheus` interceptors on both the unary and streaming paths export per-method request counts and latency histograms over the HTTP endpoint (default `127.0.0.1:9068`) (`lightwalletd`, `cmd/connstats.go`, `cmd/root.go`).

Logging narrows the distance between can and does. Most handlers log request parameters at Debug level — including `GetBlockRange`, `GetTransaction`, `SendTransaction`, `GetAddressBalance`, and `GetAddressUtxos` — and ten unary return paths log full responses at Trace level; `GetLightdInfo` and the testing-only `Ping` have no equivalent handler log (`lightwalletd`, `frontend/service.go`). The default level is Info (`cmd/root.go`), so `--log-level 5` enables the Debug request logs and can record block ranges, txids, raw transactions, and transparent addresses for the named handlers, to the default log file `./server.log`, without client consent or notice. A separate unary interceptor, enabled by `--grpc-logging-insecure` (default `false`; the flag name itself marks the hazard), logs {peer address, method, duration, error} per request, and carries the source comment “TODO: anonymize the addresses. cryptopan?” (`lightwalletd`, `common/logging/logging.go`, `cmd/root.go`).² The project’s own architecture document is candid: “Logging may capture identifiable user data. It hasn’t received any privacy analysis yet and makes no attempt at sanitization” (`lightwalletd`, `docs/architecture.md`).

At the pinned Zaino revision, the successor indexer changes none of this structurally: it serves the full `CompactTxStreamer` surface; every handler logs the method name at Info level — the literal message is `[TEST] received call`, apparent leftover test instrumentation in production code — and, under the `prometheus` feature, emits per-method request counts, duration histograms, and error counters. The metric names are `zaino.grpc.requests_total`, `zaino.grpc.request_duration_seconds`, and `zaino.grpc.errors_total`, labelled by method and code, with no peer labels; no peer-address logging exists in the dispatch path (`zaino-serve`, `src/rpc/grpc/service.rs`, `src/lib.rs`). Its README acknowledges the setting: “Due to current potential data leaks / security weaknesses ... there is a need to use anonymous transport protocols

²The interceptor is wired on the unary path only, so streaming RPCs — including `GetBlockRange`, the bulk of sync traffic — never pass through it; the flag logs a skewed subset of traffic. Whether this is intentional is not determinable from the source.

(such as Nym or Tor) to obfuscate clients’ identities from Zcash’s indexing servers (Lightwalletd, Zcashd, Zaino)”, and the serve crate’s documentation states the ingestor “has been built so that other ingestors may be added that use different transport protocols (Nym, TOR)” — a capability anticipated, not implemented: no Nym, arti, or onion code exists in the workspace (zaino, README.md; zaino-serve, src/lib.rs; negative grep over packages/). Zaino is pre-1.0; all of the above is cited at commit 0e057e22 and may move.

12.7 Deployed mitigations

Two mitigations ship in the client stack. Neither touches the shielded fetch-on-detect leak of §12.4.

Transport: Tor. Crate `zcash_client_backend` ships an arti-based Tor client behind feature `tor`. Method `Client::connect_to_lightwalletd` tunnels the tonic gRPC channel through Tor (behind the additional feature `lightwalletd-tonic-tls-webpki-roots`), and `Client::isolated_client` returns a handle whose streams never share circuits with the parent, documented for “when you want separate parts of your program to each have a `tor::Client` handle, but where you don’t want their activities to be linkable to one another over the Tor network” (`zcash_client_backend`, `src/tor.rs`, `src/tor/grpc.rs`). Circuit isolation is a tool for removing shared-circuit linkage between the submission channel of §12.5 and sync traffic; it does not exclude timing, content, or application-level correlation. We classify this as *capability-in-librustzcash*, *deployment-status-unknown*: whether any deployed wallet routes `CompactTxStreamer` traffic through it by default — and whether `SendTransaction` uses a circuit isolated from sync — is not determinable from the local repositories.

Timing: exponential jitter, transparent addresses only. The SQLite backend schedules ephemeral-address checks by sampling the next check time from an exponential distribution,

$$t_{\text{next}} := t_{\text{from}} + \Delta, \quad \Delta \sim \text{Exp}(\lambda), \quad \lambda = 1/T, \quad \mathbb{E}[\Delta] = T,$$

with the expected interval T set to $(24 \cdot 60 \cdot 60)/n$ seconds for n tracked ephemeral addresses — one expected check per address per day — over a shuffled address order, “so that we don’t always check them in the same order”. The intent is explicit in the API documentation: only one such query should be performed at a time, “to avoid linking multiple transparent addresses as belonging to the same wallet in the view of the light wallet server” (`TransparentAddressMetadata::next_check_time`). No analogous jitter exists for shielded-note enhancement (§12.4).

12.8 Summary, and the structural reading

Table 10 collects the observables.

The structural reading is more limited: an unmitigated server-visible request schedule exposes connection metadata and range boundaries, while optional enhancement, mempool, and submission calls expose more when a wallet uses them. These observations are not properties of compact-block contents alone, nor are their present strengths inevitable: Tor, request bucketing, jitter, batching, and cover traffic can alter them, although the reference implementations do not supply all of those policies. Within the deployed protocol the specified upgrade path is only a stub (ZIP 314, Reserved, Remark 12.2); one proposed alternative is architectural. Project Tachyon proposes to remove exactly this layer: under its fold the wallet no longer scans, and its designers describe synchronization services as “fully oblivious to their clients’ transaction details” (Bowe–Miers, ePrint 2025/2031, §1.3; `tachyon-zcash/tachyon`, `book/src/proof-tree.md` @ 26fdc165). The ePrint gives no formal service threat model and calls timing-correlation linkage potentially fatal. The present section is the baseline against which that design must be measured.

Observable	Reveals	Mitigation status
Peer IP, connection timing	network endpoint; activity windows, not necessarily real-world identity	Tor capability in zcash_client_backend; wallet deployment unknown
Lowest GetBlockRange start	earliest visible historic backfill; approximate account birthday	none
Range boundaries	scan state and progress; often a resume height	bucketing proposed (ZIP 307), unimplemented
Out-of-order FoundNote ranges	can narrow blocks around detected notes	none (inferred; Remark 12.4)
Wallet-driven GetTransaction	txids selected for enhancement or status checks	decoys normative (SHOULD); no reference-library policy
exclude_txid_suffixes, if mempool polling is used	mempool transactions already seen	none
SendTransaction, if this submission path is used	full transaction, pre-broadcast, linked to origin	circuit-isolation capability, deployment unknown
Transparent-address RPCs	addresses outright	exponential jitter and shuffling (ephemeral addresses only)

Table 10: What the server learns from the deployed light-client protocol, and the status of each mitigation. Sources: §12.4 through §12.7.